

red_neuronal_convolutacional_cajas en R

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```
library(keras)
library(tensorflow)
library(ggplot2)
library(gridExtra)
library(png)

# Configuración inicial
dataset_path <-
"C:/Users/Johnny/Desktop/Entorno_virtual/factory_io/dataset"
epochs <- as.integer(10)
batch_size <- as.integer(32)
target_size <- as.integer(c(128, 128))
classes <- c('con_marca', 'sin_marca')

# 1. Preprocesamiento y aumento de datos
train_datagen <- image_data_generator(
  rescale = 1/255, # Normalización de píxeles
  rotation_range = 20, # Rotación aleatoria
  width_shift_range = 0.2, # Desplazamiento horizontal aleatorio
  height_shift_range = 0.2, # Desplazamiento vertical aleatorio
  horizontal_flip = TRUE, # Volteo horizontal aleatorio
  validation_split = 0.2 # 20% de los datos para validación
)

# Generadores de datos
train_generator <- flow_images_from_directory(
  directory = dataset_path,
  generator = train_datagen,
  target_size = target_size,
  color_mode = "grayscale",
  batch_size = batch_size,
  class_mode = "binary",
  classes = classes,
  subset = "training",
  seed = 123
)

## Found 8016 images belonging to 2 classes.

validation_generator <- flow_images_from_directory(
  directory = dataset_path,
  generator = train_datagen,
  target_size = target_size,
```

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    color_mode = "grayscale",
    batch_size = batch_size,
    class_mode = "binary",
    classes = classes,
    subset = "validation",
    seed = 123
)

## Found 2004 images belonging to 2 classes.

# 2. Construir el modelo
inputs <- layer_input(shape = c(target_size, 1))

outputs <- inputs %>%
  layer_conv_2d(filters = 32, kernel_size = c(3, 3), activation = "relu")
%>%
  layer_max_pooling_2d(pool_size = c(2, 2)) %>%
  layer_conv_2d(filters = 64, kernel_size = c(3, 3), activation = "relu")
%>%
  layer_max_pooling_2d(pool_size = c(2, 2)) %>%
  layer_conv_2d(filters = 128, kernel_size = c(3, 3), activation =
"relu") %>%
  layer_max_pooling_2d(pool_size = c(2, 2)) %>%
  layer_flatten() %>%
  layer_dense(units = 128, activation = "relu") %>%
  layer_dense(units = 1, activation = "sigmoid")

model <- keras_model(inputs = inputs, outputs = outputs) # Capa de salida
para clasificación binaria

# 3. Compilar el modelo
model$compile(
  optimizer = optimizer_adam(),
  loss = "binary_crossentropy",
  metrics = list("accuracy") # Nota el uso de list() en lugar de c()
)

# Resumen del modelo
summary(model)

## <Functional name=functional, built=True>

# Callbacks
checkpoint_dir <- "F:/Desktop/Uni/IA/R/saved_models"
if (!dir.exists(checkpoint_dir)) {
  dir.create(checkpoint_dir)
}

checkpoint_callback <- callback_model_checkpoint(
  filepath = file.path(checkpoint_dir, "weights-{epoch:02d}-

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{val_accuracy:.2f}.keras"), # Cambiado a .keras
  monitor = "val_accuracy",
  save_best_only = TRUE,
  mode = "max",
  verbose = 1
)

early_stop <- callback_early_stopping(
  monitor = "val_loss",
  patience = 5,
  restore_best_weights = TRUE
)

# 4. Entrenar el modelo
history <- model$fit(
  x = train_generator,
  epochs = as.integer(epochs),
  validation_data = validation_generator,
  callbacks = list(checkpoint_callback, early_stop),
  verbose = 1L
)

## Epoch 1/10
##
## [1m 1/251[0m [37m-----[0m [1m8:50[0m 2s/step -
accuracy: 0.5000 - loss: 0.6940
## [1m 2/251[0m [37m-----[0m [1m1:11[0m 285ms/step -
accuracy: 0.5000 - loss: 0.6964
## [1m 3/251[0m [37m-----[0m [1m1:09[0m 281ms/step -
accuracy: 0.4861 - loss: 0.7011
## [1m 4/251[0m [37m-----[0m [1m1:09[0m 283ms/step -
accuracy: 0.4837 - loss: 0.7018
## [1m 5/251[0m [37m-----[0m [1m1:10[0m 285ms/step -
accuracy: 0.4845 - loss: 0.7019
## [1m 6/251[0m [37m-----[0m [1m1:09[0m 284ms/step -
accuracy: 0.4827 - loss: 0.7021
## [1m 7/251[0m [37m-----[0m [1m1:09[0m 284ms/step -
accuracy: 0.4833 - loss: 0.7020
## [1m 8/251[0m [37m-----[0m [1m1:08[0m 284ms/step -
accuracy: 0.4815 - loss: 0.7019
## [1m 9/251[0m [37m-----[0m [1m1:08[0m 285ms/step -
accuracy: 0.4793 - loss: 0.7017
## [1m 10/251[0m [37m-----[0m [1m1:08[0m 284ms/step -
accuracy: 0.4767 - loss: 0.7015
## [1m 11/251[0m [37m-----[0m [1m1:08[0m 285ms/step -
accuracy: 0.4749 - loss: 0.7013
## [1m 12/251[0m [37m-----[0m [1m1:09[0m 290ms/step -
accuracy: 0.4735 - loss: 0.7011
## [1m 13/251[0m [32m-[0m[37m-----[0m [1m1:09[0m 294ms/step
- accuracy: 0.4737 - loss: 0.7009

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## [1m 14/251[0m [32m[0m[37m[0m [1m1:10[0m 298ms/step
- accuracy: 0.4743 - loss: 0.7007
## [1m 15/251[0m [32m[0m[37m[0m [1m1:10[0m 297ms/step
- accuracy: 0.4746 - loss: 0.7005
## [1m 16/251[0m [32m[0m[37m[0m [1m1:09[0m 295ms/step
- accuracy: 0.4751 - loss: 0.7003
## [1m 17/251[0m [32m[0m[37m[0m [1m1:08[0m 294ms/step
- accuracy: 0.4752 - loss: 0.7001
## [1m 18/251[0m [32m[0m[37m[0m [1m1:08[0m 293ms/step
- accuracy: 0.4753 - loss: 0.6999
## [1m 19/251[0m [32m[0m[37m[0m [1m1:07[0m 292ms/step
- accuracy: 0.4755 - loss: 0.6998
## [1m 20/251[0m [32m[0m[37m[0m [1m1:08[0m 296ms/step
- accuracy: 0.4754 - loss: 0.6996
## [1m 21/251[0m [32m[0m[37m[0m [1m1:08[0m 297ms/step
- accuracy: 0.4754 - loss: 0.6994
## [1m 22/251[0m [32m[0m[37m[0m [1m1:08[0m 301ms/step
- accuracy: 0.4756 - loss: 0.6993
## [1m 23/251[0m [32m[0m[37m[0m [1m1:09[0m 306ms/step
- accuracy: 0.4756 - loss: 0.6992
## [1m 24/251[0m [32m[0m[37m[0m [1m1:09[0m 308ms/step
- accuracy: 0.4757 - loss: 0.6990
## [1m 25/251[0m [32m[0m[37m[0m [1m1:10[0m 310ms/step
- accuracy: 0.4756 - loss: 0.6989
## [1m 26/251[0m [32m[0m[37m[0m [1m1:09[0m 310ms/step
- accuracy: 0.4756 - loss: 0.6988
## [1m 27/251[0m [32m[0m[37m[0m [1m1:09[0m 311ms/step
- accuracy: 0.4757 - loss: 0.6987
## [1m 28/251[0m [32m[0m[37m[0m [1m1:09[0m 312ms/step
- accuracy: 0.4757 - loss: 0.6986
## [1m 29/251[0m [32m[0m[37m[0m [1m1:09[0m 312ms/step
- accuracy: 0.4760 - loss: 0.6984
## [1m 30/251[0m [32m[0m[37m[0m [1m1:09[0m 313ms/step
- accuracy: 0.4763 - loss: 0.6983
## [1m 31/251[0m [32m[0m[37m[0m [1m1:08[0m 312ms/step
- accuracy: 0.4766 - loss: 0.6982
## [1m 32/251[0m [32m[0m[37m[0m [1m1:08[0m 311ms/step
- accuracy: 0.4770 - loss: 0.6982
## [1m 33/251[0m [32m[0m[37m[0m [1m1:07[0m 311ms/step
- accuracy: 0.4774 - loss: 0.6981
## [1m 34/251[0m [32m[0m[37m[0m [1m1:07[0m 310ms/step
- accuracy: 0.4778 - loss: 0.6980
## [1m 35/251[0m [32m[0m[37m[0m [1m1:06[0m 310ms/step
- accuracy: 0.4783 - loss: 0.6979
## [1m 36/251[0m [32m[0m[37m[0m [1m1:06[0m 309ms/step
- accuracy: 0.4787 - loss: 0.6978
## [1m 37/251[0m [32m[0m[37m[0m [1m1:06[0m 312ms/step
- accuracy: 0.4791 - loss: 0.6977
## [1m 38/251[0m [32m[0m[37m[0m [1m1:06[0m 311ms/step
- accuracy: 0.4795 - loss: 0.6977
```

```
## [1m 39/251[0m [32m[0m[37m[0m [1m1:05[0m 311ms/step
- accuracy: 0.4799 - loss: 0.6976
## [1m 40/251[0m [32m[0m[37m[0m [1m1:05[0m 311ms/step
- accuracy: 0.4802 - loss: 0.6975
## [1m 41/251[0m [32m[0m[37m[0m [1m1:05[0m 310ms/step
- accuracy: 0.4805 - loss: 0.6975
## [1m 42/251[0m [32m[0m[37m[0m [1m1:05[0m 312ms/step
- accuracy: 0.4809 - loss: 0.6974
## [1m 43/251[0m [32m[0m[37m[0m [1m1:04[0m 312ms/step
- accuracy: 0.4812 - loss: 0.6973
## [1m 44/251[0m [32m[0m[37m[0m [1m1:04[0m 312ms/step
- accuracy: 0.4815 - loss: 0.6973
## [1m 45/251[0m [32m[0m[37m[0m [1m1:05[0m 316ms/step
- accuracy: 0.4819 - loss: 0.6972
## [1m 46/251[0m [32m[0m[37m[0m [1m1:04[0m 315ms/step
- accuracy: 0.4823 - loss: 0.6972
## [1m 47/251[0m [32m[0m[37m[0m [1m1:04[0m 315ms/step
- accuracy: 0.4827 - loss: 0.6971
## [1m 48/251[0m [32m[0m[37m[0m [1m1:03[0m 315ms/step
- accuracy: 0.4831 - loss: 0.6970
## [1m 49/251[0m [32m[0m[37m[0m [1m1:03[0m 315ms/step
- accuracy: 0.4834 - loss: 0.6970
## [1m 50/251[0m [32m[0m[37m[0m [1m1:03[0m 315ms/step
- accuracy: 0.4836 - loss: 0.6969
## [1m 51/251[0m [32m[0m[37m[0m [1m1:02[0m 314ms/step
- accuracy: 0.4839 - loss: 0.6969
## [1m 52/251[0m [32m[0m[37m[0m [1m1:02[0m 313ms/step
- accuracy: 0.4841 - loss: 0.6968
## [1m 53/251[0m [32m[0m[37m[0m [1m1:01[0m 313ms/step
- accuracy: 0.4843 - loss: 0.6968
## [1m 54/251[0m [32m[0m[37m[0m [1m1:01[0m 314ms/step
- accuracy: 0.4844 - loss: 0.6968
## [1m 55/251[0m [32m[0m[37m[0m [1m1:01[0m 314ms/step
- accuracy: 0.4846 - loss: 0.6967
## [1m 56/251[0m [32m[0m[37m[0m [1m1:01[0m 313ms/step
- accuracy: 0.4848 - loss: 0.6967
## [1m 57/251[0m [32m[0m[37m[0m [1m1:00[0m 313ms/step
- accuracy: 0.4849 - loss: 0.6966
## [1m 58/251[0m [32m[0m[37m[0m [1m1:00[0m 312ms/step
- accuracy: 0.4851 - loss: 0.6966
## [1m 59/251[0m [32m[0m[37m[0m [1m1:00[0m 313ms/step
- accuracy: 0.4852 - loss: 0.6966
## [1m 60/251[0m [32m[0m[37m[0m [1m1:00[0m 316ms/step
- accuracy: 0.4853 - loss: 0.6965
## [1m 61/251[0m [32m[0m[37m[0m [1m1:00[0m 317ms/step
- accuracy: 0.4854 - loss: 0.6965
## [1m 62/251[0m [32m[0m[37m[0m [1m59s[0m 317ms/step
- accuracy: 0.4856 - loss: 0.6965
## [1m 63/251[0m [32m[0m[37m[0m [1m59s[0m 317ms/step
- accuracy: 0.4857 - loss: 0.6964
```

```
## [1m 64/251[0m [32m-----[0m[37m-----[0m [1m59s[0m 318ms/step
- accuracy: 0.4858 - loss: 0.6964
## [1m 65/251[0m [32m-----[0m[37m-----[0m [1m59s[0m 318ms/step
- accuracy: 0.4860 - loss: 0.6963
## [1m 66/251[0m [32m-----[0m[37m-----[0m [1m58s[0m 318ms/step
- accuracy: 0.4862 - loss: 0.6963
## [1m 67/251[0m [32m-----[0m[37m-----[0m [1m58s[0m 318ms/step
- accuracy: 0.4863 - loss: 0.6963
## [1m 68/251[0m [32m-----[0m[37m-----[0m [1m58s[0m 319ms/step
- accuracy: 0.4865 - loss: 0.6963
## [1m 69/251[0m [32m-----[0m[37m-----[0m [1m58s[0m 319ms/step
- accuracy: 0.4867 - loss: 0.6962
## [1m 70/251[0m [32m-----[0m[37m-----[0m [1m57s[0m 320ms/step
- accuracy: 0.4868 - loss: 0.6962
## [1m 71/251[0m [32m-----[0m[37m-----[0m [1m58s[0m 324ms/step
- accuracy: 0.4870 - loss: 0.6962
## [1m 72/251[0m [32m-----[0m[37m-----[0m [1m58s[0m 325ms/step
- accuracy: 0.4871 - loss: 0.6961
## [1m 73/251[0m [32m-----[0m[37m-----[0m [1m57s[0m 324ms/step
- accuracy: 0.4873 - loss: 0.6961
## [1m 74/251[0m [32m-----[0m[37m-----[0m [1m57s[0m 325ms/step
- accuracy: 0.4875 - loss: 0.6961
## [1m 75/251[0m [32m-----[0m[37m-----[0m [1m57s[0m 326ms/step
- accuracy: 0.4877 - loss: 0.6961
## [1m 76/251[0m [32m-----[0m[37m-----[0m [1m57s[0m 327ms/step
- accuracy: 0.4879 - loss: 0.6960
## [1m 77/251[0m [32m-----[0m[37m-----[0m [1m56s[0m 327ms/step
- accuracy: 0.4880 - loss: 0.6960
## [1m 78/251[0m [32m-----[0m[37m-----[0m [1m56s[0m 329ms/step
- accuracy: 0.4882 - loss: 0.6960
## [1m 79/251[0m [32m-----[0m[37m-----[0m [1m56s[0m 329ms/step
- accuracy: 0.4883 - loss: 0.6960
## [1m 80/251[0m [32m-----[0m[37m-----[0m [1m56s[0m 328ms/step
- accuracy: 0.4885 - loss: 0.6959
## [1m 81/251[0m [32m-----[0m[37m-----[0m [1m55s[0m 328ms/step
- accuracy: 0.4886 - loss: 0.6959
## [1m 82/251[0m [32m-----[0m[37m-----[0m [1m55s[0m 328ms/step
- accuracy: 0.4888 - loss: 0.6959
## [1m 83/251[0m [32m-----[0m[37m-----[0m [1m55s[0m 329ms/step
- accuracy: 0.4889 - loss: 0.6959
## [1m 84/251[0m [32m-----[0m[37m-----[0m [1m55s[0m 330ms/step
- accuracy: 0.4890 - loss: 0.6958
## [1m 85/251[0m [32m-----[0m[37m-----[0m [1m55s[0m 332ms/step
- accuracy: 0.4891 - loss: 0.6958
## [1m 86/251[0m [32m-----[0m[37m-----[0m [1m54s[0m 333ms/step
- accuracy: 0.4892 - loss: 0.6958
## [1m 87/251[0m [32m-----[0m[37m-----[0m [1m54s[0m 333ms/step
- accuracy: 0.4893 - loss: 0.6958
## [1m 88/251[0m [32m-----[0m[37m-----[0m [1m54s[0m 332ms/step
- accuracy: 0.4894 - loss: 0.6958
```

```
## [1m 89/251[0m [32m-----[0m[37m-----[0m [1m53s[0m 332ms/step
- accuracy: 0.4895 - loss: 0.6957
## [1m 90/251[0m [32m-----[0m[37m-----[0m [1m53s[0m 332ms/step
- accuracy: 0.4897 - loss: 0.6957
## [1m 91/251[0m [32m-----[0m[37m-----[0m [1m53s[0m 333ms/step
- accuracy: 0.4898 - loss: 0.6957
## [1m 92/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 333ms/step
- accuracy: 0.4899 - loss: 0.6957
## [1m 93/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 333ms/step
- accuracy: 0.4900 - loss: 0.6957
## [1m 94/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 332ms/step
- accuracy: 0.4901 - loss: 0.6956
## [1m 95/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 333ms/step
- accuracy: 0.4903 - loss: 0.6956
## [1m 96/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 333ms/step
- accuracy: 0.4904 - loss: 0.6956
## [1m 97/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 333ms/step
- accuracy: 0.4905 - loss: 0.6956
## [1m 98/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 333ms/step
- accuracy: 0.4906 - loss: 0.6956
## [1m 99/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 333ms/step
- accuracy: 0.4907 - loss: 0.6955
## [1m100/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 332ms/step
- accuracy: 0.4908 - loss: 0.6955
## [1m101/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 332ms/step
- accuracy: 0.4909 - loss: 0.6955
## [1m102/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 331ms/step
- accuracy: 0.4910 - loss: 0.6955
## [1m103/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 331ms/step
- accuracy: 0.4911 - loss: 0.6955
## [1m104/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 331ms/step
- accuracy: 0.4912 - loss: 0.6955
## [1m105/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 330ms/step
- accuracy: 0.4913 - loss: 0.6955
## [1m106/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 329ms/step
- accuracy: 0.4913 - loss: 0.6954
## [1m107/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 329ms/step
- accuracy: 0.4914 - loss: 0.6954
## [1m108/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 329ms/step
- accuracy: 0.4915 - loss: 0.6954
## [1m109/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 329ms/step
- accuracy: 0.4915 - loss: 0.6954
## [1m110/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 328ms/step
- accuracy: 0.4916 - loss: 0.6954
## [1m111/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 329ms/step
- accuracy: 0.4917 - loss: 0.6954
## [1m112/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 329ms/step
- accuracy: 0.4918 - loss: 0.6954
## [1m113/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 329ms/step
- accuracy: 0.4919 - loss: 0.6953
```

```
## [1m114/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 328ms/step
- accuracy: 0.4919 - loss: 0.6953
## [1m115/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 328ms/step
- accuracy: 0.4920 - loss: 0.6953
## [1m116/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 328ms/step
- accuracy: 0.4920 - loss: 0.6953
## [1m117/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 329ms/step
- accuracy: 0.4921 - loss: 0.6953
## [1m118/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 329ms/step
- accuracy: 0.4921 - loss: 0.6953
## [1m119/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 328ms/step
- accuracy: 0.4921 - loss: 0.6953
## [1m120/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 328ms/step
- accuracy: 0.4922 - loss: 0.6953
## [1m121/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 328ms/step
- accuracy: 0.4922 - loss: 0.6952
## [1m122/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 328ms/step
- accuracy: 0.4923 - loss: 0.6952
## [1m123/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 328ms/step
- accuracy: 0.4923 - loss: 0.6952
## [1m124/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 328ms/step
- accuracy: 0.4924 - loss: 0.6952
## [1m125/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 327ms/step
- accuracy: 0.4924 - loss: 0.6952
## [1m126/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 327ms/step
- accuracy: 0.4925 - loss: 0.6952
## [1m127/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 327ms/step
- accuracy: 0.4925 - loss: 0.6952
## [1m128/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 327ms/step
- accuracy: 0.4926 - loss: 0.6952
## [1m129/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 327ms/step
- accuracy: 0.4926 - loss: 0.6952
## [1m130/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 326ms/step
- accuracy: 0.4927 - loss: 0.6951
## [1m131/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 326ms/step
- accuracy: 0.4927 - loss: 0.6951
## [1m132/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 326ms/step
- accuracy: 0.4928 - loss: 0.6951
## [1m133/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 326ms/step
- accuracy: 0.4929 - loss: 0.6951
## [1m134/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 325ms/step
- accuracy: 0.4929 - loss: 0.6951
## [1m135/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 325ms/step
- accuracy: 0.4930 - loss: 0.6951
## [1m136/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 325ms/step
- accuracy: 0.4930 - loss: 0.6951
## [1m137/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 325ms/step
- accuracy: 0.4930 - loss: 0.6951
## [1m138/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 325ms/step
- accuracy: 0.4931 - loss: 0.6951
```



```
## [1m139/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 326ms/step
- accuracy: 0.4931 - loss: 0.6951
## [1m140/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 326ms/step
- accuracy: 0.4932 - loss: 0.6950
## [1m141/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 326ms/step
- accuracy: 0.4932 - loss: 0.6950
## [1m142/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 325ms/step
- accuracy: 0.4933 - loss: 0.6950
## [1m143/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 325ms/step
- accuracy: 0.4933 - loss: 0.6950
## [1m144/251[0m [32m-----[0m[37m-----[0m [1m34s[0m 325ms/step
- accuracy: 0.4934 - loss: 0.6950
## [1m145/251[0m [32m-----[0m[37m-----[0m [1m34s[0m 325ms/step
- accuracy: 0.4934 - loss: 0.6950
## [1m146/251[0m [32m-----[0m[37m-----[0m [1m34s[0m 325ms/step
- accuracy: 0.4934 - loss: 0.6950
## [1m147/251[0m [32m-----[0m[37m-----[0m [1m33s[0m 325ms/step
- accuracy: 0.4935 - loss: 0.6950
## [1m148/251[0m [32m-----[0m[37m-----[0m [1m33s[0m 325ms/step
- accuracy: 0.4935 - loss: 0.6950
## [1m149/251[0m [32m-----[0m[37m-----[0m [1m33s[0m 325ms/step
- accuracy: 0.4935 - loss: 0.6950
## [1m150/251[0m [32m-----[0m[37m-----[0m [1m32s[0m 324ms/step
- accuracy: 0.4935 - loss: 0.6950
## [1m151/251[0m [32m-----[0m[37m-----[0m [1m32s[0m 324ms/step
- accuracy: 0.4936 - loss: 0.6949
## [1m152/251[0m [32m-----[0m[37m-----[0m [1m32s[0m 324ms/step
- accuracy: 0.4936 - loss: 0.6949
## [1m153/251[0m [32m-----[0m[37m-----[0m [1m31s[0m 323ms/step
- accuracy: 0.4936 - loss: 0.6949
## [1m154/251[0m [32m-----[0m[37m-----[0m [1m31s[0m 323ms/step
- accuracy: 0.4936 - loss: 0.6949
## [1m155/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 323ms/step
- accuracy: 0.4937 - loss: 0.6949
## [1m156/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 323ms/step
- accuracy: 0.4937 - loss: 0.6949
## [1m157/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 323ms/step
- accuracy: 0.4937 - loss: 0.6949
## [1m158/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 323ms/step
- accuracy: 0.4937 - loss: 0.6949
## [1m159/251[0m [32m-----[0m[37m-----[0m [1m29s[0m 323ms/step
- accuracy: 0.4938 - loss: 0.6949
## [1m160/251[0m [32m-----[0m[37m-----[0m [1m29s[0m 322ms/step
- accuracy: 0.4938 - loss: 0.6949
## [1m161/251[0m [32m-----[0m[37m-----[0m [1m28s[0m 322ms/step
- accuracy: 0.4938 - loss: 0.6949
## [1m162/251[0m [32m-----[0m[37m-----[0m [1m28s[0m 322ms/step
- accuracy: 0.4938 - loss: 0.6949
## [1m163/251[0m [32m-----[0m[37m-----[0m [1m28s[0m 322ms/step
- accuracy: 0.4939 - loss: 0.6949
```

```
## [1m164/251[0m [32m-----[0m[37m-----[0m [1m27s[0m 322ms/step
- accuracy: 0.4939 - loss: 0.6948
## [1m165/251[0m [32m-----[0m[37m-----[0m [1m27s[0m 322ms/step
- accuracy: 0.4939 - loss: 0.6948
## [1m166/251[0m [32m-----[0m[37m-----[0m [1m27s[0m 322ms/step
- accuracy: 0.4940 - loss: 0.6948
## [1m167/251[0m [32m-----[0m[37m-----[0m [1m26s[0m 321ms/step
- accuracy: 0.4940 - loss: 0.6948
## [1m168/251[0m [32m-----[0m[37m-----[0m [1m26s[0m 321ms/step
- accuracy: 0.4940 - loss: 0.6948
## [1m169/251[0m [32m-----[0m[37m-----[0m [1m26s[0m 321ms/step
- accuracy: 0.4941 - loss: 0.6948
## [1m170/251[0m [32m-----[0m[37m-----[0m [1m26s[0m 321ms/step
- accuracy: 0.4941 - loss: 0.6948
## [1m171/251[0m [32m-----[0m[37m-----[0m [1m25s[0m 321ms/step
- accuracy: 0.4941 - loss: 0.6948
## [1m172/251[0m [32m-----[0m[37m-----[0m [1m25s[0m 321ms/step
- accuracy: 0.4942 - loss: 0.6948
## [1m173/251[0m [32m-----[0m[37m-----[0m [1m25s[0m 321ms/step
- accuracy: 0.4942 - loss: 0.6948
## [1m174/251[0m [32m-----[0m[37m-----[0m [1m24s[0m 321ms/step
- accuracy: 0.4942 - loss: 0.6948
## [1m175/251[0m [32m-----[0m[37m-----[0m [1m24s[0m 321ms/step
- accuracy: 0.4942 - loss: 0.6948
## [1m176/251[0m [32m-----[0m[37m-----[0m [1m24s[0m 320ms/step
- accuracy: 0.4943 - loss: 0.6948
## [1m177/251[0m [32m-----[0m[37m-----[0m [1m23s[0m 320ms/step
- accuracy: 0.4943 - loss: 0.6948
## [1m178/251[0m [32m-----[0m[37m-----[0m [1m23s[0m 320ms/step
- accuracy: 0.4943 - loss: 0.6947
## [1m179/251[0m [32m-----[0m[37m-----[0m [1m23s[0m 320ms/step
- accuracy: 0.4944 - loss: 0.6947
## [1m180/251[0m [32m-----[0m[37m-----[0m [1m22s[0m 320ms/step
- accuracy: 0.4944 - loss: 0.6947
## [1m181/251[0m [32m-----[0m[37m-----[0m [1m22s[0m 320ms/step
- accuracy: 0.4944 - loss: 0.6947
## [1m182/251[0m [32m-----[0m[37m-----[0m [1m22s[0m 320ms/step
- accuracy: 0.4945 - loss: 0.6947
## [1m183/251[0m [32m-----[0m[37m-----[0m [1m21s[0m 320ms/step
- accuracy: 0.4945 - loss: 0.6947
## [1m184/251[0m [32m-----[0m[37m-----[0m [1m21s[0m 320ms/step
- accuracy: 0.4946 - loss: 0.6947
## [1m185/251[0m [32m-----[0m[37m-----[0m [1m21s[0m 319ms/step
- accuracy: 0.4946 - loss: 0.6947
## [1m186/251[0m [32m-----[0m[37m-----[0m [1m20s[0m 319ms/step
- accuracy: 0.4946 - loss: 0.6947
## [1m187/251[0m [32m-----[0m[37m-----[0m [1m20s[0m 319ms/step
- accuracy: 0.4947 - loss: 0.6947
## [1m188/251[0m [32m-----[0m[37m-----[0m [1m20s[0m 319ms/step
- accuracy: 0.4947 - loss: 0.6947
```

[illegible]

```
## [1m214/251[0m [32m-----[0m[37m---[0m [1m11s[0m 319ms/step  
- accuracy: 0.4954 - loss: 0.6945  
## [1m215/251[0m [32m-----[0m[37m---[0m [1m11s[0m 319ms/step  
- accuracy: 0.4954 - loss: 0.6945  
## [1m216/251[0m [32m-----[0m[37m---[0m [1m11s[0m 319ms/step  
- accuracy: 0.4955 - loss: 0.6945  
## [1m217/251[0m [32m-----[0m[37m---[0m [1m10s[0m 319ms/step  
- accuracy: 0.4955 - loss: 0.6945  
## [1m218/251[0m [32m-----[0m[37m---[0m [1m10s[0m 319ms/step  
- accuracy: 0.4955 - loss: 0.6945  
## [1m219/251[0m [32m-----[0m[37m---[0m [1m10s[0m 319ms/step  
- accuracy: 0.4955 - loss: 0.6945  
## [1m220/251[0m [32m-----[0m[37m---[0m [1m9s[0m 319ms/step -  
accuracy: 0.4955 - loss: 0.6945  
## [1m221/251[0m [32m-----[0m[37m---[0m [1m9s[0m 319ms/step -  
accuracy: 0.4956 - loss: 0.6945  
## [1m222/251[0m [32m-----[0m[37m---[0m [1m9s[0m 319ms/step -  
accuracy: 0.4956 - loss: 0.6945  
## [1m223/251[0m [32m-----[0m[37m---[0m [1m8s[0m 319ms/step -  
accuracy: 0.4956 - loss: 0.6945  
## [1m224/251[0m [32m-----[0m[37m---[0m [1m8s[0m 318ms/step -  
accuracy: 0.4956 - loss: 0.6945  
## [1m225/251[0m [32m-----[0m[37m---[0m [1m8s[0m 319ms/step -  
accuracy: 0.4956 - loss: 0.6945  
## [1m226/251[0m [32m-----[0m[37m---[0m [1m7s[0m 319ms/step -  
accuracy: 0.4957 - loss: 0.6945  
## [1m227/251[0m [32m-----[0m[37m---[0m [1m7s[0m 319ms/step -  
accuracy: 0.4957 - loss: 0.6945  
## [1m228/251[0m [32m-----[0m[37m---[0m [1m7s[0m 319ms/step -  
accuracy: 0.4957 - loss: 0.6945  
## [1m229/251[0m [32m-----[0m[37m---[0m [1m7s[0m 319ms/step -  
accuracy: 0.4957 - loss: 0.6945  
## [1m230/251[0m [32m-----[0m[37m---[0m [1m6s[0m 319ms/step -  
accuracy: 0.4957 - loss: 0.6944  
## [1m231/251[0m [32m-----[0m[37m---[0m [1m6s[0m 319ms/step -  
accuracy: 0.4958 - loss: 0.6944  
## [1m232/251[0m [32m-----[0m[37m---[0m [1m6s[0m 319ms/step -  
accuracy: 0.4958 - loss: 0.6944  
## [1m233/251[0m [32m-----[0m[37m---[0m [1m5s[0m 319ms/step -  
accuracy: 0.4958 - loss: 0.6944  
## [1m234/251[0m [32m-----[0m[37m---[0m [1m5s[0m 319ms/step -  
accuracy: 0.4958 - loss: 0.6944  
## [1m235/251[0m [32m-----[0m[37m---[0m [1m5s[0m 319ms/step -  
accuracy: 0.4959 - loss: 0.6944  
## [1m236/251[0m [32m-----[0m[37m---[0m [1m4s[0m 319ms/step -  
accuracy: 0.4959 - loss: 0.6944  
## [1m237/251[0m [32m-----[0m[37m---[0m [1m4s[0m 319ms/step -  
accuracy: 0.4959 - loss: 0.6944  
## [1m238/251[0m [32m-----[0m[37m---[0m [1m4s[0m 319ms/step -  
accuracy: 0.4960 - loss: 0.6944
```

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## [1m239/251[0m [32m-----[0m [37m[0m [1m3s[0m 318ms/step -
accuracy: 0.4960 - loss: 0.6944
## [1m240/251[0m [32m-----[0m [37m[0m [1m3s[0m 318ms/step -
accuracy: 0.4961 - loss: 0.6943
## [1m241/251[0m [32m-----[0m [37m[0m [1m3s[0m 318ms/step -
accuracy: 0.4961 - loss: 0.6943
## [1m242/251[0m [32m-----[0m [37m[0m [1m2s[0m 318ms/step -
accuracy: 0.4962 - loss: 0.6943
## [1m243/251[0m [32m-----[0m [37m[0m [1m2s[0m 317ms/step -
accuracy: 0.4962 - loss: 0.6943
## [1m244/251[0m [32m-----[0m [37m[0m [1m2s[0m 317ms/step -
accuracy: 0.4963 - loss: 0.6943
## [1m245/251[0m [32m-----[0m [37m[0m [1m1s[0m 317ms/step -
accuracy: 0.4964 - loss: 0.6942
## [1m246/251[0m [32m-----[0m [37m[0m [1m1s[0m 317ms/step -
accuracy: 0.4964 - loss: 0.6942
## [1m247/251[0m [32m-----[0m [37m[0m [1m1s[0m 317ms/step -
accuracy: 0.4965 - loss: 0.6942
## [1m248/251[0m [32m-----[0m [37m[0m [1m0s[0m 317ms/step -
accuracy: 0.4966 - loss: 0.6941
## [1m249/251[0m [32m-----[0m [37m[0m [1m0s[0m 316ms/step -
accuracy: 0.4967 - loss: 0.6941
## [1m250/251[0m [32m-----[0m [37m[0m [1m0s[0m 316ms/step -
accuracy: 0.4968 - loss: 0.6940
## [1m251/251[0m [32m-----[0m [37m[0m [1m0s[0m 316ms/step -
accuracy: 0.4969 - loss: 0.6940
## Epoch 1: val_accuracy improved from -inf to 0.79092, saving model to
F:\Desktop\Uni\IA\R\saved_models\weights-01-0.79.keras
##
## [1m251/251[0m [32m-----[0m [37m[0m [1m88s[0m 342ms/step
- accuracy: 0.4970 - loss: 0.6940 - val_accuracy: 0.7909 - val_loss:
0.4736
## Epoch 2/10
##
## [1m 1/251[0m [37m-----[0m [1m1:33[0m 375ms/step -
accuracy: 0.9375 - loss: 0.3363
## [1m 2/251[0m [37m-----[0m [1m1:14[0m 298ms/step -
accuracy: 0.8672 - loss: 0.3806
## [1m 3/251[0m [37m-----[0m [1m1:10[0m 286ms/step -
accuracy: 0.8351 - loss: 0.3957
## [1m 4/251[0m [37m-----[0m [1m1:11[0m 289ms/step -
accuracy: 0.8255 - loss: 0.4040
## [1m 5/251[0m [37m-----[0m [1m1:11[0m 292ms/step -
accuracy: 0.8229 - loss: 0.4103
## [1m 6/251[0m [37m-----[0m [1m1:10[0m 288ms/step -
accuracy: 0.8255 - loss: 0.4126
## [1m 7/251[0m [37m-----[0m [1m1:09[0m 284ms/step -
accuracy: 0.8288 - loss: 0.4144
## [1m 8/251[0m [37m-----[0m [1m1:09[0m 285ms/step -
accuracy: 0.8326 - loss: 0.4147

```

```
## [1m 9/251[0m [37m-----[0m [1m1:09[0m 286ms/step -  
accuracy: 0.8365 - loss: 0.4129  
## [1m 10/251[0m [37m-----[0m [1m1:08[0m 282ms/step -  
accuracy: 0.8398 - loss: 0.4109  
## [1m 11/251[0m [37m-----[0m [1m1:07[0m 280ms/step -  
accuracy: 0.8427 - loss: 0.4086  
## [1m 12/251[0m [37m-----[0m [1m1:06[0m 279ms/step -  
accuracy: 0.8458 - loss: 0.4051  
## [1m 13/251[0m [32m-[0m[37m-----[0m [1m1:06[0m 280ms/step  
- accuracy: 0.8492 - loss: 0.4008  
## [1m 14/251[0m [32m-[0m[37m-----[0m [1m1:06[0m 281ms/step  
- accuracy: 0.8523 - loss: 0.3961  
## [1m 15/251[0m [32m-[0m[37m-----[0m [1m1:06[0m 282ms/step  
- accuracy: 0.8551 - loss: 0.3918  
## [1m 16/251[0m [32m-[0m[37m-----[0m [1m1:06[0m 285ms/step  
- accuracy: 0.8574 - loss: 0.3880  
## [1m 17/251[0m [32m-[0m[37m-----[0m [1m1:07[0m 287ms/step  
- accuracy: 0.8587 - loss: 0.3860  
## [1m 18/251[0m [32m-[0m[37m-----[0m [1m1:06[0m 285ms/step  
- accuracy: 0.8600 - loss: 0.3839  
## [1m 19/251[0m [32m-[0m[37m-----[0m [1m1:06[0m 286ms/step  
- accuracy: 0.8606 - loss: 0.3829  
## [1m 20/251[0m [32m-[0m[37m-----[0m [1m1:05[0m 286ms/step  
- accuracy: 0.8604 - loss: 0.3830  
## [1m 21/251[0m [32m-[0m[37m-----[0m [1m1:05[0m 284ms/step  
- accuracy: 0.8601 - loss: 0.3829  
## [1m 22/251[0m [32m-[0m[37m-----[0m [1m1:04[0m 283ms/step  
- accuracy: 0.8600 - loss: 0.3826  
## [1m 23/251[0m [32m-[0m[37m-----[0m [1m1:04[0m 283ms/step  
- accuracy: 0.8601 - loss: 0.3819  
## [1m 24/251[0m [32m-[0m[37m-----[0m [1m1:03[0m 282ms/step  
- accuracy: 0.8603 - loss: 0.3811  
## [1m 25/251[0m [32m-[0m[37m-----[0m [1m1:03[0m 281ms/step  
- accuracy: 0.8607 - loss: 0.3803  
## [1m 26/251[0m [32m-[0m[37m-----[0m [1m1:02[0m 280ms/step  
- accuracy: 0.8611 - loss: 0.3795  
## [1m 27/251[0m [32m-[0m[37m-----[0m [1m1:02[0m 279ms/step  
- accuracy: 0.8615 - loss: 0.3787  
## [1m 28/251[0m [32m-[0m[37m-----[0m [1m1:02[0m 278ms/step  
- accuracy: 0.8620 - loss: 0.3779  
## [1m 29/251[0m [32m-[0m[37m-----[0m [1m1:01[0m 278ms/step  
- accuracy: 0.8624 - loss: 0.3772  
## [1m 30/251[0m [32m-[0m[37m-----[0m [1m1:01[0m 277ms/step  
- accuracy: 0.8630 - loss: 0.3763  
## [1m 31/251[0m [32m-[0m[37m-----[0m [1m1:00[0m 277ms/step  
- accuracy: 0.8637 - loss: 0.3754  
## [1m 32/251[0m [32m-[0m[37m-----[0m [1m1:00[0m 277ms/step  
- accuracy: 0.8644 - loss: 0.3744  
## [1m 33/251[0m [32m-[0m[37m-----[0m [1m1:00[0m 276ms/step  
- accuracy: 0.8651 - loss: 0.3733
```

```
## [1m 34/251[0m [32m[0m[37m[0m [1m59s[0m 276ms/step
- accuracy: 0.8659 - loss: 0.3722
## [1m 35/251[0m [32m[0m[37m[0m [1m59s[0m 277ms/step
- accuracy: 0.8666 - loss: 0.3710
## [1m 36/251[0m [32m[0m[37m[0m [1m59s[0m 276ms/step
- accuracy: 0.8673 - loss: 0.3698
## [1m 37/251[0m [32m[0m[37m[0m [1m58s[0m 276ms/step
- accuracy: 0.8681 - loss: 0.3685
## [1m 38/251[0m [32m[0m[37m[0m [1m58s[0m 275ms/step
- accuracy: 0.8688 - loss: 0.3672
## [1m 39/251[0m [32m[0m[37m[0m [1m58s[0m 275ms/step
- accuracy: 0.8696 - loss: 0.3658
## [1m 40/251[0m [32m[0m[37m[0m [1m57s[0m 275ms/step
- accuracy: 0.8703 - loss: 0.3643
## [1m 41/251[0m [32m[0m[37m[0m [1m57s[0m 274ms/step
- accuracy: 0.8711 - loss: 0.3628
## [1m 42/251[0m [32m[0m[37m[0m [1m57s[0m 274ms/step
- accuracy: 0.8718 - loss: 0.3614
## [1m 43/251[0m [32m[0m[37m[0m [1m56s[0m 274ms/step
- accuracy: 0.8726 - loss: 0.3599
## [1m 44/251[0m [32m[0m[37m[0m [1m56s[0m 274ms/step
- accuracy: 0.8733 - loss: 0.3584
## [1m 45/251[0m [32m[0m[37m[0m [1m56s[0m 274ms/step
- accuracy: 0.8740 - loss: 0.3569
## [1m 46/251[0m [32m[0m[37m[0m [1m56s[0m 274ms/step
- accuracy: 0.8747 - loss: 0.3553
## [1m 47/251[0m [32m[0m[37m[0m [1m55s[0m 274ms/step
- accuracy: 0.8754 - loss: 0.3537
## [1m 48/251[0m [32m[0m[37m[0m [1m55s[0m 274ms/step
- accuracy: 0.8762 - loss: 0.3522
## [1m 49/251[0m [32m[0m[37m[0m [1m55s[0m 275ms/step
- accuracy: 0.8768 - loss: 0.3506
## [1m 50/251[0m [32m[0m[37m[0m [1m55s[0m 275ms/step
- accuracy: 0.8775 - loss: 0.3491
## [1m 51/251[0m [32m[0m[37m[0m [1m55s[0m 276ms/step
- accuracy: 0.8781 - loss: 0.3476
## [1m 52/251[0m [32m[0m[37m[0m [1m55s[0m 277ms/step
- accuracy: 0.8787 - loss: 0.3461
## [1m 53/251[0m [32m[0m[37m[0m [1m55s[0m 279ms/step
- accuracy: 0.8794 - loss: 0.3445
## [1m 54/251[0m [32m[0m[37m[0m [1m55s[0m 280ms/step
- accuracy: 0.8800 - loss: 0.3429
## [1m 55/251[0m [32m[0m[37m[0m [1m54s[0m 280ms/step
- accuracy: 0.8807 - loss: 0.3413
## [1m 56/251[0m [32m[0m[37m[0m [1m54s[0m 281ms/step
- accuracy: 0.8813 - loss: 0.3398
## [1m 57/251[0m [32m[0m[37m[0m [1m54s[0m 282ms/step
- accuracy: 0.8819 - loss: 0.3382
## [1m 58/251[0m [32m[0m[37m[0m [1m54s[0m 284ms/step
- accuracy: 0.8825 - loss: 0.3367
```

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## [1m 59/251[0m [32m[0m[37m[0m [1m54s[0m 285ms/step
- accuracy: 0.8831 - loss: 0.3352
## [1m 60/251[0m [32m[0m[37m[0m [1m54s[0m 285ms/step
- accuracy: 0.8836 - loss: 0.3336
## [1m 61/251[0m [32m[0m[37m[0m [1m54s[0m 285ms/step
- accuracy: 0.8842 - loss: 0.3321
## [1m 62/251[0m [32m[0m[37m[0m [1m53s[0m 286ms/step
- accuracy: 0.8848 - loss: 0.3306
## [1m 63/251[0m [32m[0m[37m[0m [1m53s[0m 285ms/step
- accuracy: 0.8853 - loss: 0.3292
## [1m 64/251[0m [32m[0m[37m[0m [1m53s[0m 285ms/step
- accuracy: 0.8858 - loss: 0.3277
## [1m 65/251[0m [32m[0m[37m[0m [1m53s[0m 286ms/step
- accuracy: 0.8864 - loss: 0.3262
## [1m 66/251[0m [32m[0m[37m[0m [1m53s[0m 287ms/step
- accuracy: 0.8869 - loss: 0.3247
## [1m 67/251[0m [32m[0m[37m[0m [1m52s[0m 287ms/step
- accuracy: 0.8874 - loss: 0.3232
## [1m 68/251[0m [32m[0m[37m[0m [1m52s[0m 288ms/step
- accuracy: 0.8879 - loss: 0.3218
## [1m 69/251[0m [32m[0m[37m[0m [1m52s[0m 288ms/step
- accuracy: 0.8884 - loss: 0.3203
## [1m 70/251[0m [32m[0m[37m[0m [1m52s[0m 289ms/step
- accuracy: 0.8889 - loss: 0.3189
## [1m 71/251[0m [32m[0m[37m[0m [1m52s[0m 289ms/step
- accuracy: 0.8894 - loss: 0.3174
## [1m 72/251[0m [32m[0m[37m[0m [1m51s[0m 290ms/step
- accuracy: 0.8899 - loss: 0.3160
## [1m 73/251[0m [32m[0m[37m[0m [1m51s[0m 292ms/step
- accuracy: 0.8904 - loss: 0.3146
## [1m 74/251[0m [32m[0m[37m[0m [1m51s[0m 292ms/step
- accuracy: 0.8909 - loss: 0.3132
## [1m 75/251[0m [32m[0m[37m[0m [1m51s[0m 292ms/step
- accuracy: 0.8914 - loss: 0.3118
## [1m 76/251[0m [32m[0m[37m[0m [1m51s[0m 292ms/step
- accuracy: 0.8918 - loss: 0.3105
## [1m 77/251[0m [32m[0m[37m[0m [1m50s[0m 292ms/step
- accuracy: 0.8923 - loss: 0.3091
## [1m 78/251[0m [32m[0m[37m[0m [1m50s[0m 291ms/step
- accuracy: 0.8928 - loss: 0.3078
## [1m 79/251[0m [32m[0m[37m[0m [1m50s[0m 291ms/step
- accuracy: 0.8932 - loss: 0.3065
## [1m 80/251[0m [32m[0m[37m[0m [1m49s[0m 292ms/step
- accuracy: 0.8937 - loss: 0.3051
## [1m 81/251[0m [32m[0m[37m[0m [1m49s[0m 292ms/step
- accuracy: 0.8941 - loss: 0.3038
## [1m 82/251[0m [32m[0m[37m[0m [1m49s[0m 293ms/step
- accuracy: 0.8946 - loss: 0.3025
## [1m 83/251[0m [32m[0m[37m[0m [1m49s[0m 295ms/step
- accuracy: 0.8950 - loss: 0.3013
```



```
## [1m 84/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 295ms/step
- accuracy: 0.8955 - loss: 0.3000
## [1m 85/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 295ms/step
- accuracy: 0.8959 - loss: 0.2987
## [1m 86/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 296ms/step
- accuracy: 0.8963 - loss: 0.2975
## [1m 87/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 297ms/step
- accuracy: 0.8967 - loss: 0.2962
## [1m 88/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 297ms/step
- accuracy: 0.8972 - loss: 0.2950
## [1m 89/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 299ms/step
- accuracy: 0.8976 - loss: 0.2938
## [1m 90/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 299ms/step
- accuracy: 0.8980 - loss: 0.2926
## [1m 91/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 300ms/step
- accuracy: 0.8984 - loss: 0.2914
## [1m 92/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 300ms/step
- accuracy: 0.8987 - loss: 0.2902
## [1m 93/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 301ms/step
- accuracy: 0.8991 - loss: 0.2891
## [1m 94/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 304ms/step
- accuracy: 0.8995 - loss: 0.2880
## [1m 95/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 305ms/step
- accuracy: 0.8999 - loss: 0.2868
## [1m 96/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 307ms/step
- accuracy: 0.9002 - loss: 0.2857
## [1m 97/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 308ms/step
- accuracy: 0.9006 - loss: 0.2846
## [1m 98/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 309ms/step
- accuracy: 0.9010 - loss: 0.2835
## [1m 99/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 310ms/step
- accuracy: 0.9013 - loss: 0.2824
## [1m100/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 311ms/step
- accuracy: 0.9017 - loss: 0.2813
## [1m101/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 312ms/step
- accuracy: 0.9021 - loss: 0.2803
## [1m102/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 313ms/step
- accuracy: 0.9024 - loss: 0.2792
## [1m103/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 314ms/step
- accuracy: 0.9027 - loss: 0.2782
## [1m104/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 315ms/step
- accuracy: 0.9031 - loss: 0.2772
## [1m105/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 317ms/step
- accuracy: 0.9034 - loss: 0.2762
## [1m106/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 317ms/step
- accuracy: 0.9037 - loss: 0.2752
## [1m107/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 318ms/step
- accuracy: 0.9040 - loss: 0.2742
## [1m108/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 319ms/step
- accuracy: 0.9044 - loss: 0.2733
```

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## [1m109/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 319ms/step
- accuracy: 0.9047 - loss: 0.2723
## [1m110/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 319ms/step
- accuracy: 0.9050 - loss: 0.2714
## [1m111/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 319ms/step
- accuracy: 0.9053 - loss: 0.2704
## [1m112/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 319ms/step
- accuracy: 0.9056 - loss: 0.2695
## [1m113/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 319ms/step
- accuracy: 0.9059 - loss: 0.2686
## [1m114/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 320ms/step
- accuracy: 0.9062 - loss: 0.2677
## [1m115/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 320ms/step
- accuracy: 0.9065 - loss: 0.2668
## [1m116/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 320ms/step
- accuracy: 0.9068 - loss: 0.2659
## [1m117/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 320ms/step
- accuracy: 0.9071 - loss: 0.2650
## [1m118/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 320ms/step
- accuracy: 0.9074 - loss: 0.2641
## [1m119/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 319ms/step
- accuracy: 0.9077 - loss: 0.2632
## [1m120/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 319ms/step
- accuracy: 0.9080 - loss: 0.2624
## [1m121/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 319ms/step
- accuracy: 0.9082 - loss: 0.2615
## [1m122/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 319ms/step
- accuracy: 0.9085 - loss: 0.2607
## [1m123/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 319ms/step
- accuracy: 0.9088 - loss: 0.2599
## [1m124/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 321ms/step
- accuracy: 0.9091 - loss: 0.2591
## [1m125/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 321ms/step
- accuracy: 0.9093 - loss: 0.2582
## [1m126/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 321ms/step
- accuracy: 0.9096 - loss: 0.2574
## [1m127/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 321ms/step
- accuracy: 0.9099 - loss: 0.2566
## [1m128/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 322ms/step
- accuracy: 0.9101 - loss: 0.2558
## [1m129/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 322ms/step
- accuracy: 0.9104 - loss: 0.2551
## [1m130/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 322ms/step
- accuracy: 0.9106 - loss: 0.2543
## [1m131/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 321ms/step
- accuracy: 0.9109 - loss: 0.2535
## [1m132/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 321ms/step
- accuracy: 0.9111 - loss: 0.2528
## [1m133/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 321ms/step
- accuracy: 0.9114 - loss: 0.2520
```

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## [1m134/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 321ms/step
- accuracy: 0.9116 - loss: 0.2513
## [1m135/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 320ms/step
- accuracy: 0.9118 - loss: 0.2505
## [1m136/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 320ms/step
- accuracy: 0.9121 - loss: 0.2498
## [1m137/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 321ms/step
- accuracy: 0.9123 - loss: 0.2491
## [1m138/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 321ms/step
- accuracy: 0.9125 - loss: 0.2484
## [1m139/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 321ms/step
- accuracy: 0.9127 - loss: 0.2477
## [1m140/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 321ms/step
- accuracy: 0.9130 - loss: 0.2470
## [1m141/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 321ms/step
- accuracy: 0.9132 - loss: 0.2463
## [1m142/251[0m [32m-----[0m[37m-----[0m [1m34s[0m 321ms/step
- accuracy: 0.9134 - loss: 0.2456
## [1m143/251[0m [32m-----[0m[37m-----[0m [1m34s[0m 321ms/step
- accuracy: 0.9136 - loss: 0.2449
## [1m144/251[0m [32m-----[0m[37m-----[0m [1m34s[0m 321ms/step
- accuracy: 0.9138 - loss: 0.2442
## [1m145/251[0m [32m-----[0m[37m-----[0m [1m33s[0m 321ms/step
- accuracy: 0.9140 - loss: 0.2436
## [1m146/251[0m [32m-----[0m[37m-----[0m [1m33s[0m 320ms/step
- accuracy: 0.9143 - loss: 0.2429
## [1m147/251[0m [32m-----[0m[37m-----[0m [1m33s[0m 320ms/step
- accuracy: 0.9145 - loss: 0.2422
## [1m148/251[0m [32m-----[0m[37m-----[0m [1m32s[0m 320ms/step
- accuracy: 0.9147 - loss: 0.2416
## [1m149/251[0m [32m-----[0m[37m-----[0m [1m32s[0m 320ms/step
- accuracy: 0.9149 - loss: 0.2409
## [1m150/251[0m [32m-----[0m[37m-----[0m [1m32s[0m 319ms/step
- accuracy: 0.9151 - loss: 0.2403
## [1m151/251[0m [32m-----[0m[37m-----[0m [1m31s[0m 319ms/step
- accuracy: 0.9153 - loss: 0.2396
## [1m152/251[0m [32m-----[0m[37m-----[0m [1m31s[0m 319ms/step
- accuracy: 0.9155 - loss: 0.2390
## [1m153/251[0m [32m-----[0m[37m-----[0m [1m31s[0m 319ms/step
- accuracy: 0.9157 - loss: 0.2384
## [1m154/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 319ms/step
- accuracy: 0.9159 - loss: 0.2378
## [1m155/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 320ms/step
- accuracy: 0.9161 - loss: 0.2371
## [1m156/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 320ms/step
- accuracy: 0.9162 - loss: 0.2365
## [1m157/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 320ms/step
- accuracy: 0.9164 - loss: 0.2359
## [1m158/251[0m [32m-----[0m[37m-----[0m [1m29s[0m 320ms/step
- accuracy: 0.9166 - loss: 0.2353
```

```
## [1m159/251[0m [32m-----[0m[37m-----[0m [1m29s[0m 320ms/step
- accuracy: 0.9168 - loss: 0.2347
## [1m160/251[0m [32m-----[0m[37m-----[0m [1m29s[0m 320ms/step
- accuracy: 0.9170 - loss: 0.2341
## [1m161/251[0m [32m-----[0m[37m-----[0m [1m28s[0m 320ms/step
- accuracy: 0.9171 - loss: 0.2336
## [1m162/251[0m [32m-----[0m[37m-----[0m [1m28s[0m 320ms/step
- accuracy: 0.9173 - loss: 0.2330
## [1m163/251[0m [32m-----[0m[37m-----[0m [1m28s[0m 320ms/step
- accuracy: 0.9175 - loss: 0.2324
## [1m164/251[0m [32m-----[0m[37m-----[0m [1m27s[0m 320ms/step
- accuracy: 0.9177 - loss: 0.2318
## [1m165/251[0m [32m-----[0m[37m-----[0m [1m27s[0m 321ms/step
- accuracy: 0.9178 - loss: 0.2313
## [1m166/251[0m [32m-----[0m[37m-----[0m [1m27s[0m 320ms/step
- accuracy: 0.9180 - loss: 0.2307
## [1m167/251[0m [32m-----[0m[37m-----[0m [1m26s[0m 320ms/step
- accuracy: 0.9182 - loss: 0.2301
## [1m168/251[0m [32m-----[0m[37m-----[0m [1m26s[0m 320ms/step
- accuracy: 0.9183 - loss: 0.2296
## [1m169/251[0m [32m-----[0m[37m-----[0m [1m26s[0m 320ms/step
- accuracy: 0.9185 - loss: 0.2290
## [1m170/251[0m [32m-----[0m[37m-----[0m [1m25s[0m 320ms/step
- accuracy: 0.9187 - loss: 0.2285
## [1m171/251[0m [32m-----[0m[37m-----[0m [1m25s[0m 321ms/step
- accuracy: 0.9188 - loss: 0.2279
## [1m172/251[0m [32m-----[0m[37m-----[0m [1m25s[0m 321ms/step
- accuracy: 0.9190 - loss: 0.2274
## [1m173/251[0m [32m-----[0m[37m-----[0m [1m25s[0m 321ms/step
- accuracy: 0.9192 - loss: 0.2269
## [1m174/251[0m [32m-----[0m[37m-----[0m [1m24s[0m 321ms/step
- accuracy: 0.9193 - loss: 0.2263
## [1m175/251[0m [32m-----[0m[37m-----[0m [1m24s[0m 321ms/step
- accuracy: 0.9195 - loss: 0.2258
## [1m176/251[0m [32m-----[0m[37m-----[0m [1m24s[0m 321ms/step
- accuracy: 0.9196 - loss: 0.2253
## [1m177/251[0m [32m-----[0m[37m-----[0m [1m23s[0m 321ms/step
- accuracy: 0.9198 - loss: 0.2248
## [1m178/251[0m [32m-----[0m[37m-----[0m [1m23s[0m 322ms/step
- accuracy: 0.9199 - loss: 0.2242
## [1m179/251[0m [32m-----[0m[37m-----[0m [1m23s[0m 322ms/step
- accuracy: 0.9201 - loss: 0.2237
## [1m180/251[0m [32m-----[0m[37m-----[0m [1m22s[0m 322ms/step
- accuracy: 0.9202 - loss: 0.2232
## [1m181/251[0m [32m-----[0m[37m-----[0m [1m22s[0m 321ms/step
- accuracy: 0.9204 - loss: 0.2227
## [1m182/251[0m [32m-----[0m[37m-----[0m [1m22s[0m 321ms/step
- accuracy: 0.9205 - loss: 0.2222
## [1m183/251[0m [32m-----[0m[37m-----[0m [1m21s[0m 321ms/step
- accuracy: 0.9207 - loss: 0.2217
```

```
## [1m184/251[0m [32m-----[0m[37m-----[0m [1m21s[0m 321ms/step
- accuracy: 0.9208 - loss: 0.2212
## [1m185/251[0m [32m-----[0m[37m-----[0m [1m21s[0m 320ms/step
- accuracy: 0.9210 - loss: 0.2207
## [1m186/251[0m [32m-----[0m[37m-----[0m [1m20s[0m 320ms/step
- accuracy: 0.9211 - loss: 0.2202
## [1m187/251[0m [32m-----[0m[37m-----[0m [1m20s[0m 320ms/step
- accuracy: 0.9213 - loss: 0.2197
## [1m188/251[0m [32m-----[0m[37m-----[0m [1m20s[0m 320ms/step
- accuracy: 0.9214 - loss: 0.2193
## [1m189/251[0m [32m-----[0m[37m-----[0m [1m19s[0m 320ms/step
- accuracy: 0.9216 - loss: 0.2188
## [1m190/251[0m [32m-----[0m[37m-----[0m [1m19s[0m 320ms/step
- accuracy: 0.9217 - loss: 0.2183
## [1m191/251[0m [32m-----[0m[37m-----[0m [1m19s[0m 320ms/step
- accuracy: 0.9218 - loss: 0.2178
## [1m192/251[0m [32m-----[0m[37m-----[0m [1m18s[0m 319ms/step
- accuracy: 0.9220 - loss: 0.2174
## [1m193/251[0m [32m-----[0m[37m-----[0m [1m18s[0m 319ms/step
- accuracy: 0.9221 - loss: 0.2169
## [1m194/251[0m [32m-----[0m[37m-----[0m [1m18s[0m 319ms/step
- accuracy: 0.9223 - loss: 0.2164
## [1m195/251[0m [32m-----[0m[37m-----[0m [1m17s[0m 319ms/step
- accuracy: 0.9224 - loss: 0.2160
## [1m196/251[0m [32m-----[0m[37m-----[0m [1m17s[0m 319ms/step
- accuracy: 0.9225 - loss: 0.2155
## [1m197/251[0m [32m-----[0m[37m-----[0m [1m17s[0m 319ms/step
- accuracy: 0.9227 - loss: 0.2151
## [1m198/251[0m [32m-----[0m[37m-----[0m [1m16s[0m 319ms/step
- accuracy: 0.9228 - loss: 0.2146
## [1m199/251[0m [32m-----[0m[37m-----[0m [1m16s[0m 319ms/step
- accuracy: 0.9229 - loss: 0.2142
## [1m200/251[0m [32m-----[0m[37m-----[0m [1m16s[0m 318ms/step
- accuracy: 0.9231 - loss: 0.2137
## [1m201/251[0m [32m-----[0m[37m-----[0m [1m15s[0m 318ms/step
- accuracy: 0.9232 - loss: 0.2133
## [1m202/251[0m [32m-----[0m[37m-----[0m [1m15s[0m 318ms/step
- accuracy: 0.9233 - loss: 0.2128
## [1m203/251[0m [32m-----[0m[37m-----[0m [1m15s[0m 318ms/step
- accuracy: 0.9235 - loss: 0.2124
## [1m204/251[0m [32m-----[0m[37m-----[0m [1m14s[0m 318ms/step
- accuracy: 0.9236 - loss: 0.2120
## [1m205/251[0m [32m-----[0m[37m-----[0m [1m14s[0m 317ms/step
- accuracy: 0.9237 - loss: 0.2115
## [1m206/251[0m [32m-----[0m[37m-----[0m [1m14s[0m 317ms/step
- accuracy: 0.9238 - loss: 0.2111
## [1m207/251[0m [32m-----[0m[37m-----[0m [1m13s[0m 317ms/step
- accuracy: 0.9240 - loss: 0.2107
## [1m208/251[0m [32m-----[0m[37m-----[0m [1m13s[0m 317ms/step
- accuracy: 0.9241 - loss: 0.2103
```

```
## [1m209/251[0m [32m-----[0m[37m-----[0m [1m13s[0m 317ms/step
- accuracy: 0.9242 - loss: 0.2099
## [1m210/251[0m [32m-----[0m[37m-----[0m [1m12s[0m 317ms/step
- accuracy: 0.9243 - loss: 0.2095
## [1m211/251[0m [32m-----[0m[37m-----[0m [1m12s[0m 317ms/step
- accuracy: 0.9244 - loss: 0.2091
## [1m212/251[0m [32m-----[0m[37m-----[0m [1m12s[0m 316ms/step
- accuracy: 0.9246 - loss: 0.2087
## [1m213/251[0m [32m-----[0m[37m-----[0m [1m12s[0m 316ms/step
- accuracy: 0.9247 - loss: 0.2083
## [1m214/251[0m [32m-----[0m[37m-----[0m [1m11s[0m 316ms/step
- accuracy: 0.9248 - loss: 0.2079
## [1m215/251[0m [32m-----[0m[37m-----[0m [1m11s[0m 316ms/step
- accuracy: 0.9249 - loss: 0.2075
## [1m216/251[0m [32m-----[0m[37m-----[0m [1m11s[0m 316ms/step
- accuracy: 0.9250 - loss: 0.2071
## [1m217/251[0m [32m-----[0m[37m-----[0m [1m10s[0m 316ms/step
- accuracy: 0.9251 - loss: 0.2067
## [1m218/251[0m [32m-----[0m[37m-----[0m [1m10s[0m 316ms/step
- accuracy: 0.9252 - loss: 0.2063
## [1m219/251[0m [32m-----[0m[37m-----[0m [1m10s[0m 316ms/step
- accuracy: 0.9253 - loss: 0.2059
## [1m220/251[0m [32m-----[0m[37m-----[0m [1m9s[0m 316ms/step -
accuracy: 0.9255 - loss: 0.2055
## [1m221/251[0m [32m-----[0m[37m-----[0m [1m9s[0m 315ms/step -
accuracy: 0.9256 - loss: 0.2051
## [1m222/251[0m [32m-----[0m[37m-----[0m [1m9s[0m 315ms/step -
accuracy: 0.9257 - loss: 0.2048
## [1m223/251[0m [32m-----[0m[37m-----[0m [1m8s[0m 315ms/step -
accuracy: 0.9258 - loss: 0.2044
## [1m224/251[0m [32m-----[0m[37m-----[0m [1m8s[0m 315ms/step -
accuracy: 0.9259 - loss: 0.2040
## [1m225/251[0m [32m-----[0m[37m-----[0m [1m8s[0m 315ms/step -
accuracy: 0.9260 - loss: 0.2036
## [1m226/251[0m [32m-----[0m[37m-----[0m [1m7s[0m 315ms/step -
accuracy: 0.9261 - loss: 0.2033
## [1m227/251[0m [32m-----[0m[37m-----[0m [1m7s[0m 315ms/step -
accuracy: 0.9262 - loss: 0.2029
## [1m228/251[0m [32m-----[0m[37m-----[0m [1m7s[0m 314ms/step -
accuracy: 0.9263 - loss: 0.2025
## [1m229/251[0m [32m-----[0m[37m-----[0m [1m6s[0m 314ms/step -
accuracy: 0.9264 - loss: 0.2022
## [1m230/251[0m [32m-----[0m[37m-----[0m [1m6s[0m 314ms/step -
accuracy: 0.9266 - loss: 0.2018
## [1m231/251[0m [32m-----[0m[37m-----[0m [1m6s[0m 314ms/step -
accuracy: 0.9267 - loss: 0.2014
## [1m232/251[0m [32m-----[0m[37m-----[0m [1m5s[0m 314ms/step -
accuracy: 0.9268 - loss: 0.2011
## [1m233/251[0m [32m-----[0m[37m-----[0m [1m5s[0m 314ms/step -
accuracy: 0.9269 - loss: 0.2007
```

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## [1m234/251[0m [32m-----[0m [37m-----[0m [1m5s[0m 314ms/step -
accuracy: 0.9270 - loss: 0.2004
## [1m235/251[0m [32m-----[0m [37m-----[0m [1m5s[0m 314ms/step -
accuracy: 0.9271 - loss: 0.2000
## [1m236/251[0m [32m-----[0m [37m-----[0m [1m4s[0m 314ms/step -
accuracy: 0.9272 - loss: 0.1997
## [1m237/251[0m [32m-----[0m [37m-----[0m [1m4s[0m 314ms/step -
accuracy: 0.9273 - loss: 0.1993
## [1m238/251[0m [32m-----[0m [37m-----[0m [1m4s[0m 314ms/step -
accuracy: 0.9274 - loss: 0.1990
## [1m239/251[0m [32m-----[0m [37m-----[0m [1m3s[0m 314ms/step -
accuracy: 0.9275 - loss: 0.1986
## [1m240/251[0m [32m-----[0m [37m-----[0m [1m3s[0m 314ms/step -
accuracy: 0.9276 - loss: 0.1983
## [1m241/251[0m [32m-----[0m [37m-----[0m [1m3s[0m 314ms/step -
accuracy: 0.9277 - loss: 0.1979
## [1m242/251[0m [32m-----[0m [37m-----[0m [1m2s[0m 314ms/step -
accuracy: 0.9278 - loss: 0.1976
## [1m243/251[0m [32m-----[0m [37m-----[0m [1m2s[0m 314ms/step -
accuracy: 0.9279 - loss: 0.1972
## [1m244/251[0m [32m-----[0m [37m-----[0m [1m2s[0m 314ms/step -
accuracy: 0.9280 - loss: 0.1969
## [1m245/251[0m [32m-----[0m [37m-----[0m [1m1s[0m 314ms/step -
accuracy: 0.9281 - loss: 0.1966
## [1m246/251[0m [32m-----[0m [37m-----[0m [1m1s[0m 313ms/step -
accuracy: 0.9282 - loss: 0.1962
## [1m247/251[0m [32m-----[0m [37m-----[0m [1m1s[0m 313ms/step -
accuracy: 0.9283 - loss: 0.1959
## [1m248/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 313ms/step -
accuracy: 0.9284 - loss: 0.1956
## [1m249/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 313ms/step -
accuracy: 0.9285 - loss: 0.1952
## [1m250/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 313ms/step -
accuracy: 0.9286 - loss: 0.1949
## [1m251/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 313ms/step -
accuracy: 0.9287 - loss: 0.1946
## Epoch 2: val_accuracy improved from 0.79092 to 0.95858, saving model
to F:\Desktop\Uni\IA\R\saved_models\weights-02-0.96.keras
##
## [1m251/251[0m [32m-----[0m [37m-----[0m [1m85s[0m 339ms/step
- accuracy: 0.9288 - loss: 0.1943 - val_accuracy: 0.9586 - val_loss:
0.0812
## Epoch 3/10
##
## [1m 1/251[0m [37m-----[0m [1m1:31[0m 367ms/step -
accuracy: 0.9688 - loss: 0.0482
## [1m 2/251[0m [37m-----[0m [1m1:19[0m 319ms/step -
accuracy: 0.9688 - loss: 0.0498
## [1m 3/251[0m [37m-----[0m [1m1:14[0m 300ms/step -
accuracy: 0.9653 - loss: 0.0534

```

```
## [1m 4/251[0m [37m-----[0m [1m1:12[0m 294ms/step -  
accuracy: 0.9661 - loss: 0.0530  
## [1m 5/251[0m [37m-----[0m [1m1:17[0m 314ms/step -  
accuracy: 0.9679 - loss: 0.0515  
## [1m 6/251[0m [37m-----[0m [1m1:21[0m 334ms/step -  
accuracy: 0.9681 - loss: 0.0530  
## [1m 7/251[0m [37m-----[0m [1m1:20[0m 329ms/step -  
accuracy: 0.9675 - loss: 0.0564  
## [1m 8/251[0m [37m-----[0m [1m1:19[0m 328ms/step -  
accuracy: 0.9672 - loss: 0.0586  
## [1m 9/251[0m [37m-----[0m [1m1:22[0m 339ms/step -  
accuracy: 0.9666 - loss: 0.0607  
## [1m 10/251[0m [37m-----[0m [1m1:21[0m 339ms/step -  
accuracy: 0.9652 - loss: 0.0629  
## [1m 11/251[0m [37m-----[0m [1m1:21[0m 340ms/step -  
accuracy: 0.9643 - loss: 0.0645  
## [1m 12/251[0m [37m-----[0m [1m1:21[0m 339ms/step -  
accuracy: 0.9636 - loss: 0.0654  
## [1m 13/251[0m [32m-[0m[37m-----[0m [1m1:19[0m 336ms/step  
- accuracy: 0.9628 - loss: 0.0664  
## [1m 14/251[0m [32m-[0m[37m-----[0m [1m1:19[0m 337ms/step  
- accuracy: 0.9625 - loss: 0.0672  
## [1m 15/251[0m [32m-[0m[37m-----[0m [1m1:19[0m 339ms/step  
- accuracy: 0.9621 - loss: 0.0677  
## [1m 16/251[0m [32m-[0m[37m-----[0m [1m1:18[0m 335ms/step  
- accuracy: 0.9616 - loss: 0.0683  
## [1m 17/251[0m [32m-[0m[37m-----[0m [1m1:17[0m 332ms/step  
- accuracy: 0.9612 - loss: 0.0689  
## [1m 18/251[0m [32m-[0m[37m-----[0m [1m1:17[0m 331ms/step  
- accuracy: 0.9609 - loss: 0.0692  
## [1m 19/251[0m [32m-[0m[37m-----[0m [1m1:16[0m 328ms/step  
- accuracy: 0.9606 - loss: 0.0696  
## [1m 20/251[0m [32m-[0m[37m-----[0m [1m1:16[0m 332ms/step  
- accuracy: 0.9603 - loss: 0.0700  
## [1m 21/251[0m [32m-[0m[37m-----[0m [1m1:16[0m 334ms/step  
- accuracy: 0.9601 - loss: 0.0702  
## [1m 22/251[0m [32m-[0m[37m-----[0m [1m1:16[0m 334ms/step  
- accuracy: 0.9599 - loss: 0.0705  
## [1m 23/251[0m [32m-[0m[37m-----[0m [1m1:16[0m 335ms/step  
- accuracy: 0.9596 - loss: 0.0706  
## [1m 24/251[0m [32m-[0m[37m-----[0m [1m1:16[0m 336ms/step  
- accuracy: 0.9595 - loss: 0.0708  
## [1m 25/251[0m [32m-[0m[37m-----[0m [1m1:15[0m 335ms/step  
- accuracy: 0.9593 - loss: 0.0710  
## [1m 26/251[0m [32m-[0m[37m-----[0m [1m1:14[0m 332ms/step  
- accuracy: 0.9591 - loss: 0.0711  
## [1m 27/251[0m [32m-[0m[37m-----[0m [1m1:13[0m 330ms/step  
- accuracy: 0.9590 - loss: 0.0712  
## [1m 28/251[0m [32m-[0m[37m-----[0m [1m1:13[0m 329ms/step  
- accuracy: 0.9588 - loss: 0.0712
```



```
## [1m 29/251[0m [32m[0m[37m[0m [1m1:12[0m 327ms/step
- accuracy: 0.9587 - loss: 0.0713
## [1m 30/251[0m [32m[0m[37m[0m [1m1:11[0m 326ms/step
- accuracy: 0.9586 - loss: 0.0714
## [1m 31/251[0m [32m[0m[37m[0m [1m1:11[0m 324ms/step
- accuracy: 0.9584 - loss: 0.0715
## [1m 32/251[0m [32m[0m[37m[0m [1m1:10[0m 322ms/step
- accuracy: 0.9583 - loss: 0.0715
## [1m 33/251[0m [32m[0m[37m[0m [1m1:09[0m 320ms/step
- accuracy: 0.9582 - loss: 0.0717
## [1m 34/251[0m [32m[0m[37m[0m [1m1:09[0m 318ms/step
- accuracy: 0.9581 - loss: 0.0717
## [1m 35/251[0m [32m[0m[37m[0m [1m1:08[0m 316ms/step
- accuracy: 0.9580 - loss: 0.0718
## [1m 36/251[0m [32m[0m[37m[0m [1m1:07[0m 315ms/step
- accuracy: 0.9579 - loss: 0.0718
## [1m 37/251[0m [32m[0m[37m[0m [1m1:07[0m 313ms/step
- accuracy: 0.9579 - loss: 0.0718
## [1m 38/251[0m [32m[0m[37m[0m [1m1:06[0m 312ms/step
- accuracy: 0.9579 - loss: 0.0718
## [1m 39/251[0m [32m[0m[37m[0m [1m1:05[0m 311ms/step
- accuracy: 0.9578 - loss: 0.0718
## [1m 40/251[0m [32m[0m[37m[0m [1m1:05[0m 310ms/step
- accuracy: 0.9578 - loss: 0.0718
## [1m 41/251[0m [32m[0m[37m[0m [1m1:04[0m 309ms/step
- accuracy: 0.9578 - loss: 0.0718
## [1m 42/251[0m [32m[0m[37m[0m [1m1:04[0m 308ms/step
- accuracy: 0.9577 - loss: 0.0718
## [1m 43/251[0m [32m[0m[37m[0m [1m1:03[0m 307ms/step
- accuracy: 0.9577 - loss: 0.0719
## [1m 44/251[0m [32m[0m[37m[0m [1m1:03[0m 307ms/step
- accuracy: 0.9577 - loss: 0.0718
## [1m 45/251[0m [32m[0m[37m[0m [1m1:02[0m 306ms/step
- accuracy: 0.9578 - loss: 0.0718
## [1m 46/251[0m [32m[0m[37m[0m [1m1:02[0m 305ms/step
- accuracy: 0.9578 - loss: 0.0718
## [1m 47/251[0m [32m[0m[37m[0m [1m1:02[0m 304ms/step
- accuracy: 0.9578 - loss: 0.0718
## [1m 48/251[0m [32m[0m[37m[0m [1m1:01[0m 303ms/step
- accuracy: 0.9579 - loss: 0.0717
## [1m 49/251[0m [32m[0m[37m[0m [1m1:01[0m 303ms/step
- accuracy: 0.9579 - loss: 0.0717
## [1m 50/251[0m [32m[0m[37m[0m [1m1:00[0m 302ms/step
- accuracy: 0.9580 - loss: 0.0716
## [1m 51/251[0m [32m[0m[37m[0m [1m1:00[0m 302ms/step
- accuracy: 0.9581 - loss: 0.0715
## [1m 52/251[0m [32m[0m[37m[0m [1m59s[0m 301ms/step
- accuracy: 0.9581 - loss: 0.0714
## [1m 53/251[0m [32m[0m[37m[0m [1m59s[0m 300ms/step
- accuracy: 0.9582 - loss: 0.0713
```

```
## [1m 54/251[0m [32m[0m[37m[0m [1m59s[0m 300ms/step
- accuracy: 0.9582 - loss: 0.0712
## [1m 55/251[0m [32m[0m[37m[0m [1m58s[0m 300ms/step
- accuracy: 0.9583 - loss: 0.0711
## [1m 56/251[0m [32m[0m[37m[0m [1m58s[0m 299ms/step
- accuracy: 0.9583 - loss: 0.0711
## [1m 57/251[0m [32m[0m[37m[0m [1m57s[0m 299ms/step
- accuracy: 0.9583 - loss: 0.0710
## [1m 58/251[0m [32m[0m[37m[0m [1m57s[0m 299ms/step
- accuracy: 0.9584 - loss: 0.0710
## [1m 59/251[0m [32m[0m[37m[0m [1m57s[0m 298ms/step
- accuracy: 0.9584 - loss: 0.0709
## [1m 60/251[0m [32m[0m[37m[0m [1m56s[0m 298ms/step
- accuracy: 0.9584 - loss: 0.0708
## [1m 61/251[0m [32m[0m[37m[0m [1m56s[0m 297ms/step
- accuracy: 0.9584 - loss: 0.0707
## [1m 62/251[0m [32m[0m[37m[0m [1m56s[0m 297ms/step
- accuracy: 0.9585 - loss: 0.0707
## [1m 63/251[0m [32m[0m[37m[0m [1m55s[0m 296ms/step
- accuracy: 0.9585 - loss: 0.0706
## [1m 64/251[0m [32m[0m[37m[0m [1m55s[0m 296ms/step
- accuracy: 0.9585 - loss: 0.0705
## [1m 65/251[0m [32m[0m[37m[0m [1m54s[0m 295ms/step
- accuracy: 0.9586 - loss: 0.0704
## [1m 66/251[0m [32m[0m[37m[0m [1m54s[0m 295ms/step
- accuracy: 0.9586 - loss: 0.0704
## [1m 67/251[0m [32m[0m[37m[0m [1m54s[0m 296ms/step
- accuracy: 0.9586 - loss: 0.0703
## [1m 68/251[0m [32m[0m[37m[0m [1m54s[0m 297ms/step
- accuracy: 0.9587 - loss: 0.0702
## [1m 69/251[0m [32m[0m[37m[0m [1m53s[0m 297ms/step
- accuracy: 0.9587 - loss: 0.0701
## [1m 70/251[0m [32m[0m[37m[0m [1m53s[0m 298ms/step
- accuracy: 0.9587 - loss: 0.0700
## [1m 71/251[0m [32m[0m[37m[0m [1m53s[0m 299ms/step
- accuracy: 0.9588 - loss: 0.0700
## [1m 72/251[0m [32m[0m[37m[0m [1m53s[0m 300ms/step
- accuracy: 0.9588 - loss: 0.0699
## [1m 73/251[0m [32m[0m[37m[0m [1m53s[0m 300ms/step
- accuracy: 0.9588 - loss: 0.0699
## [1m 74/251[0m [32m[0m[37m[0m [1m53s[0m 301ms/step
- accuracy: 0.9588 - loss: 0.0698
## [1m 75/251[0m [32m[0m[37m[0m [1m53s[0m 301ms/step
- accuracy: 0.9589 - loss: 0.0697
## [1m 76/251[0m [32m[0m[37m[0m [1m52s[0m 302ms/step
- accuracy: 0.9589 - loss: 0.0696
## [1m 77/251[0m [32m[0m[37m[0m [1m52s[0m 302ms/step
- accuracy: 0.9590 - loss: 0.0696
## [1m 78/251[0m [32m[0m[37m[0m [1m52s[0m 303ms/step
- accuracy: 0.9590 - loss: 0.0695
```

```
## [1m 79/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 303ms/step
- accuracy: 0.9591 - loss: 0.0694
## [1m 80/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 303ms/step
- accuracy: 0.9591 - loss: 0.0693
## [1m 81/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 303ms/step
- accuracy: 0.9592 - loss: 0.0692
## [1m 82/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 303ms/step
- accuracy: 0.9592 - loss: 0.0691
## [1m 83/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 303ms/step
- accuracy: 0.9592 - loss: 0.0691
## [1m 84/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 303ms/step
- accuracy: 0.9593 - loss: 0.0690
## [1m 85/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 303ms/step
- accuracy: 0.9593 - loss: 0.0689
## [1m 86/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 304ms/step
- accuracy: 0.9594 - loss: 0.0688
## [1m 87/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 304ms/step
- accuracy: 0.9594 - loss: 0.0688
## [1m 88/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 304ms/step
- accuracy: 0.9594 - loss: 0.0687
## [1m 89/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 304ms/step
- accuracy: 0.9595 - loss: 0.0686
## [1m 90/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 304ms/step
- accuracy: 0.9595 - loss: 0.0686
## [1m 91/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 303ms/step
- accuracy: 0.9595 - loss: 0.0685
## [1m 92/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 303ms/step
- accuracy: 0.9596 - loss: 0.0685
## [1m 93/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 304ms/step
- accuracy: 0.9596 - loss: 0.0684
## [1m 94/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 303ms/step
- accuracy: 0.9596 - loss: 0.0683
## [1m 95/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 303ms/step
- accuracy: 0.9597 - loss: 0.0683
## [1m 96/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 303ms/step
- accuracy: 0.9597 - loss: 0.0682
## [1m 97/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 303ms/step
- accuracy: 0.9597 - loss: 0.0682
## [1m 98/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 303ms/step
- accuracy: 0.9598 - loss: 0.0681
## [1m 99/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 303ms/step
- accuracy: 0.9598 - loss: 0.0680
## [1m100/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 303ms/step
- accuracy: 0.9599 - loss: 0.0680
## [1m101/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 303ms/step
- accuracy: 0.9599 - loss: 0.0679
## [1m102/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 303ms/step
- accuracy: 0.9599 - loss: 0.0679
## [1m103/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 303ms/step
- accuracy: 0.9599 - loss: 0.0678
```

```
## [1m104/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 303ms/step
- accuracy: 0.9600 - loss: 0.0677
## [1m105/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 303ms/step
- accuracy: 0.9600 - loss: 0.0677
## [1m106/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 302ms/step
- accuracy: 0.9600 - loss: 0.0676
## [1m107/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 302ms/step
- accuracy: 0.9600 - loss: 0.0676
## [1m108/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 302ms/step
- accuracy: 0.9601 - loss: 0.0675
## [1m109/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 302ms/step
- accuracy: 0.9601 - loss: 0.0674
## [1m110/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 302ms/step
- accuracy: 0.9601 - loss: 0.0674
## [1m111/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 301ms/step
- accuracy: 0.9601 - loss: 0.0673
## [1m112/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 301ms/step
- accuracy: 0.9601 - loss: 0.0673
## [1m113/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 301ms/step
- accuracy: 0.9602 - loss: 0.0672
## [1m114/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 302ms/step
- accuracy: 0.9602 - loss: 0.0672
## [1m115/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 301ms/step
- accuracy: 0.9602 - loss: 0.0671
## [1m116/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 301ms/step
- accuracy: 0.9602 - loss: 0.0671
## [1m117/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 301ms/step
- accuracy: 0.9602 - loss: 0.0671
## [1m118/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 302ms/step
- accuracy: 0.9602 - loss: 0.0670
## [1m119/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 302ms/step
- accuracy: 0.9602 - loss: 0.0670
## [1m120/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 303ms/step
- accuracy: 0.9602 - loss: 0.0670
## [1m121/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 304ms/step
- accuracy: 0.9602 - loss: 0.0669
## [1m122/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 304ms/step
- accuracy: 0.9603 - loss: 0.0669
## [1m123/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 304ms/step
- accuracy: 0.9603 - loss: 0.0669
## [1m124/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 304ms/step
- accuracy: 0.9603 - loss: 0.0668
## [1m125/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 304ms/step
- accuracy: 0.9603 - loss: 0.0668
## [1m126/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 305ms/step
- accuracy: 0.9603 - loss: 0.0667
## [1m127/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 305ms/step
- accuracy: 0.9603 - loss: 0.0667
## [1m128/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 305ms/step
- accuracy: 0.9604 - loss: 0.0666
```

```
## [1m129/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 306ms/step
- accuracy: 0.9604 - loss: 0.0666
## [1m130/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 306ms/step
- accuracy: 0.9604 - loss: 0.0666
## [1m131/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 309ms/step
- accuracy: 0.9604 - loss: 0.0665
## [1m132/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 310ms/step
- accuracy: 0.9604 - loss: 0.0665
## [1m133/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 311ms/step
- accuracy: 0.9605 - loss: 0.0664
## [1m134/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 311ms/step
- accuracy: 0.9605 - loss: 0.0664
## [1m135/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 312ms/step
- accuracy: 0.9605 - loss: 0.0663
## [1m136/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 312ms/step
- accuracy: 0.9605 - loss: 0.0663
## [1m137/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 312ms/step
- accuracy: 0.9605 - loss: 0.0663
## [1m138/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 311ms/step
- accuracy: 0.9605 - loss: 0.0662
## [1m139/251[0m [32m-----[0m[37m-----[0m [1m34s[0m 311ms/step
- accuracy: 0.9605 - loss: 0.0662
## [1m140/251[0m [32m-----[0m[37m-----[0m [1m34s[0m 311ms/step
- accuracy: 0.9606 - loss: 0.0662
## [1m141/251[0m [32m-----[0m[37m-----[0m [1m34s[0m 312ms/step
- accuracy: 0.9606 - loss: 0.0662
## [1m142/251[0m [32m-----[0m[37m-----[0m [1m33s[0m 312ms/step
- accuracy: 0.9606 - loss: 0.0661
## [1m143/251[0m [32m-----[0m[37m-----[0m [1m33s[0m 312ms/step
- accuracy: 0.9606 - loss: 0.0661
## [1m144/251[0m [32m-----[0m[37m-----[0m [1m33s[0m 312ms/step
- accuracy: 0.9606 - loss: 0.0661
## [1m145/251[0m [32m-----[0m[37m-----[0m [1m33s[0m 312ms/step
- accuracy: 0.9606 - loss: 0.0660
## [1m146/251[0m [32m-----[0m[37m-----[0m [1m32s[0m 312ms/step
- accuracy: 0.9606 - loss: 0.0660
## [1m147/251[0m [32m-----[0m[37m-----[0m [1m32s[0m 312ms/step
- accuracy: 0.9606 - loss: 0.0660
## [1m148/251[0m [32m-----[0m[37m-----[0m [1m32s[0m 311ms/step
- accuracy: 0.9606 - loss: 0.0660
## [1m149/251[0m [32m-----[0m[37m-----[0m [1m31s[0m 311ms/step
- accuracy: 0.9607 - loss: 0.0659
## [1m150/251[0m [32m-----[0m[37m-----[0m [1m31s[0m 311ms/step
- accuracy: 0.9607 - loss: 0.0659
## [1m151/251[0m [32m-----[0m[37m-----[0m [1m31s[0m 311ms/step
- accuracy: 0.9607 - loss: 0.0659
## [1m152/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 311ms/step
- accuracy: 0.9607 - loss: 0.0659
## [1m153/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 312ms/step
- accuracy: 0.9607 - loss: 0.0658
```

```
## [1m154/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 312ms/step
- accuracy: 0.9607 - loss: 0.0658
## [1m155/251[0m [32m-----[0m[37m-----[0m [1m29s[0m 312ms/step
- accuracy: 0.9607 - loss: 0.0658
## [1m156/251[0m [32m-----[0m[37m-----[0m [1m29s[0m 311ms/step
- accuracy: 0.9607 - loss: 0.0657
## [1m157/251[0m [32m-----[0m[37m-----[0m [1m29s[0m 311ms/step
- accuracy: 0.9607 - loss: 0.0657
## [1m158/251[0m [32m-----[0m[37m-----[0m [1m28s[0m 311ms/step
- accuracy: 0.9607 - loss: 0.0657
## [1m159/251[0m [32m-----[0m[37m-----[0m [1m28s[0m 311ms/step
- accuracy: 0.9608 - loss: 0.0656
## [1m160/251[0m [32m-----[0m[37m-----[0m [1m28s[0m 311ms/step
- accuracy: 0.9608 - loss: 0.0656
## [1m161/251[0m [32m-----[0m[37m-----[0m [1m28s[0m 311ms/step
- accuracy: 0.9608 - loss: 0.0656
## [1m162/251[0m [32m-----[0m[37m-----[0m [1m27s[0m 312ms/step
- accuracy: 0.9608 - loss: 0.0655
## [1m163/251[0m [32m-----[0m[37m-----[0m [1m27s[0m 312ms/step
- accuracy: 0.9608 - loss: 0.0655
## [1m164/251[0m [32m-----[0m[37m-----[0m [1m27s[0m 312ms/step
- accuracy: 0.9608 - loss: 0.0655
## [1m165/251[0m [32m-----[0m[37m-----[0m [1m26s[0m 312ms/step
- accuracy: 0.9608 - loss: 0.0655
## [1m166/251[0m [32m-----[0m[37m-----[0m [1m26s[0m 312ms/step
- accuracy: 0.9608 - loss: 0.0654
## [1m167/251[0m [32m-----[0m[37m-----[0m [1m26s[0m 312ms/step
- accuracy: 0.9609 - loss: 0.0654
## [1m168/251[0m [32m-----[0m[37m-----[0m [1m25s[0m 312ms/step
- accuracy: 0.9609 - loss: 0.0654
## [1m169/251[0m [32m-----[0m[37m-----[0m [1m25s[0m 312ms/step
- accuracy: 0.9609 - loss: 0.0654
## [1m170/251[0m [32m-----[0m[37m-----[0m [1m25s[0m 312ms/step
- accuracy: 0.9609 - loss: 0.0653
## [1m171/251[0m [32m-----[0m[37m-----[0m [1m24s[0m 312ms/step
- accuracy: 0.9609 - loss: 0.0653
## [1m172/251[0m [32m-----[0m[37m-----[0m [1m24s[0m 312ms/step
- accuracy: 0.9609 - loss: 0.0653
## [1m173/251[0m [32m-----[0m[37m-----[0m [1m24s[0m 312ms/step
- accuracy: 0.9609 - loss: 0.0653
## [1m174/251[0m [32m-----[0m[37m-----[0m [1m23s[0m 312ms/step
- accuracy: 0.9609 - loss: 0.0652
## [1m175/251[0m [32m-----[0m[37m-----[0m [1m23s[0m 312ms/step
- accuracy: 0.9609 - loss: 0.0652
## [1m176/251[0m [32m-----[0m[37m-----[0m [1m23s[0m 311ms/step
- accuracy: 0.9610 - loss: 0.0652
## [1m177/251[0m [32m-----[0m[37m-----[0m [1m23s[0m 311ms/step
- accuracy: 0.9610 - loss: 0.0652
## [1m178/251[0m [32m-----[0m[37m-----[0m [1m22s[0m 311ms/step
- accuracy: 0.9610 - loss: 0.0651
```

[illegible]

[illegible]


```
## [1m229/251[0m [32m-----[0m [37m-----[0m [1m6s[0m 309ms/step -  
accuracy: 0.9616 - loss: 0.0639  
## [1m230/251[0m [32m-----[0m [37m-----[0m [1m6s[0m 309ms/step -  
accuracy: 0.9616 - loss: 0.0639  
## [1m231/251[0m [32m-----[0m [37m-----[0m [1m6s[0m 309ms/step -  
accuracy: 0.9616 - loss: 0.0639  
## [1m232/251[0m [32m-----[0m [37m-----[0m [1m5s[0m 308ms/step -  
accuracy: 0.9616 - loss: 0.0639  
## [1m233/251[0m [32m-----[0m [37m-----[0m [1m5s[0m 308ms/step -  
accuracy: 0.9616 - loss: 0.0639  
## [1m234/251[0m [32m-----[0m [37m-----[0m [1m5s[0m 308ms/step -  
accuracy: 0.9616 - loss: 0.0638  
## [1m235/251[0m [32m-----[0m [37m-----[0m [1m4s[0m 308ms/step -  
accuracy: 0.9616 - loss: 0.0638  
## [1m236/251[0m [32m-----[0m [37m-----[0m [1m4s[0m 308ms/step -  
accuracy: 0.9617 - loss: 0.0638  
## [1m237/251[0m [32m-----[0m [37m-----[0m [1m4s[0m 308ms/step -  
accuracy: 0.9617 - loss: 0.0638  
## [1m238/251[0m [32m-----[0m [37m-----[0m [1m4s[0m 308ms/step -  
accuracy: 0.9617 - loss: 0.0638  
## [1m239/251[0m [32m-----[0m [37m-----[0m [1m3s[0m 308ms/step -  
accuracy: 0.9617 - loss: 0.0638  
## [1m240/251[0m [32m-----[0m [37m-----[0m [1m3s[0m 308ms/step -  
accuracy: 0.9617 - loss: 0.0638  
## [1m241/251[0m [32m-----[0m [37m-----[0m [1m3s[0m 307ms/step -  
accuracy: 0.9617 - loss: 0.0637  
## [1m242/251[0m [32m-----[0m [37m-----[0m [1m2s[0m 307ms/step -  
accuracy: 0.9617 - loss: 0.0637  
## [1m243/251[0m [32m-----[0m [37m-----[0m [1m2s[0m 307ms/step -  
accuracy: 0.9617 - loss: 0.0637  
## [1m244/251[0m [32m-----[0m [37m-----[0m [1m2s[0m 307ms/step -  
accuracy: 0.9617 - loss: 0.0637  
## [1m245/251[0m [32m-----[0m [37m-----[0m [1m1s[0m 307ms/step -  
accuracy: 0.9617 - loss: 0.0637  
## [1m246/251[0m [32m-----[0m [37m-----[0m [1m1s[0m 307ms/step -  
accuracy: 0.9617 - loss: 0.0637  
## [1m247/251[0m [32m-----[0m [37m-----[0m [1m1s[0m 307ms/step -  
accuracy: 0.9617 - loss: 0.0637  
## [1m248/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 307ms/step -  
accuracy: 0.9617 - loss: 0.0637  
## [1m249/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 307ms/step -  
accuracy: 0.9617 - loss: 0.0636  
## [1m250/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 307ms/step -  
accuracy: 0.9617 - loss: 0.0636  
## [1m251/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 307ms/step -  
accuracy: 0.9618 - loss: 0.0636  
## Epoch 3: val_accuracy improved from 0.95858 to 0.97305, saving model  
to F:\Desktop\Uni\IA\R\saved_models\weights-03-0.97.keras  
##  
## [1m251/251[0m [32m-----[0m [37m-----[0m [1m84s[0m 336ms/step
```

```
- accuracy: 0.9618 - loss: 0.0636 - val_accuracy: 0.9731 - val_loss: 0.0487
## Epoch 4/10
##
## [1m 1/251[0m [37m-----[0m [1m1:31[0m 367ms/step -
accuracy: 0.9688 - loss: 0.0695
## [1m 2/251[0m [37m-----[0m [1m1:19[0m 320ms/step -
accuracy: 0.9688 - loss: 0.0642
## [1m 3/251[0m [37m-----[0m [1m1:15[0m 303ms/step -
accuracy: 0.9688 - loss: 0.0624
## [1m 4/251[0m [37m-----[0m [1m1:13[0m 296ms/step -
accuracy: 0.9707 - loss: 0.0591
## [1m 5/251[0m [37m-----[0m [1m1:14[0m 301ms/step -
accuracy: 0.9716 - loss: 0.0575
## [1m 6/251[0m [37m-----[0m [1m1:16[0m 312ms/step -
accuracy: 0.9720 - loss: 0.0562
## [1m 7/251[0m [37m-----[0m [1m1:16[0m 312ms/step -
accuracy: 0.9728 - loss: 0.0546
## [1m 8/251[0m [37m-----[0m [1m1:15[0m 309ms/step -
accuracy: 0.9733 - loss: 0.0538
## [1m 9/251[0m [37m-----[0m [1m1:15[0m 311ms/step -
accuracy: 0.9731 - loss: 0.0544
## [1m 10/251[0m [37m-----[0m [1m1:25[0m 353ms/step -
accuracy: 0.9733 - loss: 0.0544
## [1m 11/251[0m [37m-----[0m [1m1:28[0m 370ms/step -
accuracy: 0.9734 - loss: 0.0542
## [1m 12/251[0m [37m-----[0m [1m1:28[0m 370ms/step -
accuracy: 0.9735 - loss: 0.0539
## [1m 13/251[0m [32m-[0m[37m-----[0m [1m1:27[0m 369ms/step
- accuracy: 0.9735 - loss: 0.0537
## [1m 14/251[0m [32m-[0m[37m-----[0m [1m1:26[0m 366ms/step
- accuracy: 0.9731 - loss: 0.0537
## [1m 15/251[0m [32m-[0m[37m-----[0m [1m1:25[0m 364ms/step
- accuracy: 0.9730 - loss: 0.0536
## [1m 16/251[0m [32m-[0m[37m-----[0m [1m1:24[0m 360ms/step
- accuracy: 0.9730 - loss: 0.0535
## [1m 17/251[0m [32m-[0m[37m-----[0m [1m1:23[0m 356ms/step
- accuracy: 0.9730 - loss: 0.0533
## [1m 18/251[0m [32m-[0m[37m-----[0m [1m1:23[0m 359ms/step
- accuracy: 0.9731 - loss: 0.0532
## [1m 19/251[0m [32m-[0m[37m-----[0m [1m1:23[0m 361ms/step
- accuracy: 0.9730 - loss: 0.0532
## [1m 20/251[0m [32m-[0m[37m-----[0m [1m1:22[0m 358ms/step
- accuracy: 0.9731 - loss: 0.0532
## [1m 21/251[0m [32m-[0m[37m-----[0m [1m1:22[0m 357ms/step
- accuracy: 0.9731 - loss: 0.0531
## [1m 22/251[0m [32m-[0m[37m-----[0m [1m1:21[0m 355ms/step
- accuracy: 0.9732 - loss: 0.0530
## [1m 23/251[0m [32m-[0m[37m-----[0m [1m1:21[0m 356ms/step
- accuracy: 0.9731 - loss: 0.0531
```

```
## [1m 24/251[0m [32m[0m[37m[0m [1m1:20[0m 353ms/step
- accuracy: 0.9730 - loss: 0.0531
## [1m 25/251[0m [32m[0m[37m[0m [1m1:19[0m 351ms/step
- accuracy: 0.9729 - loss: 0.0532
## [1m 26/251[0m [32m[0m[37m[0m [1m1:19[0m 351ms/step
- accuracy: 0.9728 - loss: 0.0533
## [1m 27/251[0m [32m[0m[37m[0m [1m1:18[0m 350ms/step
- accuracy: 0.9727 - loss: 0.0534
## [1m 28/251[0m [32m[0m[37m[0m [1m1:17[0m 350ms/step
- accuracy: 0.9726 - loss: 0.0535
## [1m 29/251[0m [32m[0m[37m[0m [1m1:17[0m 347ms/step
- accuracy: 0.9725 - loss: 0.0535
## [1m 30/251[0m [32m[0m[37m[0m [1m1:16[0m 345ms/step
- accuracy: 0.9725 - loss: 0.0536
## [1m 31/251[0m [32m[0m[37m[0m [1m1:15[0m 343ms/step
- accuracy: 0.9725 - loss: 0.0536
## [1m 32/251[0m [32m[0m[37m[0m [1m1:14[0m 342ms/step
- accuracy: 0.9724 - loss: 0.0536
## [1m 33/251[0m [32m[0m[37m[0m [1m1:14[0m 341ms/step
- accuracy: 0.9724 - loss: 0.0536
## [1m 34/251[0m [32m[0m[37m[0m [1m1:14[0m 343ms/step
- accuracy: 0.9724 - loss: 0.0536
## [1m 35/251[0m [32m[0m[37m[0m [1m1:14[0m 345ms/step
- accuracy: 0.9723 - loss: 0.0537
## [1m 36/251[0m [32m[0m[37m[0m [1m1:14[0m 345ms/step
- accuracy: 0.9723 - loss: 0.0537
## [1m 37/251[0m [32m[0m[37m[0m [1m1:13[0m 346ms/step
- accuracy: 0.9723 - loss: 0.0537
## [1m 38/251[0m [32m[0m[37m[0m [1m1:14[0m 347ms/step
- accuracy: 0.9723 - loss: 0.0537
## [1m 39/251[0m [32m[0m[37m[0m [1m1:14[0m 350ms/step
- accuracy: 0.9723 - loss: 0.0537
## [1m 40/251[0m [32m[0m[37m[0m [1m1:14[0m 352ms/step
- accuracy: 0.9723 - loss: 0.0537
## [1m 41/251[0m [32m[0m[37m[0m [1m1:13[0m 351ms/step
- accuracy: 0.9722 - loss: 0.0537
## [1m 42/251[0m [32m[0m[37m[0m [1m1:13[0m 350ms/step
- accuracy: 0.9722 - loss: 0.0537
## [1m 43/251[0m [32m[0m[37m[0m [1m1:12[0m 348ms/step
- accuracy: 0.9722 - loss: 0.0537
## [1m 44/251[0m [32m[0m[37m[0m [1m1:11[0m 347ms/step
- accuracy: 0.9723 - loss: 0.0536
## [1m 45/251[0m [32m[0m[37m[0m [1m1:11[0m 346ms/step
- accuracy: 0.9723 - loss: 0.0536
## [1m 46/251[0m [32m[0m[37m[0m [1m1:10[0m 345ms/step
- accuracy: 0.9723 - loss: 0.0535
## [1m 47/251[0m [32m[0m[37m[0m [1m1:10[0m 343ms/step
- accuracy: 0.9723 - loss: 0.0535
## [1m 48/251[0m [32m[0m[37m[0m [1m1:09[0m 342ms/step
- accuracy: 0.9723 - loss: 0.0534
```

```
## [1m 49/251[0m [32m[0m[37m[0m [1m1:08[0m 341ms/step
- accuracy: 0.9723 - loss: 0.0534
## [1m 50/251[0m [32m[0m[37m[0m [1m1:08[0m 340ms/step
- accuracy: 0.9723 - loss: 0.0534
## [1m 51/251[0m [32m[0m[37m[0m [1m1:07[0m 339ms/step
- accuracy: 0.9723 - loss: 0.0534
## [1m 52/251[0m [32m[0m[37m[0m [1m1:07[0m 339ms/step
- accuracy: 0.9723 - loss: 0.0535
## [1m 53/251[0m [32m[0m[37m[0m [1m1:07[0m 339ms/step
- accuracy: 0.9723 - loss: 0.0535
## [1m 54/251[0m [32m[0m[37m[0m [1m1:06[0m 338ms/step
- accuracy: 0.9723 - loss: 0.0536
## [1m 55/251[0m [32m[0m[37m[0m [1m1:06[0m 338ms/step
- accuracy: 0.9722 - loss: 0.0536
## [1m 56/251[0m [32m[0m[37m[0m [1m1:06[0m 339ms/step
- accuracy: 0.9722 - loss: 0.0537
## [1m 57/251[0m [32m[0m[37m[0m [1m1:05[0m 339ms/step
- accuracy: 0.9722 - loss: 0.0537
## [1m 58/251[0m [32m[0m[37m[0m [1m1:05[0m 340ms/step
- accuracy: 0.9721 - loss: 0.0538
## [1m 59/251[0m [32m[0m[37m[0m [1m1:05[0m 340ms/step
- accuracy: 0.9721 - loss: 0.0539
## [1m 60/251[0m [32m[0m[37m[0m [1m1:05[0m 341ms/step
- accuracy: 0.9720 - loss: 0.0539
## [1m 61/251[0m [32m[0m[37m[0m [1m1:04[0m 341ms/step
- accuracy: 0.9720 - loss: 0.0540
## [1m 62/251[0m [32m[0m[37m[0m [1m1:04[0m 341ms/step
- accuracy: 0.9720 - loss: 0.0540
## [1m 63/251[0m [32m[0m[37m[0m [1m1:04[0m 340ms/step
- accuracy: 0.9720 - loss: 0.0541
## [1m 64/251[0m [32m[0m[37m[0m [1m1:03[0m 340ms/step
- accuracy: 0.9719 - loss: 0.0541
## [1m 65/251[0m [32m[0m[37m[0m [1m1:03[0m 341ms/step
- accuracy: 0.9719 - loss: 0.0541
## [1m 66/251[0m [32m[0m[37m[0m [1m1:02[0m 340ms/step
- accuracy: 0.9719 - loss: 0.0541
## [1m 67/251[0m [32m[0m[37m[0m [1m1:02[0m 340ms/step
- accuracy: 0.9719 - loss: 0.0541
## [1m 68/251[0m [32m[0m[37m[0m [1m1:02[0m 340ms/step
- accuracy: 0.9719 - loss: 0.0542
## [1m 69/251[0m [32m[0m[37m[0m [1m1:01[0m 339ms/step
- accuracy: 0.9719 - loss: 0.0542
## [1m 70/251[0m [32m[0m[37m[0m [1m1:01[0m 339ms/step
- accuracy: 0.9719 - loss: 0.0542
## [1m 71/251[0m [32m[0m[37m[0m [1m1:00[0m 338ms/step
- accuracy: 0.9719 - loss: 0.0541
## [1m 72/251[0m [32m[0m[37m[0m [1m1:00[0m 337ms/step
- accuracy: 0.9719 - loss: 0.0542
## [1m 73/251[0m [32m[0m[37m[0m [1m59s[0m 337ms/step
- accuracy: 0.9719 - loss: 0.0542
```

```
## [1m 74/251[0m [32m-----[0m[37m-----[0m [1m59s[0m 335ms/step
- accuracy: 0.9718 - loss: 0.0542
## [1m 75/251[0m [32m-----[0m[37m-----[0m [1m58s[0m 335ms/step
- accuracy: 0.9718 - loss: 0.0543
## [1m 76/251[0m [32m-----[0m[37m-----[0m [1m58s[0m 334ms/step
- accuracy: 0.9718 - loss: 0.0543
## [1m 77/251[0m [32m-----[0m[37m-----[0m [1m57s[0m 333ms/step
- accuracy: 0.9718 - loss: 0.0543
## [1m 78/251[0m [32m-----[0m[37m-----[0m [1m57s[0m 334ms/step
- accuracy: 0.9718 - loss: 0.0544
## [1m 79/251[0m [32m-----[0m[37m-----[0m [1m57s[0m 333ms/step
- accuracy: 0.9717 - loss: 0.0545
## [1m 80/251[0m [32m-----[0m[37m-----[0m [1m56s[0m 333ms/step
- accuracy: 0.9717 - loss: 0.0546
## [1m 81/251[0m [32m-----[0m[37m-----[0m [1m56s[0m 333ms/step
- accuracy: 0.9717 - loss: 0.0546
## [1m 82/251[0m [32m-----[0m[37m-----[0m [1m56s[0m 332ms/step
- accuracy: 0.9717 - loss: 0.0547
## [1m 83/251[0m [32m-----[0m[37m-----[0m [1m55s[0m 332ms/step
- accuracy: 0.9717 - loss: 0.0548
## [1m 84/251[0m [32m-----[0m[37m-----[0m [1m55s[0m 331ms/step
- accuracy: 0.9716 - loss: 0.0548
## [1m 85/251[0m [32m-----[0m[37m-----[0m [1m54s[0m 330ms/step
- accuracy: 0.9716 - loss: 0.0549
## [1m 86/251[0m [32m-----[0m[37m-----[0m [1m54s[0m 330ms/step
- accuracy: 0.9716 - loss: 0.0550
## [1m 87/251[0m [32m-----[0m[37m-----[0m [1m54s[0m 329ms/step
- accuracy: 0.9715 - loss: 0.0551
## [1m 88/251[0m [32m-----[0m[37m-----[0m [1m53s[0m 329ms/step
- accuracy: 0.9715 - loss: 0.0551
## [1m 89/251[0m [32m-----[0m[37m-----[0m [1m53s[0m 329ms/step
- accuracy: 0.9714 - loss: 0.0552
## [1m 90/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 329ms/step
- accuracy: 0.9714 - loss: 0.0553
## [1m 91/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 329ms/step
- accuracy: 0.9714 - loss: 0.0554
## [1m 92/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 329ms/step
- accuracy: 0.9713 - loss: 0.0555
## [1m 93/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 330ms/step
- accuracy: 0.9713 - loss: 0.0555
## [1m 94/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 330ms/step
- accuracy: 0.9712 - loss: 0.0556
## [1m 95/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 330ms/step
- accuracy: 0.9712 - loss: 0.0557
## [1m 96/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 329ms/step
- accuracy: 0.9712 - loss: 0.0557
## [1m 97/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 330ms/step
- accuracy: 0.9711 - loss: 0.0558
## [1m 98/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 331ms/step
- accuracy: 0.9711 - loss: 0.0558
```

```
## [1m 99/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 331ms/step
- accuracy: 0.9711 - loss: 0.0558
## [1m100/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 331ms/step
- accuracy: 0.9711 - loss: 0.0559
## [1m101/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 331ms/step
- accuracy: 0.9710 - loss: 0.0559
## [1m102/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 330ms/step
- accuracy: 0.9710 - loss: 0.0559
## [1m103/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 330ms/step
- accuracy: 0.9710 - loss: 0.0560
## [1m104/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 330ms/step
- accuracy: 0.9710 - loss: 0.0560
## [1m105/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 329ms/step
- accuracy: 0.9710 - loss: 0.0560
## [1m106/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 329ms/step
- accuracy: 0.9710 - loss: 0.0560
## [1m107/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 329ms/step
- accuracy: 0.9710 - loss: 0.0561
## [1m108/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 329ms/step
- accuracy: 0.9710 - loss: 0.0561
## [1m109/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 328ms/step
- accuracy: 0.9710 - loss: 0.0561
## [1m110/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 328ms/step
- accuracy: 0.9710 - loss: 0.0561
## [1m111/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 329ms/step
- accuracy: 0.9709 - loss: 0.0561
## [1m112/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 328ms/step
- accuracy: 0.9709 - loss: 0.0562
## [1m113/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 328ms/step
- accuracy: 0.9709 - loss: 0.0562
## [1m114/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 328ms/step
- accuracy: 0.9709 - loss: 0.0562
## [1m115/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 328ms/step
- accuracy: 0.9709 - loss: 0.0563
## [1m116/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 327ms/step
- accuracy: 0.9709 - loss: 0.0563
## [1m117/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 327ms/step
- accuracy: 0.9708 - loss: 0.0563
## [1m118/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 327ms/step
- accuracy: 0.9708 - loss: 0.0563
## [1m119/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 327ms/step
- accuracy: 0.9708 - loss: 0.0563
## [1m120/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 327ms/step
- accuracy: 0.9708 - loss: 0.0564
## [1m121/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 327ms/step
- accuracy: 0.9708 - loss: 0.0564
## [1m122/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 326ms/step
- accuracy: 0.9708 - loss: 0.0564
## [1m123/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 326ms/step
- accuracy: 0.9707 - loss: 0.0564
```

```
## [1m124/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 326ms/step
- accuracy: 0.9707 - loss: 0.0565
## [1m125/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 325ms/step
- accuracy: 0.9707 - loss: 0.0565
## [1m126/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 325ms/step
- accuracy: 0.9707 - loss: 0.0565
## [1m127/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 325ms/step
- accuracy: 0.9707 - loss: 0.0565
## [1m128/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 325ms/step
- accuracy: 0.9707 - loss: 0.0566
## [1m129/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 325ms/step
- accuracy: 0.9706 - loss: 0.0566
## [1m130/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 324ms/step
- accuracy: 0.9706 - loss: 0.0566
## [1m131/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 324ms/step
- accuracy: 0.9706 - loss: 0.0566
## [1m132/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 324ms/step
- accuracy: 0.9706 - loss: 0.0566
## [1m133/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 323ms/step
- accuracy: 0.9706 - loss: 0.0567
## [1m134/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 323ms/step
- accuracy: 0.9705 - loss: 0.0567
## [1m135/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 323ms/step
- accuracy: 0.9705 - loss: 0.0567
## [1m136/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 323ms/step
- accuracy: 0.9705 - loss: 0.0567
## [1m137/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 323ms/step
- accuracy: 0.9705 - loss: 0.0567
## [1m138/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 323ms/step
- accuracy: 0.9705 - loss: 0.0568
## [1m139/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 323ms/step
- accuracy: 0.9705 - loss: 0.0568
## [1m140/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 322ms/step
- accuracy: 0.9705 - loss: 0.0568
## [1m141/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 322ms/step
- accuracy: 0.9704 - loss: 0.0568
## [1m142/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 322ms/step
- accuracy: 0.9704 - loss: 0.0568
## [1m143/251[0m [32m-----[0m[37m-----[0m [1m34s[0m 322ms/step
- accuracy: 0.9704 - loss: 0.0568
## [1m144/251[0m [32m-----[0m[37m-----[0m [1m34s[0m 321ms/step
- accuracy: 0.9704 - loss: 0.0568
## [1m145/251[0m [32m-----[0m[37m-----[0m [1m34s[0m 321ms/step
- accuracy: 0.9704 - loss: 0.0569
## [1m146/251[0m [32m-----[0m[37m-----[0m [1m33s[0m 321ms/step
- accuracy: 0.9703 - loss: 0.0569
## [1m147/251[0m [32m-----[0m[37m-----[0m [1m33s[0m 321ms/step
- accuracy: 0.9703 - loss: 0.0569
## [1m148/251[0m [32m-----[0m[37m-----[0m [1m33s[0m 321ms/step
- accuracy: 0.9703 - loss: 0.0569
```

```
## [1m149/251[0m [32m-----[0m[37m-----[0m [1m32s[0m 320ms/step
- accuracy: 0.9703 - loss: 0.0569
## [1m150/251[0m [32m-----[0m[37m-----[0m [1m32s[0m 320ms/step
- accuracy: 0.9703 - loss: 0.0570
## [1m151/251[0m [32m-----[0m[37m-----[0m [1m31s[0m 320ms/step
- accuracy: 0.9702 - loss: 0.0570
## [1m152/251[0m [32m-----[0m[37m-----[0m [1m31s[0m 320ms/step
- accuracy: 0.9702 - loss: 0.0570
## [1m153/251[0m [32m-----[0m[37m-----[0m [1m31s[0m 319ms/step
- accuracy: 0.9702 - loss: 0.0570
## [1m154/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 319ms/step
- accuracy: 0.9702 - loss: 0.0570
## [1m155/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 319ms/step
- accuracy: 0.9702 - loss: 0.0570
## [1m156/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 318ms/step
- accuracy: 0.9702 - loss: 0.0570
## [1m157/251[0m [32m-----[0m[37m-----[0m [1m29s[0m 318ms/step
- accuracy: 0.9701 - loss: 0.0571
## [1m158/251[0m [32m-----[0m[37m-----[0m [1m29s[0m 318ms/step
- accuracy: 0.9701 - loss: 0.0571
## [1m159/251[0m [32m-----[0m[37m-----[0m [1m29s[0m 318ms/step
- accuracy: 0.9701 - loss: 0.0571
## [1m160/251[0m [32m-----[0m[37m-----[0m [1m28s[0m 318ms/step
- accuracy: 0.9701 - loss: 0.0571
## [1m161/251[0m [32m-----[0m[37m-----[0m [1m28s[0m 318ms/step
- accuracy: 0.9701 - loss: 0.0571
## [1m162/251[0m [32m-----[0m[37m-----[0m [1m28s[0m 317ms/step
- accuracy: 0.9701 - loss: 0.0571
## [1m163/251[0m [32m-----[0m[37m-----[0m [1m27s[0m 317ms/step
- accuracy: 0.9701 - loss: 0.0571
## [1m164/251[0m [32m-----[0m[37m-----[0m [1m27s[0m 317ms/step
- accuracy: 0.9700 - loss: 0.0571
## [1m165/251[0m [32m-----[0m[37m-----[0m [1m27s[0m 318ms/step
- accuracy: 0.9700 - loss: 0.0571
## [1m166/251[0m [32m-----[0m[37m-----[0m [1m27s[0m 318ms/step
- accuracy: 0.9700 - loss: 0.0572
## [1m167/251[0m [32m-----[0m[37m-----[0m [1m26s[0m 317ms/step
- accuracy: 0.9700 - loss: 0.0572
## [1m168/251[0m [32m-----[0m[37m-----[0m [1m26s[0m 318ms/step
- accuracy: 0.9700 - loss: 0.0572
## [1m169/251[0m [32m-----[0m[37m-----[0m [1m26s[0m 318ms/step
- accuracy: 0.9700 - loss: 0.0572
## [1m170/251[0m [32m-----[0m[37m-----[0m [1m25s[0m 318ms/step
- accuracy: 0.9700 - loss: 0.0572
## [1m171/251[0m [32m-----[0m[37m-----[0m [1m25s[0m 318ms/step
- accuracy: 0.9700 - loss: 0.0572
## [1m172/251[0m [32m-----[0m[37m-----[0m [1m25s[0m 318ms/step
- accuracy: 0.9700 - loss: 0.0572
## [1m173/251[0m [32m-----[0m[37m-----[0m [1m24s[0m 317ms/step
- accuracy: 0.9699 - loss: 0.0572
```


[illegible]

[illegible]

[illegible]

```

## [1m249/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 316ms/step -
accuracy: 0.9697 - loss: 0.0566
## [1m250/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 316ms/step -
accuracy: 0.9697 - loss: 0.0566
## [1m251/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 316ms/step -
accuracy: 0.9697 - loss: 0.0566
## Epoch 4: val_accuracy improved from 0.97305 to 0.97655, saving model
to F:\Desktop\Uni\IA\R\saved_models\weights-04-0.98.keras
##
## [1m251/251[0m [32m-----[0m [37m-----[0m [1m86s[0m 343ms/step
- accuracy: 0.9697 - loss: 0.0566 - val_accuracy: 0.9765 - val_loss:
0.0441
## Epoch 5/10
##
## [1m 1/251[0m [37m-----[0m [1m1:30[0m 361ms/step -
accuracy: 1.0000 - loss: 0.0056
## [1m 2/251[0m [37m-----[0m [1m1:18[0m 317ms/step -
accuracy: 0.9922 - loss: 0.0176
## [1m 3/251[0m [37m-----[0m [1m1:14[0m 300ms/step -
accuracy: 0.9844 - loss: 0.0304
## [1m 4/251[0m [37m-----[0m [1m1:14[0m 303ms/step -
accuracy: 0.9824 - loss: 0.0338
## [1m 5/251[0m [37m-----[0m [1m1:14[0m 305ms/step -
accuracy: 0.9809 - loss: 0.0361
## [1m 6/251[0m [37m-----[0m [1m1:12[0m 297ms/step -
accuracy: 0.9798 - loss: 0.0381
## [1m 7/251[0m [37m-----[0m [1m1:11[0m 293ms/step -
accuracy: 0.9795 - loss: 0.0389
## [1m 8/251[0m [37m-----[0m [1m1:10[0m 290ms/step -
accuracy: 0.9791 - loss: 0.0396
## [1m 9/251[0m [37m-----[0m [1m1:09[0m 288ms/step -
accuracy: 0.9791 - loss: 0.0398
## [1m 10/251[0m [37m-----[0m [1m1:09[0m 289ms/step -
accuracy: 0.9787 - loss: 0.0402
## [1m 11/251[0m [37m-----[0m [1m1:08[0m 287ms/step -
accuracy: 0.9783 - loss: 0.0404
## [1m 12/251[0m [37m-----[0m [1m1:09[0m 289ms/step -
accuracy: 0.9782 - loss: 0.0403
## [1m 13/251[0m [32m-----[0m [37m-----[0m [1m1:09[0m 291ms/step
- accuracy: 0.9782 - loss: 0.0400
## [1m 14/251[0m [32m-----[0m [37m-----[0m [1m1:08[0m 290ms/step
- accuracy: 0.9778 - loss: 0.0401
## [1m 15/251[0m [32m-----[0m [37m-----[0m [1m1:08[0m 289ms/step
- accuracy: 0.9774 - loss: 0.0403
## [1m 16/251[0m [32m-----[0m [37m-----[0m [1m1:07[0m 289ms/step
- accuracy: 0.9768 - loss: 0.0406
## [1m 17/251[0m [32m-----[0m [37m-----[0m [1m1:08[0m 291ms/step
- accuracy: 0.9764 - loss: 0.0409
## [1m 18/251[0m [32m-----[0m [37m-----[0m [1m1:08[0m 293ms/step
- accuracy: 0.9758 - loss: 0.0414

```

```
## [1m 19/251[0m [32m[0m[37m[0m [1m1:07[0m 293ms/step
- accuracy: 0.9754 - loss: 0.0417
## [1m 20/251[0m [32m[0m[37m[0m [1m1:07[0m 293ms/step
- accuracy: 0.9750 - loss: 0.0420
## [1m 21/251[0m [32m[0m[37m[0m [1m1:07[0m 292ms/step
- accuracy: 0.9748 - loss: 0.0422
## [1m 22/251[0m [32m[0m[37m[0m [1m1:06[0m 292ms/step
- accuracy: 0.9746 - loss: 0.0424
## [1m 23/251[0m [32m[0m[37m[0m [1m1:06[0m 291ms/step
- accuracy: 0.9743 - loss: 0.0427
## [1m 24/251[0m [32m[0m[37m[0m [1m1:05[0m 290ms/step
- accuracy: 0.9741 - loss: 0.0429
## [1m 25/251[0m [32m[0m[37m[0m [1m1:05[0m 289ms/step
- accuracy: 0.9738 - loss: 0.0432
## [1m 26/251[0m [32m[0m[37m[0m [1m1:05[0m 290ms/step
- accuracy: 0.9735 - loss: 0.0436
## [1m 27/251[0m [32m[0m[37m[0m [1m1:04[0m 290ms/step
- accuracy: 0.9733 - loss: 0.0439
## [1m 28/251[0m [32m[0m[37m[0m [1m1:04[0m 289ms/step
- accuracy: 0.9729 - loss: 0.0443
## [1m 29/251[0m [32m[0m[37m[0m [1m1:03[0m 285ms/step
- accuracy: 0.9726 - loss: 0.0447
## [1m 30/251[0m [32m[0m[37m[0m [1m1:03[0m 286ms/step
- accuracy: 0.9723 - loss: 0.0450
## [1m 31/251[0m [32m[0m[37m[0m [1m1:02[0m 286ms/step
- accuracy: 0.9720 - loss: 0.0454
## [1m 32/251[0m [32m[0m[37m[0m [1m1:02[0m 287ms/step
- accuracy: 0.9718 - loss: 0.0457
## [1m 33/251[0m [32m[0m[37m[0m [1m1:02[0m 288ms/step
- accuracy: 0.9716 - loss: 0.0461
## [1m 34/251[0m [32m[0m[37m[0m [1m1:02[0m 288ms/step
- accuracy: 0.9714 - loss: 0.0464
## [1m 35/251[0m [32m[0m[37m[0m [1m1:02[0m 289ms/step
- accuracy: 0.9712 - loss: 0.0468
## [1m 36/251[0m [32m[0m[37m[0m [1m1:02[0m 290ms/step
- accuracy: 0.9710 - loss: 0.0471
## [1m 37/251[0m [32m[0m[37m[0m [1m1:01[0m 290ms/step
- accuracy: 0.9709 - loss: 0.0474
## [1m 38/251[0m [32m[0m[37m[0m [1m1:01[0m 289ms/step
- accuracy: 0.9707 - loss: 0.0477
## [1m 39/251[0m [32m[0m[37m[0m [1m1:01[0m 289ms/step
- accuracy: 0.9705 - loss: 0.0480
## [1m 40/251[0m [32m[0m[37m[0m [1m1:01[0m 289ms/step
- accuracy: 0.9703 - loss: 0.0483
## [1m 41/251[0m [32m[0m[37m[0m [1m1:01[0m 291ms/step
- accuracy: 0.9702 - loss: 0.0486
## [1m 42/251[0m [32m[0m[37m[0m [1m1:00[0m 291ms/step
- accuracy: 0.9700 - loss: 0.0489
## [1m 43/251[0m [32m[0m[37m[0m [1m1:00[0m 290ms/step
- accuracy: 0.9698 - loss: 0.0492
```

```
## [1m 44/251[0m [32m[0m[37m[0m [1m1:00[0m 290ms/step
- accuracy: 0.9697 - loss: 0.0494
## [1m 45/251[0m [32m[0m[37m[0m [1m59s[0m 289ms/step
- accuracy: 0.9696 - loss: 0.0496
## [1m 46/251[0m [32m[0m[37m[0m [1m59s[0m 289ms/step
- accuracy: 0.9695 - loss: 0.0498
## [1m 47/251[0m [32m[0m[37m[0m [1m59s[0m 289ms/step
- accuracy: 0.9694 - loss: 0.0500
## [1m 48/251[0m [32m[0m[37m[0m [1m58s[0m 290ms/step
- accuracy: 0.9694 - loss: 0.0501
## [1m 49/251[0m [32m[0m[37m[0m [1m58s[0m 290ms/step
- accuracy: 0.9693 - loss: 0.0504
## [1m 50/251[0m [32m[0m[37m[0m [1m58s[0m 290ms/step
- accuracy: 0.9692 - loss: 0.0506
## [1m 51/251[0m [32m[0m[37m[0m [1m57s[0m 289ms/step
- accuracy: 0.9692 - loss: 0.0508
## [1m 52/251[0m [32m[0m[37m[0m [1m57s[0m 289ms/step
- accuracy: 0.9691 - loss: 0.0510
## [1m 53/251[0m [32m[0m[37m[0m [1m57s[0m 289ms/step
- accuracy: 0.9690 - loss: 0.0512
## [1m 54/251[0m [32m[0m[37m[0m [1m56s[0m 289ms/step
- accuracy: 0.9690 - loss: 0.0514
## [1m 55/251[0m [32m[0m[37m[0m [1m56s[0m 289ms/step
- accuracy: 0.9690 - loss: 0.0515
## [1m 56/251[0m [32m[0m[37m[0m [1m56s[0m 289ms/step
- accuracy: 0.9689 - loss: 0.0517
## [1m 57/251[0m [32m[0m[37m[0m [1m56s[0m 289ms/step
- accuracy: 0.9689 - loss: 0.0518
## [1m 58/251[0m [32m[0m[37m[0m [1m55s[0m 289ms/step
- accuracy: 0.9689 - loss: 0.0520
## [1m 59/251[0m [32m[0m[37m[0m [1m55s[0m 289ms/step
- accuracy: 0.9688 - loss: 0.0521
## [1m 60/251[0m [32m[0m[37m[0m [1m55s[0m 289ms/step
- accuracy: 0.9688 - loss: 0.0522
## [1m 61/251[0m [32m[0m[37m[0m [1m54s[0m 288ms/step
- accuracy: 0.9688 - loss: 0.0523
## [1m 62/251[0m [32m[0m[37m[0m [1m54s[0m 289ms/step
- accuracy: 0.9687 - loss: 0.0524
## [1m 63/251[0m [32m[0m[37m[0m [1m54s[0m 290ms/step
- accuracy: 0.9687 - loss: 0.0525
## [1m 64/251[0m [32m[0m[37m[0m [1m54s[0m 290ms/step
- accuracy: 0.9687 - loss: 0.0526
## [1m 65/251[0m [32m[0m[37m[0m [1m53s[0m 290ms/step
- accuracy: 0.9686 - loss: 0.0527
## [1m 66/251[0m [32m[0m[37m[0m [1m53s[0m 292ms/step
- accuracy: 0.9686 - loss: 0.0528
## [1m 67/251[0m [32m[0m[37m[0m [1m53s[0m 293ms/step
- accuracy: 0.9686 - loss: 0.0529
## [1m 68/251[0m [32m[0m[37m[0m [1m53s[0m 295ms/step
- accuracy: 0.9685 - loss: 0.0530
```

```
## [1m 69/251[0m [32m-----[0m[37m-----[0m [1m53s[0m 295ms/step
- accuracy: 0.9685 - loss: 0.0531
## [1m 70/251[0m [32m-----[0m[37m-----[0m [1m53s[0m 294ms/step
- accuracy: 0.9685 - loss: 0.0532
## [1m 71/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 294ms/step
- accuracy: 0.9685 - loss: 0.0533
## [1m 72/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 294ms/step
- accuracy: 0.9685 - loss: 0.0533
## [1m 73/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 294ms/step
- accuracy: 0.9685 - loss: 0.0534
## [1m 74/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 294ms/step
- accuracy: 0.9685 - loss: 0.0534
## [1m 75/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 294ms/step
- accuracy: 0.9685 - loss: 0.0535
## [1m 76/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 294ms/step
- accuracy: 0.9685 - loss: 0.0536
## [1m 77/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 293ms/step
- accuracy: 0.9685 - loss: 0.0536
## [1m 78/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 293ms/step
- accuracy: 0.9684 - loss: 0.0537
## [1m 79/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 293ms/step
- accuracy: 0.9684 - loss: 0.0537
## [1m 80/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 293ms/step
- accuracy: 0.9685 - loss: 0.0538
## [1m 81/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 294ms/step
- accuracy: 0.9685 - loss: 0.0538
## [1m 82/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 293ms/step
- accuracy: 0.9685 - loss: 0.0538
## [1m 83/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 294ms/step
- accuracy: 0.9685 - loss: 0.0539
## [1m 84/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 294ms/step
- accuracy: 0.9684 - loss: 0.0539
## [1m 85/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 293ms/step
- accuracy: 0.9684 - loss: 0.0540
## [1m 86/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 293ms/step
- accuracy: 0.9684 - loss: 0.0540
## [1m 87/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 293ms/step
- accuracy: 0.9684 - loss: 0.0540
## [1m 88/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 293ms/step
- accuracy: 0.9684 - loss: 0.0541
## [1m 89/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 293ms/step
- accuracy: 0.9684 - loss: 0.0541
## [1m 90/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 293ms/step
- accuracy: 0.9684 - loss: 0.0542
## [1m 91/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 293ms/step
- accuracy: 0.9684 - loss: 0.0542
## [1m 92/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 293ms/step
- accuracy: 0.9684 - loss: 0.0542
## [1m 93/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 293ms/step
- accuracy: 0.9684 - loss: 0.0542
```

```
## [1m 94/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 293ms/step
- accuracy: 0.9683 - loss: 0.0543
## [1m 95/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 293ms/step
- accuracy: 0.9683 - loss: 0.0543
## [1m 96/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 294ms/step
- accuracy: 0.9683 - loss: 0.0543
## [1m 97/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 294ms/step
- accuracy: 0.9684 - loss: 0.0543
## [1m 98/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 294ms/step
- accuracy: 0.9684 - loss: 0.0543
## [1m 99/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 295ms/step
- accuracy: 0.9684 - loss: 0.0544
## [1m100/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 296ms/step
- accuracy: 0.9684 - loss: 0.0544
## [1m101/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 297ms/step
- accuracy: 0.9684 - loss: 0.0544
## [1m102/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 297ms/step
- accuracy: 0.9684 - loss: 0.0544
## [1m103/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 298ms/step
- accuracy: 0.9684 - loss: 0.0544
## [1m104/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 298ms/step
- accuracy: 0.9684 - loss: 0.0544
## [1m105/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 298ms/step
- accuracy: 0.9684 - loss: 0.0544
## [1m106/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 298ms/step
- accuracy: 0.9684 - loss: 0.0544
## [1m107/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 297ms/step
- accuracy: 0.9684 - loss: 0.0544
## [1m108/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 297ms/step
- accuracy: 0.9684 - loss: 0.0544
## [1m109/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 297ms/step
- accuracy: 0.9684 - loss: 0.0544
## [1m110/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 297ms/step
- accuracy: 0.9685 - loss: 0.0544
## [1m111/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 297ms/step
- accuracy: 0.9685 - loss: 0.0544
## [1m112/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 297ms/step
- accuracy: 0.9685 - loss: 0.0544
## [1m113/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 296ms/step
- accuracy: 0.9685 - loss: 0.0544
## [1m114/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 297ms/step
- accuracy: 0.9685 - loss: 0.0544
## [1m115/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 296ms/step
- accuracy: 0.9685 - loss: 0.0544
## [1m116/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 296ms/step
- accuracy: 0.9685 - loss: 0.0544
## [1m117/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 296ms/step
- accuracy: 0.9685 - loss: 0.0544
## [1m118/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 296ms/step
- accuracy: 0.9685 - loss: 0.0544
```


[illegible]

[illegible]

[illegible]

```
## [1m194/251[0m [32m-----[0m [0m [1m17s[0m 301ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m195/251[0m [32m-----[0m [0m [1m16s[0m 303ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m196/251[0m [32m-----[0m [0m [1m16s[0m 304ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m197/251[0m [32m-----[0m [0m [1m16s[0m 304ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m198/251[0m [32m-----[0m [0m [1m16s[0m 305ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m199/251[0m [32m-----[0m [0m [1m15s[0m 305ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m200/251[0m [32m-----[0m [0m [1m15s[0m 304ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m201/251[0m [32m-----[0m [0m [1m15s[0m 304ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m202/251[0m [32m-----[0m [0m [1m14s[0m 304ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m203/251[0m [32m-----[0m [0m [1m14s[0m 304ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m204/251[0m [32m-----[0m [0m [1m14s[0m 305ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m205/251[0m [32m-----[0m [0m [1m14s[0m 305ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m206/251[0m [32m-----[0m [0m [1m13s[0m 305ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m207/251[0m [32m-----[0m [0m [1m13s[0m 304ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m208/251[0m [32m-----[0m [0m [1m13s[0m 304ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m209/251[0m [32m-----[0m [0m [1m12s[0m 304ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m210/251[0m [32m-----[0m [0m [1m12s[0m 304ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m211/251[0m [32m-----[0m [0m [1m12s[0m 304ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m212/251[0m [32m-----[0m [0m [1m11s[0m 304ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m213/251[0m [32m-----[0m [0m [1m11s[0m 304ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m214/251[0m [32m-----[0m [0m [1m11s[0m 304ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m215/251[0m [32m-----[0m [0m [1m10s[0m 304ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m216/251[0m [32m-----[0m [0m [1m10s[0m 304ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m217/251[0m [32m-----[0m [0m [1m10s[0m 304ms/step
- accuracy: 0.9685 - loss: 0.0545
## [1m218/251[0m [32m-----[0m [0m [1m10s[0m 304ms/step
- accuracy: 0.9685 - loss: 0.0545
```

[illegible]

```

## [1m244/251[0m [32m-----[0m [37m-----[0m [1m2s[0m 304ms/step -
accuracy: 0.9686 - loss: 0.0543
## [1m245/251[0m [32m-----[0m [37m-----[0m [1m1s[0m 304ms/step -
accuracy: 0.9686 - loss: 0.0543
## [1m246/251[0m [32m-----[0m [37m-----[0m [1m1s[0m 304ms/step -
accuracy: 0.9686 - loss: 0.0543
## [1m247/251[0m [32m-----[0m [37m-----[0m [1m1s[0m 304ms/step -
accuracy: 0.9686 - loss: 0.0543
## [1m248/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 304ms/step -
accuracy: 0.9686 - loss: 0.0543
## [1m249/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 304ms/step -
accuracy: 0.9686 - loss: 0.0543
## [1m250/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 304ms/step -
accuracy: 0.9686 - loss: 0.0543
## [1m251/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 304ms/step -
accuracy: 0.9686 - loss: 0.0543
## Epoch 5: val_accuracy did not improve from 0.97655
##
## [1m251/251[0m [32m-----[0m [37m-----[0m [1m83s[0m 329ms/step
- accuracy: 0.9686 - loss: 0.0543 - val_accuracy: 0.9750 - val_loss:
0.0476
## Epoch 6/10
##
## [1m 1/251[0m [37m-----[0m [1m1:33[0m 375ms/step -
accuracy: 0.9375 - loss: 0.0764
## [1m 2/251[0m [37m-----[0m [1m1:16[0m 308ms/step -
accuracy: 0.9375 - loss: 0.0805
## [1m 3/251[0m [37m-----[0m [1m1:15[0m 304ms/step -
accuracy: 0.9340 - loss: 0.0831
## [1m 4/251[0m [37m-----[0m [1m1:16[0m 308ms/step -
accuracy: 0.9329 - loss: 0.0825
## [1m 5/251[0m [37m-----[0m [1m1:15[0m 306ms/step -
accuracy: 0.9351 - loss: 0.0795
## [1m 6/251[0m [37m-----[0m [1m1:15[0m 307ms/step -
accuracy: 0.9364 - loss: 0.0771
## [1m 7/251[0m [37m-----[0m [1m1:14[0m 306ms/step -
accuracy: 0.9384 - loss: 0.0741
## [1m 8/251[0m [37m-----[0m [1m1:14[0m 305ms/step -
accuracy: 0.9408 - loss: 0.0709
## [1m 9/251[0m [37m-----[0m [1m1:13[0m 305ms/step -
accuracy: 0.9427 - loss: 0.0683
## [1m 10/251[0m [37m-----[0m [1m1:14[0m 307ms/step -
accuracy: 0.9444 - loss: 0.0661
## [1m 11/251[0m [37m-----[0m [1m1:14[0m 310ms/step -
accuracy: 0.9456 - loss: 0.0646
## [1m 12/251[0m [37m-----[0m [1m1:13[0m 309ms/step -
accuracy: 0.9468 - loss: 0.0631
## [1m 13/251[0m [32m-----[0m [37m-----[0m [1m1:13[0m 308ms/step
- accuracy: 0.9480 - loss: 0.0618
## [1m 14/251[0m [32m-----[0m [37m-----[0m [1m1:12[0m 305ms/step

```

```
- accuracy: 0.9490 - loss: 0.0607
## [1m 15/251[0m [32m[0m[37m[0m [1m1:11[0m 305ms/step
- accuracy: 0.9497 - loss: 0.0599
## [1m 16/251[0m [32m[0m[37m[0m [1m1:12[0m 307ms/step
- accuracy: 0.9503 - loss: 0.0593
## [1m 17/251[0m [32m[0m[37m[0m [1m1:11[0m 307ms/step
- accuracy: 0.9509 - loss: 0.0587
## [1m 18/251[0m [32m[0m[37m[0m [1m1:11[0m 305ms/step
- accuracy: 0.9514 - loss: 0.0580
## [1m 19/251[0m [32m[0m[37m[0m [1m1:12[0m 312ms/step
- accuracy: 0.9519 - loss: 0.0575
## [1m 20/251[0m [32m[0m[37m[0m [1m1:11[0m 312ms/step
- accuracy: 0.9524 - loss: 0.0569
## [1m 21/251[0m [32m[0m[37m[0m [1m1:11[0m 311ms/step
- accuracy: 0.9528 - loss: 0.0564
## [1m 22/251[0m [32m[0m[37m[0m [1m1:11[0m 312ms/step
- accuracy: 0.9533 - loss: 0.0558
## [1m 23/251[0m [32m[0m[37m[0m [1m1:10[0m 311ms/step
- accuracy: 0.9537 - loss: 0.0555
## [1m 24/251[0m [32m[0m[37m[0m [1m1:10[0m 310ms/step
- accuracy: 0.9540 - loss: 0.0562
## [1m 25/251[0m [32m[0m[37m[0m [1m1:10[0m 310ms/step
- accuracy: 0.9543 - loss: 0.0568
## [1m 26/251[0m [32m[0m[37m[0m [1m1:09[0m 310ms/step
- accuracy: 0.9545 - loss: 0.0573
## [1m 27/251[0m [32m[0m[37m[0m [1m1:09[0m 309ms/step
- accuracy: 0.9548 - loss: 0.0578
## [1m 28/251[0m [32m[0m[37m[0m [1m1:08[0m 308ms/step
- accuracy: 0.9551 - loss: 0.0582
## [1m 29/251[0m [32m[0m[37m[0m [1m1:08[0m 309ms/step
- accuracy: 0.9553 - loss: 0.0584
## [1m 30/251[0m [32m[0m[37m[0m [1m1:08[0m 311ms/step
- accuracy: 0.9556 - loss: 0.0588
## [1m 31/251[0m [32m[0m[37m[0m [1m1:08[0m 311ms/step
- accuracy: 0.9558 - loss: 0.0591
## [1m 32/251[0m [32m[0m[37m[0m [1m1:08[0m 313ms/step
- accuracy: 0.9560 - loss: 0.0594
## [1m 33/251[0m [32m[0m[37m[0m [1m1:08[0m 313ms/step
- accuracy: 0.9563 - loss: 0.0596
## [1m 34/251[0m [32m[0m[37m[0m [1m1:08[0m 314ms/step
- accuracy: 0.9565 - loss: 0.0598
## [1m 35/251[0m [32m[0m[37m[0m [1m1:07[0m 313ms/step
- accuracy: 0.9568 - loss: 0.0599
## [1m 36/251[0m [32m[0m[37m[0m [1m1:08[0m 318ms/step
- accuracy: 0.9570 - loss: 0.0600
## [1m 37/251[0m [32m[0m[37m[0m [1m1:07[0m 318ms/step
- accuracy: 0.9573 - loss: 0.0601
## [1m 38/251[0m [32m[0m[37m[0m [1m1:07[0m 318ms/step
- accuracy: 0.9575 - loss: 0.0602
## [1m 39/251[0m [32m[0m[37m[0m [1m1:07[0m 318ms/step
```

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- accuracy: 0.9577 - loss: 0.0603
## [1m 40/251[0m [32m-----[0m[37m-----[0m [1m1:07[0m 318ms/step
- accuracy: 0.9579 - loss: 0.0604
## [1m 41/251[0m [32m-----[0m[37m-----[0m [1m1:06[0m 318ms/step
- accuracy: 0.9581 - loss: 0.0604
## [1m 42/251[0m [32m-----[0m[37m-----[0m [1m1:06[0m 318ms/step
- accuracy: 0.9583 - loss: 0.0604
## [1m 43/251[0m [32m-----[0m[37m-----[0m [1m1:05[0m 317ms/step
- accuracy: 0.9585 - loss: 0.0604
## [1m 44/251[0m [32m-----[0m[37m-----[0m [1m1:05[0m 317ms/step
- accuracy: 0.9588 - loss: 0.0603
## [1m 45/251[0m [32m-----[0m[37m-----[0m [1m1:05[0m 317ms/step
- accuracy: 0.9590 - loss: 0.0603
## [1m 46/251[0m [32m-----[0m[37m-----[0m [1m1:04[0m 317ms/step
- accuracy: 0.9592 - loss: 0.0603
## [1m 47/251[0m [32m-----[0m[37m-----[0m [1m1:04[0m 317ms/step
- accuracy: 0.9594 - loss: 0.0602
## [1m 48/251[0m [32m-----[0m[37m-----[0m [1m1:04[0m 316ms/step
- accuracy: 0.9596 - loss: 0.0602
## [1m 49/251[0m [32m-----[0m[37m-----[0m [1m1:03[0m 316ms/step
- accuracy: 0.9598 - loss: 0.0602
## [1m 50/251[0m [32m-----[0m[37m-----[0m [1m1:03[0m 317ms/step
- accuracy: 0.9600 - loss: 0.0602
## [1m 51/251[0m [32m-----[0m[37m-----[0m [1m1:03[0m 316ms/step
- accuracy: 0.9602 - loss: 0.0601
## [1m 52/251[0m [32m-----[0m[37m-----[0m [1m1:03[0m 318ms/step
- accuracy: 0.9604 - loss: 0.0600
## [1m 53/251[0m [32m-----[0m[37m-----[0m [1m1:02[0m 318ms/step
- accuracy: 0.9605 - loss: 0.0600
## [1m 54/251[0m [32m-----[0m[37m-----[0m [1m1:02[0m 318ms/step
- accuracy: 0.9607 - loss: 0.0599
## [1m 55/251[0m [32m-----[0m[37m-----[0m [1m1:02[0m 317ms/step
- accuracy: 0.9609 - loss: 0.0598
## [1m 56/251[0m [32m-----[0m[37m-----[0m [1m1:01[0m 317ms/step
- accuracy: 0.9611 - loss: 0.0598
## [1m 57/251[0m [32m-----[0m[37m-----[0m [1m1:01[0m 317ms/step
- accuracy: 0.9613 - loss: 0.0597
## [1m 58/251[0m [32m-----[0m[37m-----[0m [1m1:01[0m 317ms/step
- accuracy: 0.9614 - loss: 0.0596
## [1m 59/251[0m [32m-----[0m[37m-----[0m [1m1:00[0m 317ms/step
- accuracy: 0.9616 - loss: 0.0596
## [1m 60/251[0m [32m-----[0m[37m-----[0m [1m1:00[0m 317ms/step
- accuracy: 0.9617 - loss: 0.0595
## [1m 61/251[0m [32m-----[0m[37m-----[0m [1m1:00[0m 317ms/step
- accuracy: 0.9619 - loss: 0.0595
## [1m 62/251[0m [32m-----[0m[37m-----[0m [1m59s[0m 316ms/step
- accuracy: 0.9620 - loss: 0.0594
## [1m 63/251[0m [32m-----[0m[37m-----[0m [1m59s[0m 316ms/step
- accuracy: 0.9621 - loss: 0.0594
## [1m 64/251[0m [32m-----[0m[37m-----[0m [1m59s[0m 316ms/step
```



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- accuracy: 0.9622 - loss: 0.0593
## [1m 65/251[0m [32m-----[0m[37m-----[0m [1m58s[0m 315ms/step
- accuracy: 0.9623 - loss: 0.0593
## [1m 66/251[0m [32m-----[0m[37m-----[0m [1m58s[0m 316ms/step
- accuracy: 0.9624 - loss: 0.0592
## [1m 67/251[0m [32m-----[0m[37m-----[0m [1m58s[0m 316ms/step
- accuracy: 0.9625 - loss: 0.0592
## [1m 68/251[0m [32m-----[0m[37m-----[0m [1m57s[0m 316ms/step
- accuracy: 0.9626 - loss: 0.0591
## [1m 69/251[0m [32m-----[0m[37m-----[0m [1m57s[0m 316ms/step
- accuracy: 0.9626 - loss: 0.0591
## [1m 70/251[0m [32m-----[0m[37m-----[0m [1m57s[0m 317ms/step
- accuracy: 0.9627 - loss: 0.0591
## [1m 71/251[0m [32m-----[0m[37m-----[0m [1m57s[0m 318ms/step
- accuracy: 0.9628 - loss: 0.0590
## [1m 72/251[0m [32m-----[0m[37m-----[0m [1m56s[0m 317ms/step
- accuracy: 0.9629 - loss: 0.0590
## [1m 73/251[0m [32m-----[0m[37m-----[0m [1m56s[0m 317ms/step
- accuracy: 0.9629 - loss: 0.0590
## [1m 74/251[0m [32m-----[0m[37m-----[0m [1m56s[0m 317ms/step
- accuracy: 0.9630 - loss: 0.0590
## [1m 75/251[0m [32m-----[0m[37m-----[0m [1m55s[0m 317ms/step
- accuracy: 0.9630 - loss: 0.0589
## [1m 76/251[0m [32m-----[0m[37m-----[0m [1m55s[0m 317ms/step
- accuracy: 0.9631 - loss: 0.0589
## [1m 77/251[0m [32m-----[0m[37m-----[0m [1m55s[0m 317ms/step
- accuracy: 0.9632 - loss: 0.0588
## [1m 78/251[0m [32m-----[0m[37m-----[0m [1m54s[0m 317ms/step
- accuracy: 0.9633 - loss: 0.0588
## [1m 79/251[0m [32m-----[0m[37m-----[0m [1m54s[0m 318ms/step
- accuracy: 0.9633 - loss: 0.0587
## [1m 80/251[0m [32m-----[0m[37m-----[0m [1m54s[0m 318ms/step
- accuracy: 0.9634 - loss: 0.0587
## [1m 81/251[0m [32m-----[0m[37m-----[0m [1m54s[0m 318ms/step
- accuracy: 0.9635 - loss: 0.0586
## [1m 82/251[0m [32m-----[0m[37m-----[0m [1m53s[0m 317ms/step
- accuracy: 0.9636 - loss: 0.0585
## [1m 83/251[0m [32m-----[0m[37m-----[0m [1m53s[0m 317ms/step
- accuracy: 0.9636 - loss: 0.0585
## [1m 84/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 317ms/step
- accuracy: 0.9637 - loss: 0.0584
## [1m 85/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 318ms/step
- accuracy: 0.9638 - loss: 0.0583
## [1m 86/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 318ms/step
- accuracy: 0.9639 - loss: 0.0583
## [1m 87/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 318ms/step
- accuracy: 0.9639 - loss: 0.0582
## [1m 88/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 318ms/step
- accuracy: 0.9640 - loss: 0.0581
## [1m 89/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 318ms/step
```

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- accuracy: 0.9641 - loss: 0.0581
## [1m 90/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 318ms/step
- accuracy: 0.9642 - loss: 0.0580
## [1m 91/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 318ms/step
- accuracy: 0.9642 - loss: 0.0580
## [1m 92/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 318ms/step
- accuracy: 0.9643 - loss: 0.0579
## [1m 93/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 318ms/step
- accuracy: 0.9644 - loss: 0.0578
## [1m 94/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 318ms/step
- accuracy: 0.9644 - loss: 0.0578
## [1m 95/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 318ms/step
- accuracy: 0.9645 - loss: 0.0577
## [1m 96/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 318ms/step
- accuracy: 0.9646 - loss: 0.0577
## [1m 97/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 317ms/step
- accuracy: 0.9646 - loss: 0.0576
## [1m 98/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 316ms/step
- accuracy: 0.9647 - loss: 0.0576
## [1m 99/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 316ms/step
- accuracy: 0.9647 - loss: 0.0575
## [1m100/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 316ms/step
- accuracy: 0.9648 - loss: 0.0575
## [1m101/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 316ms/step
- accuracy: 0.9648 - loss: 0.0574
## [1m102/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 316ms/step
- accuracy: 0.9649 - loss: 0.0574
## [1m103/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 316ms/step
- accuracy: 0.9650 - loss: 0.0573
## [1m104/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 316ms/step
- accuracy: 0.9650 - loss: 0.0573
## [1m105/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 316ms/step
- accuracy: 0.9651 - loss: 0.0572
## [1m106/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 316ms/step
- accuracy: 0.9651 - loss: 0.0571
## [1m107/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 317ms/step
- accuracy: 0.9652 - loss: 0.0571
## [1m108/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 317ms/step
- accuracy: 0.9652 - loss: 0.0570
## [1m109/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 317ms/step
- accuracy: 0.9653 - loss: 0.0570
## [1m110/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 317ms/step
- accuracy: 0.9653 - loss: 0.0569
## [1m111/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 317ms/step
- accuracy: 0.9654 - loss: 0.0569
## [1m112/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 317ms/step
- accuracy: 0.9655 - loss: 0.0568
## [1m113/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 317ms/step
- accuracy: 0.9655 - loss: 0.0568
## [1m114/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 317ms/step
```

```
- accuracy: 0.9656 - loss: 0.0567
## [1m115/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 317ms/step
- accuracy: 0.9656 - loss: 0.0567
## [1m116/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 318ms/step
- accuracy: 0.9657 - loss: 0.0566
## [1m117/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 318ms/step
- accuracy: 0.9657 - loss: 0.0566
## [1m118/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 317ms/step
- accuracy: 0.9658 - loss: 0.0565
## [1m119/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 317ms/step
- accuracy: 0.9658 - loss: 0.0565
## [1m120/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 317ms/step
- accuracy: 0.9659 - loss: 0.0564
## [1m121/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 317ms/step
- accuracy: 0.9659 - loss: 0.0564
## [1m122/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 317ms/step
- accuracy: 0.9660 - loss: 0.0564
## [1m123/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 317ms/step
- accuracy: 0.9660 - loss: 0.0563
## [1m124/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 317ms/step
- accuracy: 0.9661 - loss: 0.0563
## [1m125/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 317ms/step
- accuracy: 0.9661 - loss: 0.0562
## [1m126/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 317ms/step
- accuracy: 0.9661 - loss: 0.0562
## [1m127/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 317ms/step
- accuracy: 0.9662 - loss: 0.0562
## [1m128/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 317ms/step
- accuracy: 0.9662 - loss: 0.0561
## [1m129/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 317ms/step
- accuracy: 0.9663 - loss: 0.0561
## [1m130/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 317ms/step
- accuracy: 0.9663 - loss: 0.0561
## [1m131/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 317ms/step
- accuracy: 0.9663 - loss: 0.0560
## [1m132/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 318ms/step
- accuracy: 0.9664 - loss: 0.0560
## [1m133/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 318ms/step
- accuracy: 0.9664 - loss: 0.0560
## [1m134/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 318ms/step
- accuracy: 0.9664 - loss: 0.0559
## [1m135/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 318ms/step
- accuracy: 0.9665 - loss: 0.0559
## [1m136/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 319ms/step
- accuracy: 0.9665 - loss: 0.0559
## [1m137/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 320ms/step
- accuracy: 0.9665 - loss: 0.0559
## [1m138/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 321ms/step
- accuracy: 0.9666 - loss: 0.0558
## [1m139/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 321ms/step
```

```
- accuracy: 0.9666 - loss: 0.0558
## [1m140/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 321ms/step
- accuracy: 0.9666 - loss: 0.0558
## [1m141/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 321ms/step
- accuracy: 0.9666 - loss: 0.0558
## [1m142/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 321ms/step
- accuracy: 0.9667 - loss: 0.0558
## [1m143/251[0m [32m-----[0m[37m-----[0m [1m34s[0m 321ms/step
- accuracy: 0.9667 - loss: 0.0557
## [1m144/251[0m [32m-----[0m[37m-----[0m [1m34s[0m 322ms/step
- accuracy: 0.9667 - loss: 0.0557
## [1m145/251[0m [32m-----[0m[37m-----[0m [1m34s[0m 322ms/step
- accuracy: 0.9668 - loss: 0.0557
## [1m146/251[0m [32m-----[0m[37m-----[0m [1m33s[0m 322ms/step
- accuracy: 0.9668 - loss: 0.0557
## [1m147/251[0m [32m-----[0m[37m-----[0m [1m33s[0m 322ms/step
- accuracy: 0.9668 - loss: 0.0556
## [1m148/251[0m [32m-----[0m[37m-----[0m [1m33s[0m 322ms/step
- accuracy: 0.9668 - loss: 0.0556
## [1m149/251[0m [32m-----[0m[37m-----[0m [1m32s[0m 322ms/step
- accuracy: 0.9669 - loss: 0.0556
## [1m150/251[0m [32m-----[0m[37m-----[0m [1m32s[0m 322ms/step
- accuracy: 0.9669 - loss: 0.0556
## [1m151/251[0m [32m-----[0m[37m-----[0m [1m32s[0m 322ms/step
- accuracy: 0.9669 - loss: 0.0556
## [1m152/251[0m [32m-----[0m[37m-----[0m [1m31s[0m 322ms/step
- accuracy: 0.9670 - loss: 0.0555
## [1m153/251[0m [32m-----[0m[37m-----[0m [1m31s[0m 323ms/step
- accuracy: 0.9670 - loss: 0.0555
## [1m154/251[0m [32m-----[0m[37m-----[0m [1m31s[0m 323ms/step
- accuracy: 0.9670 - loss: 0.0555
## [1m155/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 323ms/step
- accuracy: 0.9670 - loss: 0.0555
## [1m156/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 323ms/step
- accuracy: 0.9671 - loss: 0.0555
## [1m157/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 322ms/step
- accuracy: 0.9671 - loss: 0.0554
## [1m158/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 323ms/step
- accuracy: 0.9671 - loss: 0.0554
## [1m159/251[0m [32m-----[0m[37m-----[0m [1m29s[0m 323ms/step
- accuracy: 0.9671 - loss: 0.0554
## [1m160/251[0m [32m-----[0m[37m-----[0m [1m29s[0m 323ms/step
- accuracy: 0.9672 - loss: 0.0554
## [1m161/251[0m [32m-----[0m[37m-----[0m [1m29s[0m 323ms/step
- accuracy: 0.9672 - loss: 0.0554
## [1m162/251[0m [32m-----[0m[37m-----[0m [1m28s[0m 323ms/step
- accuracy: 0.9672 - loss: 0.0554
## [1m163/251[0m [32m-----[0m[37m-----[0m [1m28s[0m 323ms/step
- accuracy: 0.9672 - loss: 0.0554
## [1m164/251[0m [32m-----[0m[37m-----[0m [1m28s[0m 323ms/step
```

[illegible]

[illegible]

```
- accuracy: 0.9683 - loss: 0.0545
## [1m215/251[0m [32m-----[0m[37m-----[0m [1m11s[0m 327ms/step
- accuracy: 0.9683 - loss: 0.0545
## [1m216/251[0m [32m-----[0m[37m-----[0m [1m11s[0m 328ms/step
- accuracy: 0.9684 - loss: 0.0544
## [1m217/251[0m [32m-----[0m[37m-----[0m [1m11s[0m 328ms/step
- accuracy: 0.9684 - loss: 0.0544
## [1m218/251[0m [32m-----[0m[37m-----[0m [1m10s[0m 328ms/step
- accuracy: 0.9684 - loss: 0.0544
## [1m219/251[0m [32m-----[0m[37m-----[0m [1m10s[0m 328ms/step
- accuracy: 0.9684 - loss: 0.0544
## [1m220/251[0m [32m-----[0m[37m-----[0m [1m10s[0m 327ms/step
- accuracy: 0.9684 - loss: 0.0544
## [1m221/251[0m [32m-----[0m[37m-----[0m [1m9s[0m 328ms/step -
accuracy: 0.9684 - loss: 0.0544
## [1m222/251[0m [32m-----[0m[37m-----[0m [1m9s[0m 328ms/step -
accuracy: 0.9685 - loss: 0.0543
## [1m223/251[0m [32m-----[0m[37m-----[0m [1m9s[0m 328ms/step -
accuracy: 0.9685 - loss: 0.0543
## [1m224/251[0m [32m-----[0m[37m-----[0m [1m8s[0m 328ms/step -
accuracy: 0.9685 - loss: 0.0543
## [1m225/251[0m [32m-----[0m[37m-----[0m [1m8s[0m 328ms/step -
accuracy: 0.9685 - loss: 0.0543
## [1m226/251[0m [32m-----[0m[37m-----[0m [1m8s[0m 328ms/step -
accuracy: 0.9685 - loss: 0.0543
## [1m227/251[0m [32m-----[0m[37m-----[0m [1m7s[0m 328ms/step -
accuracy: 0.9686 - loss: 0.0543
## [1m228/251[0m [32m-----[0m[37m-----[0m [1m7s[0m 328ms/step -
accuracy: 0.9686 - loss: 0.0542
## [1m229/251[0m [32m-----[0m[37m-----[0m [1m7s[0m 328ms/step -
accuracy: 0.9686 - loss: 0.0542
## [1m230/251[0m [32m-----[0m[37m-----[0m [1m6s[0m 328ms/step -
accuracy: 0.9686 - loss: 0.0542
## [1m231/251[0m [32m-----[0m[37m-----[0m [1m6s[0m 328ms/step -
accuracy: 0.9686 - loss: 0.0542
## [1m232/251[0m [32m-----[0m[37m-----[0m [1m6s[0m 328ms/step -
accuracy: 0.9686 - loss: 0.0542
## [1m233/251[0m [32m-----[0m[37m-----[0m [1m5s[0m 328ms/step -
accuracy: 0.9687 - loss: 0.0542
## [1m234/251[0m [32m-----[0m[37m-----[0m [1m5s[0m 328ms/step -
accuracy: 0.9687 - loss: 0.0541
## [1m235/251[0m [32m-----[0m[37m-----[0m [1m5s[0m 329ms/step -
accuracy: 0.9687 - loss: 0.0541
## [1m236/251[0m [32m-----[0m[37m-----[0m [1m4s[0m 329ms/step -
accuracy: 0.9687 - loss: 0.0541
## [1m237/251[0m [32m-----[0m[37m-----[0m [1m4s[0m 329ms/step -
accuracy: 0.9687 - loss: 0.0541
## [1m238/251[0m [32m-----[0m[37m-----[0m [1m4s[0m 329ms/step -
accuracy: 0.9687 - loss: 0.0541
## [1m239/251[0m [32m-----[0m[37m-----[0m [1m3s[0m 329ms/step -
```

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accuracy: 0.9688 - loss: 0.0541
## [1m240/251[0m [32m-----[0m [37m-----[0m [1m3s[0m 328ms/step -
accuracy: 0.9688 - loss: 0.0540
## [1m241/251[0m [32m-----[0m [37m-----[0m [1m3s[0m 328ms/step -
accuracy: 0.9688 - loss: 0.0540
## [1m242/251[0m [32m-----[0m [37m-----[0m [1m2s[0m 328ms/step -
accuracy: 0.9688 - loss: 0.0540
## [1m243/251[0m [32m-----[0m [37m-----[0m [1m2s[0m 328ms/step -
accuracy: 0.9688 - loss: 0.0540
## [1m244/251[0m [32m-----[0m [37m-----[0m [1m2s[0m 328ms/step -
accuracy: 0.9688 - loss: 0.0540
## [1m245/251[0m [32m-----[0m [37m-----[0m [1m1s[0m 328ms/step -
accuracy: 0.9689 - loss: 0.0540
## [1m246/251[0m [32m-----[0m [37m-----[0m [1m1s[0m 328ms/step -
accuracy: 0.9689 - loss: 0.0539
## [1m247/251[0m [32m-----[0m [37m-----[0m [1m1s[0m 328ms/step -
accuracy: 0.9689 - loss: 0.0539
## [1m248/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 328ms/step -
accuracy: 0.9689 - loss: 0.0539
## [1m249/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 328ms/step -
accuracy: 0.9689 - loss: 0.0539
## [1m250/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 328ms/step -
accuracy: 0.9689 - loss: 0.0539
## [1m251/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 328ms/step -
accuracy: 0.9690 - loss: 0.0539
## Epoch 6: val_accuracy did not improve from 0.97655
##
## [1m251/251[0m [32m-----[0m [37m-----[0m [1m89s[0m 354ms/step
- accuracy: 0.9690 - loss: 0.0538 - val_accuracy: 0.9646 - val_loss:
0.0673
## Epoch 7/10
##
## [1m 1/251[0m [37m-----[0m [1m1:30[0m 361ms/step -
accuracy: 0.9688 - loss: 0.2142
## [1m 2/251[0m [37m-----[0m [1m1:33[0m 375ms/step -
accuracy: 0.9688 - loss: 0.1664
## [1m 3/251[0m [37m-----[0m [1m1:22[0m 331ms/step -
accuracy: 0.9653 - loss: 0.1534
## [1m 4/251[0m [37m-----[0m [1m1:19[0m 322ms/step -
accuracy: 0.9642 - loss: 0.1409
## [1m 5/251[0m [37m-----[0m [1m1:18[0m 318ms/step -
accuracy: 0.9639 - loss: 0.1309
## [1m 6/251[0m [37m-----[0m [1m1:17[0m 318ms/step -
accuracy: 0.9638 - loss: 0.1235
## [1m 7/251[0m [37m-----[0m [1m1:16[0m 312ms/step -
accuracy: 0.9645 - loss: 0.1168
## [1m 8/251[0m [37m-----[0m [1m1:13[0m 303ms/step -
accuracy: 0.9655 - loss: 0.1106
## [1m 9/251[0m [37m-----[0m [1m1:13[0m 303ms/step -
accuracy: 0.9667 - loss: 0.1051

```



```
## [1m 10/251[0m [37m-----[0m [1m1:13[0m 303ms/step -  
accuracy: 0.9675 - loss: 0.1005  
## [1m 11/251[0m [37m-----[0m [1m1:12[0m 301ms/step -  
accuracy: 0.9684 - loss: 0.0963  
## [1m 12/251[0m [37m-----[0m [1m1:11[0m 300ms/step -  
accuracy: 0.9693 - loss: 0.0925  
## [1m 13/251[0m [32m-[0m[37m-----[0m [1m1:11[0m 300ms/step  
- accuracy: 0.9698 - loss: 0.0893  
## [1m 14/251[0m [32m-[0m[37m-----[0m [1m1:10[0m 298ms/step  
- accuracy: 0.9704 - loss: 0.0865  
## [1m 15/251[0m [32m-[0m[37m-----[0m [1m1:09[0m 296ms/step  
- accuracy: 0.9709 - loss: 0.0839  
## [1m 16/251[0m [32m-[0m[37m-----[0m [1m1:09[0m 295ms/step  
- accuracy: 0.9714 - loss: 0.0817  
## [1m 17/251[0m [32m-[0m[37m-----[0m [1m1:08[0m 294ms/step  
- accuracy: 0.9718 - loss: 0.0797  
## [1m 18/251[0m [32m-[0m[37m-----[0m [1m1:08[0m 292ms/step  
- accuracy: 0.9721 - loss: 0.0780  
## [1m 19/251[0m [32m-[0m[37m-----[0m [1m1:07[0m 291ms/step  
- accuracy: 0.9725 - loss: 0.0764  
## [1m 20/251[0m [32m-[0m[37m-----[0m [1m1:07[0m 292ms/step  
- accuracy: 0.9727 - loss: 0.0754  
## [1m 21/251[0m [32m-[0m[37m-----[0m [1m1:08[0m 297ms/step  
- accuracy: 0.9729 - loss: 0.0744  
## [1m 22/251[0m [32m-[0m[37m-----[0m [1m1:07[0m 295ms/step  
- accuracy: 0.9730 - loss: 0.0735  
## [1m 23/251[0m [32m-[0m[37m-----[0m [1m1:07[0m 297ms/step  
- accuracy: 0.9732 - loss: 0.0727  
## [1m 24/251[0m [32m-[0m[37m-----[0m [1m1:07[0m 298ms/step  
- accuracy: 0.9733 - loss: 0.0718  
## [1m 25/251[0m [32m-[0m[37m-----[0m [1m1:07[0m 298ms/step  
- accuracy: 0.9735 - loss: 0.0710  
## [1m 26/251[0m [32m-[0m[37m-----[0m [1m1:07[0m 299ms/step  
- accuracy: 0.9737 - loss: 0.0702  
## [1m 27/251[0m [32m-[0m[37m-----[0m [1m1:07[0m 301ms/step  
- accuracy: 0.9738 - loss: 0.0695  
## [1m 28/251[0m [32m-[0m[37m-----[0m [1m1:06[0m 300ms/step  
- accuracy: 0.9738 - loss: 0.0689  
## [1m 29/251[0m [32m-[0m[37m-----[0m [1m1:06[0m 300ms/step  
- accuracy: 0.9738 - loss: 0.0684  
## [1m 30/251[0m [32m-[0m[37m-----[0m [1m1:08[0m 309ms/step  
- accuracy: 0.9739 - loss: 0.0679  
## [1m 31/251[0m [32m-[0m[37m-----[0m [1m1:07[0m 309ms/step  
- accuracy: 0.9739 - loss: 0.0674  
## [1m 32/251[0m [32m-[0m[37m-----[0m [1m1:07[0m 309ms/step  
- accuracy: 0.9739 - loss: 0.0670  
## [1m 33/251[0m [32m-[0m[37m-----[0m [1m1:07[0m 309ms/step  
- accuracy: 0.9739 - loss: 0.0665  
## [1m 34/251[0m [32m-[0m[37m-----[0m [1m1:07[0m 309ms/step  
- accuracy: 0.9738 - loss: 0.0662
```

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## [1m 35/251[0m [32m[0m[37m[0m [1m1:06[0m 309ms/step
- accuracy: 0.9737 - loss: 0.0658
## [1m 36/251[0m [32m[0m[37m[0m [1m1:06[0m 308ms/step
- accuracy: 0.9737 - loss: 0.0654
## [1m 37/251[0m [32m[0m[37m[0m [1m1:05[0m 307ms/step
- accuracy: 0.9736 - loss: 0.0651
## [1m 38/251[0m [32m[0m[37m[0m [1m1:05[0m 306ms/step
- accuracy: 0.9736 - loss: 0.0647
## [1m 39/251[0m [32m[0m[37m[0m [1m1:04[0m 306ms/step
- accuracy: 0.9736 - loss: 0.0644
## [1m 40/251[0m [32m[0m[37m[0m [1m1:04[0m 306ms/step
- accuracy: 0.9735 - loss: 0.0641
## [1m 41/251[0m [32m[0m[37m[0m [1m1:04[0m 305ms/step
- accuracy: 0.9735 - loss: 0.0639
## [1m 42/251[0m [32m[0m[37m[0m [1m1:03[0m 306ms/step
- accuracy: 0.9734 - loss: 0.0636
## [1m 43/251[0m [32m[0m[37m[0m [1m1:03[0m 306ms/step
- accuracy: 0.9734 - loss: 0.0633
## [1m 44/251[0m [32m[0m[37m[0m [1m1:03[0m 306ms/step
- accuracy: 0.9734 - loss: 0.0630
## [1m 45/251[0m [32m[0m[37m[0m [1m1:03[0m 306ms/step
- accuracy: 0.9734 - loss: 0.0627
## [1m 46/251[0m [32m[0m[37m[0m [1m1:02[0m 307ms/step
- accuracy: 0.9733 - loss: 0.0625
## [1m 47/251[0m [32m[0m[37m[0m [1m1:02[0m 306ms/step
- accuracy: 0.9733 - loss: 0.0623
## [1m 48/251[0m [32m[0m[37m[0m [1m1:02[0m 308ms/step
- accuracy: 0.9733 - loss: 0.0620
## [1m 49/251[0m [32m[0m[37m[0m [1m1:02[0m 309ms/step
- accuracy: 0.9733 - loss: 0.0618
## [1m 50/251[0m [32m[0m[37m[0m [1m1:02[0m 310ms/step
- accuracy: 0.9733 - loss: 0.0616
## [1m 51/251[0m [32m[0m[37m[0m [1m1:02[0m 310ms/step
- accuracy: 0.9733 - loss: 0.0613
## [1m 52/251[0m [32m[0m[37m[0m [1m1:01[0m 311ms/step
- accuracy: 0.9733 - loss: 0.0611
## [1m 53/251[0m [32m[0m[37m[0m [1m1:01[0m 312ms/step
- accuracy: 0.9733 - loss: 0.0609
## [1m 54/251[0m [32m[0m[37m[0m [1m1:01[0m 312ms/step
- accuracy: 0.9733 - loss: 0.0607
## [1m 55/251[0m [32m[0m[37m[0m [1m1:01[0m 312ms/step
- accuracy: 0.9733 - loss: 0.0605
## [1m 56/251[0m [32m[0m[37m[0m [1m1:00[0m 312ms/step
- accuracy: 0.9733 - loss: 0.0603
## [1m 57/251[0m [32m[0m[37m[0m [1m1:00[0m 312ms/step
- accuracy: 0.9733 - loss: 0.0601
## [1m 58/251[0m [32m[0m[37m[0m [1m1:00[0m 312ms/step
- accuracy: 0.9733 - loss: 0.0599
## [1m 59/251[0m [32m[0m[37m[0m [1m59s[0m 311ms/step
- accuracy: 0.9733 - loss: 0.0597
```

```
## [1m 60/251[0m [32m-----[0m[37m-----[0m [1m59s[0m 313ms/step
- accuracy: 0.9733 - loss: 0.0595
## [1m 61/251[0m [32m-----[0m[37m-----[0m [1m59s[0m 314ms/step
- accuracy: 0.9733 - loss: 0.0593
## [1m 62/251[0m [32m-----[0m[37m-----[0m [1m59s[0m 314ms/step
- accuracy: 0.9734 - loss: 0.0591
## [1m 63/251[0m [32m-----[0m[37m-----[0m [1m58s[0m 313ms/step
- accuracy: 0.9734 - loss: 0.0589
## [1m 64/251[0m [32m-----[0m[37m-----[0m [1m58s[0m 315ms/step
- accuracy: 0.9734 - loss: 0.0587
## [1m 65/251[0m [32m-----[0m[37m-----[0m [1m58s[0m 315ms/step
- accuracy: 0.9734 - loss: 0.0586
## [1m 66/251[0m [32m-----[0m[37m-----[0m [1m58s[0m 315ms/step
- accuracy: 0.9734 - loss: 0.0584
## [1m 67/251[0m [32m-----[0m[37m-----[0m [1m58s[0m 315ms/step
- accuracy: 0.9734 - loss: 0.0583
## [1m 68/251[0m [32m-----[0m[37m-----[0m [1m57s[0m 316ms/step
- accuracy: 0.9733 - loss: 0.0581
## [1m 69/251[0m [32m-----[0m[37m-----[0m [1m57s[0m 316ms/step
- accuracy: 0.9733 - loss: 0.0580
## [1m 70/251[0m [32m-----[0m[37m-----[0m [1m57s[0m 316ms/step
- accuracy: 0.9733 - loss: 0.0579
## [1m 71/251[0m [32m-----[0m[37m-----[0m [1m57s[0m 317ms/step
- accuracy: 0.9733 - loss: 0.0578
## [1m 72/251[0m [32m-----[0m[37m-----[0m [1m56s[0m 317ms/step
- accuracy: 0.9733 - loss: 0.0576
## [1m 73/251[0m [32m-----[0m[37m-----[0m [1m56s[0m 318ms/step
- accuracy: 0.9733 - loss: 0.0575
## [1m 74/251[0m [32m-----[0m[37m-----[0m [1m56s[0m 318ms/step
- accuracy: 0.9733 - loss: 0.0574
## [1m 75/251[0m [32m-----[0m[37m-----[0m [1m56s[0m 319ms/step
- accuracy: 0.9733 - loss: 0.0573
## [1m 76/251[0m [32m-----[0m[37m-----[0m [1m55s[0m 317ms/step
- accuracy: 0.9733 - loss: 0.0572
## [1m 77/251[0m [32m-----[0m[37m-----[0m [1m55s[0m 317ms/step
- accuracy: 0.9733 - loss: 0.0571
## [1m 78/251[0m [32m-----[0m[37m-----[0m [1m54s[0m 317ms/step
- accuracy: 0.9733 - loss: 0.0570
## [1m 79/251[0m [32m-----[0m[37m-----[0m [1m54s[0m 317ms/step
- accuracy: 0.9733 - loss: 0.0569
## [1m 80/251[0m [32m-----[0m[37m-----[0m [1m54s[0m 318ms/step
- accuracy: 0.9732 - loss: 0.0568
## [1m 81/251[0m [32m-----[0m[37m-----[0m [1m54s[0m 318ms/step
- accuracy: 0.9732 - loss: 0.0567
## [1m 82/251[0m [32m-----[0m[37m-----[0m [1m53s[0m 318ms/step
- accuracy: 0.9732 - loss: 0.0566
## [1m 83/251[0m [32m-----[0m[37m-----[0m [1m53s[0m 318ms/step
- accuracy: 0.9732 - loss: 0.0566
## [1m 84/251[0m [32m-----[0m[37m-----[0m [1m53s[0m 319ms/step
- accuracy: 0.9732 - loss: 0.0565
```

```
## [1m 85/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 319ms/step
- accuracy: 0.9732 - loss: 0.0564
## [1m 86/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 319ms/step
- accuracy: 0.9732 - loss: 0.0563
## [1m 87/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 319ms/step
- accuracy: 0.9732 - loss: 0.0562
## [1m 88/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 320ms/step
- accuracy: 0.9732 - loss: 0.0562
## [1m 89/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 319ms/step
- accuracy: 0.9732 - loss: 0.0561
## [1m 90/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 320ms/step
- accuracy: 0.9732 - loss: 0.0560
## [1m 91/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 320ms/step
- accuracy: 0.9732 - loss: 0.0559
## [1m 92/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 319ms/step
- accuracy: 0.9732 - loss: 0.0559
## [1m 93/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 319ms/step
- accuracy: 0.9732 - loss: 0.0558
## [1m 94/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 319ms/step
- accuracy: 0.9731 - loss: 0.0557
## [1m 95/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 318ms/step
- accuracy: 0.9731 - loss: 0.0557
## [1m 96/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 318ms/step
- accuracy: 0.9731 - loss: 0.0556
## [1m 97/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 318ms/step
- accuracy: 0.9731 - loss: 0.0556
## [1m 98/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 318ms/step
- accuracy: 0.9731 - loss: 0.0555
## [1m 99/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 317ms/step
- accuracy: 0.9731 - loss: 0.0554
## [1m100/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 317ms/step
- accuracy: 0.9731 - loss: 0.0554
## [1m101/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 316ms/step
- accuracy: 0.9731 - loss: 0.0553
## [1m102/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 316ms/step
- accuracy: 0.9731 - loss: 0.0553
## [1m103/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 316ms/step
- accuracy: 0.9731 - loss: 0.0552
## [1m104/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 316ms/step
- accuracy: 0.9731 - loss: 0.0552
## [1m105/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 316ms/step
- accuracy: 0.9731 - loss: 0.0552
## [1m106/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 316ms/step
- accuracy: 0.9731 - loss: 0.0551
## [1m107/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 315ms/step
- accuracy: 0.9731 - loss: 0.0551
## [1m108/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 315ms/step
- accuracy: 0.9731 - loss: 0.0550
## [1m109/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 314ms/step
- accuracy: 0.9731 - loss: 0.0550
```

```
## [1m110/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 314ms/step
- accuracy: 0.9731 - loss: 0.0549
## [1m111/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 315ms/step
- accuracy: 0.9731 - loss: 0.0549
## [1m112/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 315ms/step
- accuracy: 0.9732 - loss: 0.0548
## [1m113/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 316ms/step
- accuracy: 0.9732 - loss: 0.0548
## [1m114/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 317ms/step
- accuracy: 0.9732 - loss: 0.0547
## [1m115/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 318ms/step
- accuracy: 0.9732 - loss: 0.0547
## [1m116/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 318ms/step
- accuracy: 0.9732 - loss: 0.0547
## [1m117/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 318ms/step
- accuracy: 0.9732 - loss: 0.0546
## [1m118/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 318ms/step
- accuracy: 0.9732 - loss: 0.0546
## [1m119/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 318ms/step
- accuracy: 0.9732 - loss: 0.0545
## [1m120/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 319ms/step
- accuracy: 0.9732 - loss: 0.0545
## [1m121/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 319ms/step
- accuracy: 0.9732 - loss: 0.0545
## [1m122/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 319ms/step
- accuracy: 0.9732 - loss: 0.0544
## [1m123/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 319ms/step
- accuracy: 0.9732 - loss: 0.0544
## [1m124/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 319ms/step
- accuracy: 0.9732 - loss: 0.0544
## [1m125/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 320ms/step
- accuracy: 0.9732 - loss: 0.0543
## [1m126/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 320ms/step
- accuracy: 0.9732 - loss: 0.0543
## [1m127/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 319ms/step
- accuracy: 0.9731 - loss: 0.0543
## [1m128/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 320ms/step
- accuracy: 0.9731 - loss: 0.0542
## [1m129/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 319ms/step
- accuracy: 0.9731 - loss: 0.0542
## [1m130/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 319ms/step
- accuracy: 0.9731 - loss: 0.0542
## [1m131/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 319ms/step
- accuracy: 0.9731 - loss: 0.0542
## [1m132/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 319ms/step
- accuracy: 0.9731 - loss: 0.0541
## [1m133/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 319ms/step
- accuracy: 0.9731 - loss: 0.0541
## [1m134/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 320ms/step
- accuracy: 0.9731 - loss: 0.0541
```

[illegible]

[illegible]

[illegible]

[illegible]

```

## [1m235/251[0m [32m-----[0m [37m-----[0m [1m5s[0m 321ms/step -
accuracy: 0.9730 - loss: 0.0525
## [1m236/251[0m [32m-----[0m [37m-----[0m [1m4s[0m 321ms/step -
accuracy: 0.9730 - loss: 0.0525
## [1m237/251[0m [32m-----[0m [37m-----[0m [1m4s[0m 321ms/step -
accuracy: 0.9730 - loss: 0.0524
## [1m238/251[0m [32m-----[0m [37m-----[0m [1m4s[0m 321ms/step -
accuracy: 0.9729 - loss: 0.0524
## [1m239/251[0m [32m-----[0m [37m-----[0m [1m3s[0m 321ms/step -
accuracy: 0.9729 - loss: 0.0524
## [1m240/251[0m [32m-----[0m [37m-----[0m [1m3s[0m 321ms/step -
accuracy: 0.9729 - loss: 0.0524
## [1m241/251[0m [32m-----[0m [37m-----[0m [1m3s[0m 321ms/step -
accuracy: 0.9729 - loss: 0.0524
## [1m242/251[0m [32m-----[0m [37m-----[0m [1m2s[0m 321ms/step -
accuracy: 0.9729 - loss: 0.0524
## [1m243/251[0m [32m-----[0m [37m-----[0m [1m2s[0m 321ms/step -
accuracy: 0.9729 - loss: 0.0524
## [1m244/251[0m [32m-----[0m [37m-----[0m [1m2s[0m 320ms/step -
accuracy: 0.9729 - loss: 0.0524
## [1m245/251[0m [32m-----[0m [37m-----[0m [1m1s[0m 320ms/step -
accuracy: 0.9729 - loss: 0.0524
## [1m246/251[0m [32m-----[0m [37m-----[0m [1m1s[0m 320ms/step -
accuracy: 0.9729 - loss: 0.0524
## [1m247/251[0m [32m-----[0m [37m-----[0m [1m1s[0m 320ms/step -
accuracy: 0.9729 - loss: 0.0524
## [1m248/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 320ms/step -
accuracy: 0.9729 - loss: 0.0524
## [1m249/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 320ms/step -
accuracy: 0.9729 - loss: 0.0524
## [1m250/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 320ms/step -
accuracy: 0.9729 - loss: 0.0524
## [1m251/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 320ms/step -
accuracy: 0.9729 - loss: 0.0524
## Epoch 7: val_accuracy did not improve from 0.97655
##
## [1m251/251[0m [32m-----[0m [37m-----[0m [1m87s[0m 346ms/step
- accuracy: 0.9729 - loss: 0.0523 - val_accuracy: 0.9750 - val_loss:
0.0459
## Epoch 8/10
##
## [1m 1/251[0m [37m-----[0m [1m1:34[0m 378ms/step -
accuracy: 0.9688 - loss: 0.0405
## [1m 2/251[0m [37m-----[0m [1m1:16[0m 307ms/step -
accuracy: 0.9609 - loss: 0.0549
## [1m 3/251[0m [37m-----[0m [1m1:15[0m 305ms/step -
accuracy: 0.9635 - loss: 0.0521
## [1m 4/251[0m [37m-----[0m [1m1:16[0m 309ms/step -
accuracy: 0.9648 - loss: 0.0498
## [1m 5/251[0m [37m-----[0m [1m1:17[0m 315ms/step -

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```
accuracy: 0.9644 - loss: 0.0498
## [1m 6/251[0m [37m-----f0m [1m1:20[0m 330ms/step -
accuracy: 0.9651 - loss: 0.0497
## [1m 7/251[0m [37m-----f0m [1m1:19[0m 324ms/step -
accuracy: 0.9650 - loss: 0.0497
## [1m 8/251[0m [37m-----f0m [1m1:21[0m 336ms/step -
accuracy: 0.9655 - loss: 0.0499
## [1m 9/251[0m [37m-----f0m [1m1:23[0m 343ms/step -
accuracy: 0.9654 - loss: 0.0501
## [1m 10/251[0m [37m-----f0m [1m1:21[0m 338ms/step -
accuracy: 0.9658 - loss: 0.0498
## [1m 11/251[0m [37m-----f0m [1m1:20[0m 336ms/step -
accuracy: 0.9658 - loss: 0.0497
## [1m 12/251[0m [37m-----f0m [1m1:19[0m 332ms/step -
accuracy: 0.9660 - loss: 0.0496
## [1m 13/251[0m [32m-f0m[37m-----f0m [1m1:18[0m 332ms/step
- accuracy: 0.9662 - loss: 0.0497
## [1m 14/251[0m [32m-f0m[37m-----f0m [1m1:17[0m 328ms/step
- accuracy: 0.9664 - loss: 0.0497
## [1m 15/251[0m [32m-f0m[37m-----f0m [1m1:16[0m 326ms/step
- accuracy: 0.9667 - loss: 0.0495
## [1m 16/251[0m [32m-f0m[37m-----f0m [1m1:17[0m 328ms/step
- accuracy: 0.9670 - loss: 0.0494
## [1m 17/251[0m [32m-f0m[37m-----f0m [1m1:16[0m 327ms/step
- accuracy: 0.9673 - loss: 0.0492
## [1m 18/251[0m [32m-f0m[37m-----f0m [1m1:16[0m 328ms/step
- accuracy: 0.9677 - loss: 0.0491
## [1m 19/251[0m [32m-f0m[37m-----f0m [1m1:15[0m 327ms/step
- accuracy: 0.9680 - loss: 0.0489
## [1m 20/251[0m [32m-f0m[37m-----f0m [1m1:15[0m 326ms/step
- accuracy: 0.9683 - loss: 0.0488
## [1m 21/251[0m [32m-f0m[37m-----f0m [1m1:14[0m 325ms/step
- accuracy: 0.9687 - loss: 0.0486
## [1m 22/251[0m [32m-f0m[37m-----f0m [1m1:14[0m 324ms/step
- accuracy: 0.9690 - loss: 0.0485
## [1m 23/251[0m [32m-f0m[37m-----f0m [1m1:13[0m 323ms/step
- accuracy: 0.9694 - loss: 0.0484
## [1m 24/251[0m [32m-f0m[37m-----f0m [1m1:14[0m 327ms/step
- accuracy: 0.9696 - loss: 0.0484
## [1m 25/251[0m [32m-f0m[37m-----f0m [1m1:13[0m 325ms/step
- accuracy: 0.9698 - loss: 0.0484
## [1m 26/251[0m [32m-f0m[37m-----f0m [1m1:13[0m 325ms/step
- accuracy: 0.9699 - loss: 0.0484
## [1m 27/251[0m [32m-f0m[37m-----f0m [1m1:12[0m 324ms/step
- accuracy: 0.9701 - loss: 0.0484
## [1m 28/251[0m [32m-f0m[37m-----f0m [1m1:11[0m 322ms/step
- accuracy: 0.9703 - loss: 0.0483
## [1m 29/251[0m [32m-f0m[37m-----f0m [1m1:10[0m 320ms/step
- accuracy: 0.9705 - loss: 0.0483
## [1m 30/251[0m [32m-f0m[37m-----f0m [1m1:10[0m 318ms/step
```

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- accuracy: 0.9706 - loss: 0.0482
## [1m 31/251[0m [32m[0m[37m[0m [1m1:10[0m 320ms/step
- accuracy: 0.9708 - loss: 0.0481
## [1m 32/251[0m [32m[0m[37m[0m [1m1:09[0m 319ms/step
- accuracy: 0.9710 - loss: 0.0480
## [1m 33/251[0m [32m[0m[37m[0m [1m1:09[0m 319ms/step
- accuracy: 0.9711 - loss: 0.0479
## [1m 34/251[0m [32m[0m[37m[0m [1m1:09[0m 318ms/step
- accuracy: 0.9713 - loss: 0.0479
## [1m 35/251[0m [32m[0m[37m[0m [1m1:08[0m 317ms/step
- accuracy: 0.9714 - loss: 0.0478
## [1m 36/251[0m [32m[0m[37m[0m [1m1:07[0m 316ms/step
- accuracy: 0.9716 - loss: 0.0478
## [1m 37/251[0m [32m[0m[37m[0m [1m1:07[0m 318ms/step
- accuracy: 0.9717 - loss: 0.0478
## [1m 38/251[0m [32m[0m[37m[0m [1m1:08[0m 322ms/step
- accuracy: 0.9718 - loss: 0.0478
## [1m 39/251[0m [32m[0m[37m[0m [1m1:09[0m 326ms/step
- accuracy: 0.9719 - loss: 0.0479
## [1m 40/251[0m [32m[0m[37m[0m [1m1:09[0m 328ms/step
- accuracy: 0.9720 - loss: 0.0479
## [1m 41/251[0m [32m[0m[37m[0m [1m1:09[0m 331ms/step
- accuracy: 0.9720 - loss: 0.0480
## [1m 42/251[0m [32m[0m[37m[0m [1m1:09[0m 331ms/step
- accuracy: 0.9721 - loss: 0.0481
## [1m 43/251[0m [32m[0m[37m[0m [1m1:09[0m 333ms/step
- accuracy: 0.9721 - loss: 0.0482
## [1m 44/251[0m [32m[0m[37m[0m [1m1:09[0m 337ms/step
- accuracy: 0.9722 - loss: 0.0482
## [1m 45/251[0m [32m[0m[37m[0m [1m1:09[0m 338ms/step
- accuracy: 0.9722 - loss: 0.0482
## [1m 46/251[0m [32m[0m[37m[0m [1m1:09[0m 338ms/step
- accuracy: 0.9723 - loss: 0.0482
## [1m 47/251[0m [32m[0m[37m[0m [1m1:08[0m 338ms/step
- accuracy: 0.9724 - loss: 0.0482
## [1m 48/251[0m [32m[0m[37m[0m [1m1:08[0m 338ms/step
- accuracy: 0.9724 - loss: 0.0482
## [1m 49/251[0m [32m[0m[37m[0m [1m1:08[0m 338ms/step
- accuracy: 0.9725 - loss: 0.0482
## [1m 50/251[0m [32m[0m[37m[0m [1m1:07[0m 338ms/step
- accuracy: 0.9726 - loss: 0.0482
## [1m 51/251[0m [32m[0m[37m[0m [1m1:07[0m 338ms/step
- accuracy: 0.9727 - loss: 0.0481
## [1m 52/251[0m [32m[0m[37m[0m [1m1:07[0m 338ms/step
- accuracy: 0.9727 - loss: 0.0481
## [1m 53/251[0m [32m[0m[37m[0m [1m1:06[0m 338ms/step
- accuracy: 0.9727 - loss: 0.0481
## [1m 54/251[0m [32m[0m[37m[0m [1m1:06[0m 339ms/step
- accuracy: 0.9728 - loss: 0.0481
## [1m 55/251[0m [32m[0m[37m[0m [1m1:06[0m 340ms/step
```

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- accuracy: 0.9728 - loss: 0.0481
## [1m 56/251[0m [32m-----[0m[37m-----[0m [1m1:06[0m 340ms/step
- accuracy: 0.9729 - loss: 0.0481
## [1m 57/251[0m [32m-----[0m[37m-----[0m [1m1:06[0m 341ms/step
- accuracy: 0.9729 - loss: 0.0481
## [1m 58/251[0m [32m-----[0m[37m-----[0m [1m1:05[0m 342ms/step
- accuracy: 0.9729 - loss: 0.0481
## [1m 59/251[0m [32m-----[0m[37m-----[0m [1m1:05[0m 343ms/step
- accuracy: 0.9729 - loss: 0.0481
## [1m 60/251[0m [32m-----[0m[37m-----[0m [1m1:05[0m 343ms/step
- accuracy: 0.9729 - loss: 0.0482
## [1m 61/251[0m [32m-----[0m[37m-----[0m [1m1:05[0m 345ms/step
- accuracy: 0.9730 - loss: 0.0482
## [1m 62/251[0m [32m-----[0m[37m-----[0m [1m1:05[0m 347ms/step
- accuracy: 0.9730 - loss: 0.0483
## [1m 63/251[0m [32m-----[0m[37m-----[0m [1m1:05[0m 348ms/step
- accuracy: 0.9730 - loss: 0.0483
## [1m 64/251[0m [32m-----[0m[37m-----[0m [1m1:05[0m 348ms/step
- accuracy: 0.9730 - loss: 0.0484
## [1m 65/251[0m [32m-----[0m[37m-----[0m [1m1:04[0m 348ms/step
- accuracy: 0.9730 - loss: 0.0484
## [1m 66/251[0m [32m-----[0m[37m-----[0m [1m1:04[0m 347ms/step
- accuracy: 0.9730 - loss: 0.0485
## [1m 67/251[0m [32m-----[0m[37m-----[0m [1m1:03[0m 347ms/step
- accuracy: 0.9730 - loss: 0.0485
## [1m 68/251[0m [32m-----[0m[37m-----[0m [1m1:03[0m 348ms/step
- accuracy: 0.9730 - loss: 0.0486
## [1m 69/251[0m [32m-----[0m[37m-----[0m [1m1:03[0m 349ms/step
- accuracy: 0.9730 - loss: 0.0486
## [1m 70/251[0m [32m-----[0m[37m-----[0m [1m1:03[0m 350ms/step
- accuracy: 0.9729 - loss: 0.0487
## [1m 71/251[0m [32m-----[0m[37m-----[0m [1m1:02[0m 348ms/step
- accuracy: 0.9729 - loss: 0.0487
## [1m 72/251[0m [32m-----[0m[37m-----[0m [1m1:02[0m 349ms/step
- accuracy: 0.9729 - loss: 0.0488
## [1m 73/251[0m [32m-----[0m[37m-----[0m [1m1:02[0m 349ms/step
- accuracy: 0.9729 - loss: 0.0488
## [1m 74/251[0m [32m-----[0m[37m-----[0m [1m1:01[0m 349ms/step
- accuracy: 0.9729 - loss: 0.0489
## [1m 75/251[0m [32m-----[0m[37m-----[0m [1m1:01[0m 348ms/step
- accuracy: 0.9729 - loss: 0.0489
## [1m 76/251[0m [32m-----[0m[37m-----[0m [1m1:00[0m 348ms/step
- accuracy: 0.9729 - loss: 0.0489
## [1m 77/251[0m [32m-----[0m[37m-----[0m [1m1:00[0m 348ms/step
- accuracy: 0.9729 - loss: 0.0490
## [1m 78/251[0m [32m-----[0m[37m-----[0m [1m1:00[0m 348ms/step
- accuracy: 0.9729 - loss: 0.0490
## [1m 79/251[0m [32m-----[0m[37m-----[0m [1m59s[0m 347ms/step
- accuracy: 0.9729 - loss: 0.0490
## [1m 80/251[0m [32m-----[0m[37m-----[0m [1m59s[0m 347ms/step
```

```
- accuracy: 0.9729 - loss: 0.0490
## [1m 81/251[0m [32m-----[0m[37m-----[0m [1m58s[0m 346ms/step
- accuracy: 0.9729 - loss: 0.0490
## [1m 82/251[0m [32m-----[0m[37m-----[0m [1m58s[0m 346ms/step
- accuracy: 0.9729 - loss: 0.0491
## [1m 83/251[0m [32m-----[0m[37m-----[0m [1m58s[0m 346ms/step
- accuracy: 0.9728 - loss: 0.0491
## [1m 84/251[0m [32m-----[0m[37m-----[0m [1m57s[0m 347ms/step
- accuracy: 0.9728 - loss: 0.0492
## [1m 85/251[0m [32m-----[0m[37m-----[0m [1m57s[0m 347ms/step
- accuracy: 0.9728 - loss: 0.0492
## [1m 86/251[0m [32m-----[0m[37m-----[0m [1m57s[0m 347ms/step
- accuracy: 0.9727 - loss: 0.0493
## [1m 87/251[0m [32m-----[0m[37m-----[0m [1m56s[0m 347ms/step
- accuracy: 0.9727 - loss: 0.0493
## [1m 88/251[0m [32m-----[0m[37m-----[0m [1m56s[0m 347ms/step
- accuracy: 0.9726 - loss: 0.0494
## [1m 89/251[0m [32m-----[0m[37m-----[0m [1m56s[0m 347ms/step
- accuracy: 0.9726 - loss: 0.0494
## [1m 90/251[0m [32m-----[0m[37m-----[0m [1m55s[0m 346ms/step
- accuracy: 0.9725 - loss: 0.0495
## [1m 91/251[0m [32m-----[0m[37m-----[0m [1m55s[0m 346ms/step
- accuracy: 0.9725 - loss: 0.0496
## [1m 92/251[0m [32m-----[0m[37m-----[0m [1m55s[0m 346ms/step
- accuracy: 0.9724 - loss: 0.0496
## [1m 93/251[0m [32m-----[0m[37m-----[0m [1m54s[0m 347ms/step
- accuracy: 0.9724 - loss: 0.0497
## [1m 94/251[0m [32m-----[0m[37m-----[0m [1m54s[0m 346ms/step
- accuracy: 0.9723 - loss: 0.0497
## [1m 95/251[0m [32m-----[0m[37m-----[0m [1m53s[0m 346ms/step
- accuracy: 0.9723 - loss: 0.0498
## [1m 96/251[0m [32m-----[0m[37m-----[0m [1m53s[0m 346ms/step
- accuracy: 0.9722 - loss: 0.0498
## [1m 97/251[0m [32m-----[0m[37m-----[0m [1m53s[0m 345ms/step
- accuracy: 0.9722 - loss: 0.0499
## [1m 98/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 345ms/step
- accuracy: 0.9721 - loss: 0.0499
## [1m 99/251[0m [32m-----[0m[37m-----[0m [1m52s[0m 344ms/step
- accuracy: 0.9721 - loss: 0.0499
## [1m100/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 344ms/step
- accuracy: 0.9721 - loss: 0.0500
## [1m101/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 344ms/step
- accuracy: 0.9720 - loss: 0.0500
## [1m102/251[0m [32m-----[0m[37m-----[0m [1m51s[0m 343ms/step
- accuracy: 0.9720 - loss: 0.0501
## [1m103/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 342ms/step
- accuracy: 0.9720 - loss: 0.0501
## [1m104/251[0m [32m-----[0m[37m-----[0m [1m50s[0m 342ms/step
- accuracy: 0.9719 - loss: 0.0502
## [1m105/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 342ms/step
```

```
- accuracy: 0.9719 - loss: 0.0502
## [1m106/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 341ms/step
- accuracy: 0.9719 - loss: 0.0502
## [1m107/251[0m [32m-----[0m[37m-----[0m [1m49s[0m 341ms/step
- accuracy: 0.9719 - loss: 0.0503
## [1m108/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 341ms/step
- accuracy: 0.9718 - loss: 0.0503
## [1m109/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 340ms/step
- accuracy: 0.9718 - loss: 0.0503
## [1m110/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 340ms/step
- accuracy: 0.9718 - loss: 0.0503
## [1m111/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 340ms/step
- accuracy: 0.9718 - loss: 0.0504
## [1m112/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 340ms/step
- accuracy: 0.9717 - loss: 0.0504
## [1m113/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 339ms/step
- accuracy: 0.9717 - loss: 0.0504
## [1m114/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 339ms/step
- accuracy: 0.9717 - loss: 0.0505
## [1m115/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 339ms/step
- accuracy: 0.9717 - loss: 0.0505
## [1m116/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 339ms/step
- accuracy: 0.9717 - loss: 0.0505
## [1m117/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 338ms/step
- accuracy: 0.9717 - loss: 0.0505
## [1m118/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 338ms/step
- accuracy: 0.9716 - loss: 0.0505
## [1m119/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 338ms/step
- accuracy: 0.9716 - loss: 0.0505
## [1m120/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 338ms/step
- accuracy: 0.9716 - loss: 0.0506
## [1m121/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 338ms/step
- accuracy: 0.9716 - loss: 0.0506
## [1m122/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 339ms/step
- accuracy: 0.9716 - loss: 0.0506
## [1m123/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 339ms/step
- accuracy: 0.9716 - loss: 0.0506
## [1m124/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 339ms/step
- accuracy: 0.9716 - loss: 0.0506
## [1m125/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 339ms/step
- accuracy: 0.9715 - loss: 0.0507
## [1m126/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 339ms/step
- accuracy: 0.9715 - loss: 0.0507
## [1m127/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 339ms/step
- accuracy: 0.9715 - loss: 0.0507
## [1m128/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 339ms/step
- accuracy: 0.9715 - loss: 0.0507
## [1m129/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 339ms/step
- accuracy: 0.9715 - loss: 0.0508
## [1m130/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 340ms/step
```

```
- accuracy: 0.9714 - loss: 0.0508
## [1m131/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 340ms/step
- accuracy: 0.9714 - loss: 0.0508
## [1m132/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 340ms/step
- accuracy: 0.9714 - loss: 0.0508
## [1m133/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 340ms/step
- accuracy: 0.9714 - loss: 0.0509
## [1m134/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 340ms/step
- accuracy: 0.9713 - loss: 0.0509
## [1m135/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 340ms/step
- accuracy: 0.9713 - loss: 0.0509
## [1m136/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 340ms/step
- accuracy: 0.9713 - loss: 0.0509
## [1m137/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 340ms/step
- accuracy: 0.9713 - loss: 0.0509
## [1m138/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 340ms/step
- accuracy: 0.9713 - loss: 0.0510
## [1m139/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 341ms/step
- accuracy: 0.9712 - loss: 0.0510
## [1m140/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 340ms/step
- accuracy: 0.9712 - loss: 0.0510
## [1m141/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 340ms/step
- accuracy: 0.9712 - loss: 0.0511
## [1m142/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 340ms/step
- accuracy: 0.9712 - loss: 0.0511
## [1m143/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 340ms/step
- accuracy: 0.9712 - loss: 0.0511
## [1m144/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 339ms/step
- accuracy: 0.9711 - loss: 0.0511
## [1m145/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 339ms/step
- accuracy: 0.9711 - loss: 0.0511
## [1m146/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 339ms/step
- accuracy: 0.9711 - loss: 0.0512
## [1m147/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 339ms/step
- accuracy: 0.9711 - loss: 0.0512
## [1m148/251[0m [32m-----[0m[37m-----[0m [1m34s[0m 339ms/step
- accuracy: 0.9711 - loss: 0.0512
## [1m149/251[0m [32m-----[0m[37m-----[0m [1m34s[0m 339ms/step
- accuracy: 0.9710 - loss: 0.0512
## [1m150/251[0m [32m-----[0m[37m-----[0m [1m34s[0m 339ms/step
- accuracy: 0.9710 - loss: 0.0512
## [1m151/251[0m [32m-----[0m[37m-----[0m [1m33s[0m 339ms/step
- accuracy: 0.9710 - loss: 0.0512
## [1m152/251[0m [32m-----[0m[37m-----[0m [1m33s[0m 338ms/step
- accuracy: 0.9710 - loss: 0.0512
## [1m153/251[0m [32m-----[0m[37m-----[0m [1m33s[0m 339ms/step
- accuracy: 0.9710 - loss: 0.0513
## [1m154/251[0m [32m-----[0m[37m-----[0m [1m32s[0m 339ms/step
- accuracy: 0.9709 - loss: 0.0513
## [1m155/251[0m [32m-----[0m[37m-----[0m [1m32s[0m 339ms/step
```



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- accuracy: 0.9709 - loss: 0.0513
## [1m156/251[0m [32m-----[0m[37m-----[0m [1m32s[0m 339ms/step
- accuracy: 0.9709 - loss: 0.0513
## [1m157/251[0m [32m-----[0m[37m-----[0m [1m31s[0m 339ms/step
- accuracy: 0.9709 - loss: 0.0513
## [1m158/251[0m [32m-----[0m[37m-----[0m [1m31s[0m 339ms/step
- accuracy: 0.9709 - loss: 0.0513
## [1m159/251[0m [32m-----[0m[37m-----[0m [1m31s[0m 338ms/step
- accuracy: 0.9709 - loss: 0.0513
## [1m160/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 338ms/step
- accuracy: 0.9708 - loss: 0.0514
## [1m161/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 338ms/step
- accuracy: 0.9708 - loss: 0.0514
## [1m162/251[0m [32m-----[0m[37m-----[0m [1m30s[0m 338ms/step
- accuracy: 0.9708 - loss: 0.0514
## [1m163/251[0m [32m-----[0m[37m-----[0m [1m29s[0m 337ms/step
- accuracy: 0.9708 - loss: 0.0514
## [1m164/251[0m [32m-----[0m[37m-----[0m [1m29s[0m 337ms/step
- accuracy: 0.9708 - loss: 0.0514
## [1m165/251[0m [32m-----[0m[37m-----[0m [1m28s[0m 337ms/step
- accuracy: 0.9708 - loss: 0.0514
## [1m166/251[0m [32m-----[0m[37m-----[0m [1m28s[0m 337ms/step
- accuracy: 0.9707 - loss: 0.0514
## [1m167/251[0m [32m-----[0m[37m-----[0m [1m28s[0m 336ms/step
- accuracy: 0.9707 - loss: 0.0514
## [1m168/251[0m [32m-----[0m[37m-----[0m [1m27s[0m 336ms/step
- accuracy: 0.9707 - loss: 0.0514
## [1m169/251[0m [32m-----[0m[37m-----[0m [1m27s[0m 336ms/step
- accuracy: 0.9707 - loss: 0.0514
## [1m170/251[0m [32m-----[0m[37m-----[0m [1m27s[0m 336ms/step
- accuracy: 0.9707 - loss: 0.0515
## [1m171/251[0m [32m-----[0m[37m-----[0m [1m26s[0m 336ms/step
- accuracy: 0.9707 - loss: 0.0515
## [1m172/251[0m [32m-----[0m[37m-----[0m [1m26s[0m 336ms/step
- accuracy: 0.9707 - loss: 0.0515
## [1m173/251[0m [32m-----[0m[37m-----[0m [1m26s[0m 336ms/step
- accuracy: 0.9706 - loss: 0.0515
## [1m174/251[0m [32m-----[0m[37m-----[0m [1m25s[0m 336ms/step
- accuracy: 0.9706 - loss: 0.0515
## [1m175/251[0m [32m-----[0m[37m-----[0m [1m25s[0m 336ms/step
- accuracy: 0.9706 - loss: 0.0515
## [1m176/251[0m [32m-----[0m[37m-----[0m [1m25s[0m 336ms/step
- accuracy: 0.9706 - loss: 0.0515
## [1m177/251[0m [32m-----[0m[37m-----[0m [1m24s[0m 336ms/step
- accuracy: 0.9706 - loss: 0.0515
## [1m178/251[0m [32m-----[0m[37m-----[0m [1m24s[0m 336ms/step
- accuracy: 0.9706 - loss: 0.0515
## [1m179/251[0m [32m-----[0m[37m-----[0m [1m24s[0m 335ms/step
- accuracy: 0.9706 - loss: 0.0515
## [1m180/251[0m [32m-----[0m[37m-----[0m [1m23s[0m 335ms/step
```

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- accuracy: 0.9706 - loss: 0.0515
## [1m181/251[0m [32m-----[0m[37m-----[0m [1m23s[0m 335ms/step
- accuracy: 0.9706 - loss: 0.0515
## [1m182/251[0m [32m-----[0m[37m-----[0m [1m23s[0m 334ms/step
- accuracy: 0.9706 - loss: 0.0515
## [1m183/251[0m [32m-----[0m[37m-----[0m [1m22s[0m 334ms/step
- accuracy: 0.9705 - loss: 0.0515
## [1m184/251[0m [32m-----[0m[37m-----[0m [1m22s[0m 334ms/step
- accuracy: 0.9705 - loss: 0.0515
## [1m185/251[0m [32m-----[0m[37m-----[0m [1m22s[0m 333ms/step
- accuracy: 0.9705 - loss: 0.0515
## [1m186/251[0m [32m-----[0m[37m-----[0m [1m21s[0m 333ms/step
- accuracy: 0.9705 - loss: 0.0515
## [1m187/251[0m [32m-----[0m[37m-----[0m [1m21s[0m 333ms/step
- accuracy: 0.9705 - loss: 0.0515
## [1m188/251[0m [32m-----[0m[37m-----[0m [1m20s[0m 332ms/step
- accuracy: 0.9705 - loss: 0.0515
## [1m189/251[0m [32m-----[0m[37m-----[0m [1m20s[0m 332ms/step
- accuracy: 0.9705 - loss: 0.0515
## [1m190/251[0m [32m-----[0m[37m-----[0m [1m20s[0m 332ms/step
- accuracy: 0.9705 - loss: 0.0515
## [1m191/251[0m [32m-----[0m[37m-----[0m [1m19s[0m 331ms/step
- accuracy: 0.9705 - loss: 0.0514
## [1m192/251[0m [32m-----[0m[37m-----[0m [1m19s[0m 331ms/step
- accuracy: 0.9705 - loss: 0.0514
## [1m193/251[0m [32m-----[0m[37m-----[0m [1m19s[0m 331ms/step
- accuracy: 0.9705 - loss: 0.0514
## [1m194/251[0m [32m-----[0m[37m-----[0m [1m18s[0m 330ms/step
- accuracy: 0.9705 - loss: 0.0514
## [1m195/251[0m [32m-----[0m[37m-----[0m [1m18s[0m 330ms/step
- accuracy: 0.9705 - loss: 0.0514
## [1m196/251[0m [32m-----[0m[37m-----[0m [1m18s[0m 330ms/step
- accuracy: 0.9705 - loss: 0.0514
## [1m197/251[0m [32m-----[0m[37m-----[0m [1m17s[0m 330ms/step
- accuracy: 0.9705 - loss: 0.0514
## [1m198/251[0m [32m-----[0m[37m-----[0m [1m17s[0m 329ms/step
- accuracy: 0.9705 - loss: 0.0514
## [1m199/251[0m [32m-----[0m[37m-----[0m [1m17s[0m 329ms/step
- accuracy: 0.9705 - loss: 0.0514
## [1m200/251[0m [32m-----[0m[37m-----[0m [1m16s[0m 328ms/step
- accuracy: 0.9705 - loss: 0.0514
## [1m201/251[0m [32m-----[0m[37m-----[0m [1m16s[0m 328ms/step
- accuracy: 0.9705 - loss: 0.0514
## [1m202/251[0m [32m-----[0m[37m-----[0m [1m16s[0m 328ms/step
- accuracy: 0.9705 - loss: 0.0514
## [1m203/251[0m [32m-----[0m[37m-----[0m [1m15s[0m 328ms/step
- accuracy: 0.9705 - loss: 0.0514
## [1m204/251[0m [32m-----[0m[37m-----[0m [1m15s[0m 327ms/step
- accuracy: 0.9705 - loss: 0.0513
## [1m205/251[0m [32m-----[0m[37m-----[0m [1m15s[0m 327ms/step
```

[illegible]

```
accuracy: 0.9704 - loss: 0.0513
## [1m231/251[0m [32m-----[0m [37m-----[0m [1m6s[0m 321ms/step -
accuracy: 0.9704 - loss: 0.0513
## [1m232/251[0m [32m-----[0m [37m-----[0m [1m6s[0m 321ms/step -
accuracy: 0.9704 - loss: 0.0513
## [1m233/251[0m [32m-----[0m [37m-----[0m [1m5s[0m 320ms/step -
accuracy: 0.9704 - loss: 0.0513
## [1m234/251[0m [32m-----[0m [37m-----[0m [1m5s[0m 320ms/step -
accuracy: 0.9704 - loss: 0.0512
## [1m235/251[0m [32m-----[0m [37m-----[0m [1m5s[0m 320ms/step -
accuracy: 0.9704 - loss: 0.0512
## [1m236/251[0m [32m-----[0m [37m-----[0m [1m4s[0m 320ms/step -
accuracy: 0.9704 - loss: 0.0512
## [1m237/251[0m [32m-----[0m [37m-----[0m [1m4s[0m 319ms/step -
accuracy: 0.9704 - loss: 0.0512
## [1m238/251[0m [32m-----[0m [37m-----[0m [1m4s[0m 319ms/step -
accuracy: 0.9704 - loss: 0.0512
## [1m239/251[0m [32m-----[0m [37m-----[0m [1m3s[0m 319ms/step -
accuracy: 0.9704 - loss: 0.0512
## [1m240/251[0m [32m-----[0m [37m-----[0m [1m3s[0m 319ms/step -
accuracy: 0.9704 - loss: 0.0512
## [1m241/251[0m [32m-----[0m [37m-----[0m [1m3s[0m 318ms/step -
accuracy: 0.9704 - loss: 0.0512
## [1m242/251[0m [32m-----[0m [37m-----[0m [1m2s[0m 318ms/step -
accuracy: 0.9704 - loss: 0.0512
## [1m243/251[0m [32m-----[0m [37m-----[0m [1m2s[0m 318ms/step -
accuracy: 0.9704 - loss: 0.0512
## [1m244/251[0m [32m-----[0m [37m-----[0m [1m2s[0m 318ms/step -
accuracy: 0.9704 - loss: 0.0512
## [1m245/251[0m [32m-----[0m [37m-----[0m [1m1s[0m 318ms/step -
accuracy: 0.9704 - loss: 0.0512
## [1m246/251[0m [32m-----[0m [37m-----[0m [1m1s[0m 318ms/step -
accuracy: 0.9704 - loss: 0.0512
## [1m247/251[0m [32m-----[0m [37m-----[0m [1m1s[0m 318ms/step -
accuracy: 0.9704 - loss: 0.0512
## [1m248/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 318ms/step -
accuracy: 0.9704 - loss: 0.0512
## [1m249/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 318ms/step -
accuracy: 0.9704 - loss: 0.0512
## [1m250/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 317ms/step -
accuracy: 0.9704 - loss: 0.0512
## [1m251/251[0m [32m-----[0m [37m-----[0m [1m0s[0m 317ms/step -
accuracy: 0.9704 - loss: 0.0512
## Epoch 8: val_accuracy did not improve from 0.97655
##
## [1m251/251[0m [32m-----[0m [37m-----[0m [1m86s[0m 343ms/step
- accuracy: 0.9704 - loss: 0.0512 - val_accuracy: 0.9701 - val_loss:
0.0471
## Epoch 9/10
##
```

```
## [1m 1/251[0m [37m-----[0m [1m1:46[0m 425ms/step -  
accuracy: 0.9688 - loss: 0.0810  
## [1m 2/251[0m [37m-----[0m [1m1:21[0m 328ms/step -  
accuracy: 0.9766 - loss: 0.0709  
## [1m 3/251[0m [37m-----[0m [1m1:16[0m 308ms/step -  
accuracy: 0.9774 - loss: 0.0656  
## [1m 4/251[0m [37m-----[0m [1m1:13[0m 296ms/step -  
accuracy: 0.9792 - loss: 0.0602  
## [1m 5/251[0m [37m-----[0m [1m1:10[0m 287ms/step -  
accuracy: 0.9783 - loss: 0.0572  
## [1m 6/251[0m [37m-----[0m [1m1:08[0m 281ms/step -  
accuracy: 0.9767 - loss: 0.0565  
## [1m 7/251[0m [37m-----[0m [1m1:08[0m 279ms/step -  
accuracy: 0.9737 - loss: 0.0576  
## [1m 8/251[0m [37m-----[0m [1m1:08[0m 280ms/step -  
accuracy: 0.9716 - loss: 0.0579  
## [1m 9/251[0m [37m-----[0m [1m1:07[0m 279ms/step -  
accuracy: 0.9694 - loss: 0.0585  
## [1m 10/251[0m [37m-----[0m [1m1:07[0m 279ms/step -  
accuracy: 0.9674 - loss: 0.0591  
## [1m 11/251[0m [37m-----[0m [1m1:06[0m 278ms/step -  
accuracy: 0.9657 - loss: 0.0596  
## [1m 12/251[0m [37m-----[0m [1m1:05[0m 276ms/step -  
accuracy: 0.9638 - loss: 0.0606  
## [1m 13/251[0m [32m-[0m[37m-----[0m [1m1:05[0m 276ms/step  
- accuracy: 0.9625 - loss: 0.0613  
## [1m 14/251[0m [32m-[0m[37m-----[0m [1m1:05[0m 275ms/step  
- accuracy: 0.9617 - loss: 0.0616  
## [1m 15/251[0m [32m-[0m[37m-----[0m [1m1:05[0m 277ms/step  
- accuracy: 0.9612 - loss: 0.0616  
## [1m 16/251[0m [32m-[0m[37m-----[0m [1m1:05[0m 280ms/step  
- accuracy: 0.9608 - loss: 0.0615  
## [1m 17/251[0m [32m-[0m[37m-----[0m [1m1:05[0m 279ms/step  
- accuracy: 0.9603 - loss: 0.0614  
## [1m 18/251[0m [32m-[0m[37m-----[0m [1m1:04[0m 279ms/step  
- accuracy: 0.9600 - loss: 0.0612  
## [1m 19/251[0m [32m-[0m[37m-----[0m [1m1:04[0m 279ms/step  
- accuracy: 0.9599 - loss: 0.0609  
## [1m 20/251[0m [32m-[0m[37m-----[0m [1m1:04[0m 278ms/step  
- accuracy: 0.9598 - loss: 0.0605  
## [1m 21/251[0m [32m-[0m[37m-----[0m [1m1:04[0m 279ms/step  
- accuracy: 0.9599 - loss: 0.0601  
## [1m 22/251[0m [32m-[0m[37m-----[0m [1m1:04[0m 282ms/step  
- accuracy: 0.9600 - loss: 0.0597  
## [1m 23/251[0m [32m-[0m[37m-----[0m [1m1:04[0m 284ms/step  
- accuracy: 0.9601 - loss: 0.0593  
## [1m 24/251[0m [32m-[0m[37m-----[0m [1m1:04[0m 285ms/step  
- accuracy: 0.9602 - loss: 0.0588  
## [1m 25/251[0m [32m-[0m[37m-----[0m [1m1:04[0m 285ms/step  
- accuracy: 0.9604 - loss: 0.0584
```

```
## [1m 26/251[0m [32m[0m[37m[0m [1m1:04[0m 285ms/step
- accuracy: 0.9606 - loss: 0.0580
## [1m 27/251[0m [32m[0m[37m[0m [1m1:03[0m 284ms/step
- accuracy: 0.9608 - loss: 0.0577
## [1m 28/251[0m [32m[0m[37m[0m [1m1:03[0m 284ms/step
- accuracy: 0.9609 - loss: 0.0574
## [1m 29/251[0m [32m[0m[37m[0m [1m1:03[0m 285ms/step
- accuracy: 0.9611 - loss: 0.0571
## [1m 30/251[0m [32m[0m[37m[0m [1m1:02[0m 284ms/step
- accuracy: 0.9611 - loss: 0.0570
## [1m 31/251[0m [32m[0m[37m[0m [1m1:02[0m 284ms/step
- accuracy: 0.9612 - loss: 0.0568
## [1m 32/251[0m [32m[0m[37m[0m [1m1:02[0m 285ms/step
- accuracy: 0.9613 - loss: 0.0566
## [1m 33/251[0m [32m[0m[37m[0m [1m1:02[0m 285ms/step
- accuracy: 0.9614 - loss: 0.0564
## [1m 34/251[0m [32m[0m[37m[0m [1m1:01[0m 285ms/step
- accuracy: 0.9616 - loss: 0.0562
## [1m 35/251[0m [32m[0m[37m[0m [1m1:01[0m 285ms/step
- accuracy: 0.9617 - loss: 0.0560
## [1m 36/251[0m [32m[0m[37m[0m [1m1:01[0m 285ms/step
- accuracy: 0.9618 - loss: 0.0559
## [1m 37/251[0m [32m[0m[37m[0m [1m1:00[0m 285ms/step
- accuracy: 0.9619 - loss: 0.0557
## [1m 38/251[0m [32m[0m[37m[0m [1m1:00[0m 284ms/step
- accuracy: 0.9620 - loss: 0.0556
## [1m 39/251[0m [32m[0m[37m[0m [1m1:00[0m 284ms/step
- accuracy: 0.9621 - loss: 0.0554
## [1m 40/251[0m [32m[0m[37m[0m [1m59s[0m 283ms/step
- accuracy: 0.9622 - loss: 0.0553
## [1m 41/251[0m [32m[0m[37m[0m [1m59s[0m 283ms/step
- accuracy: 0.9623 - loss: 0.0551
## [1m 42/251[0m [32m[0m[37m[0m [1m58s[0m 282ms/step
- accuracy: 0.9624 - loss: 0.0549
## [1m 43/251[0m [32m[0m[37m[0m [1m58s[0m 282ms/step
- accuracy: 0.9626 - loss: 0.0548
## [1m 44/251[0m [32m[0m[37m[0m [1m58s[0m 282ms/step
- accuracy: 0.9627 - loss: 0.0547
## [1m 45/251[0m [32m[0m[37m[0m [1m57s[0m 281ms/step
- accuracy: 0.9628 - loss: 0.0545
## [1m 46/251[0m [32m[0m[37m[0m [1m57s[0m 281ms/step
- accuracy: 0.9629 - loss: 0.0544
## [1m 47/251[0m [32m[0m[37m[0m [1m57s[0m 280ms/step
- accuracy: 0.9630 - loss: 0.0543
## [1m 48/251[0m [32m[0m[37m[0m [1m56s[0m 280ms/step
- accuracy: 0.9630 - loss: 0.0542
## [1m 49/251[0m [32m[0m[37m[0m [1m56s[0m 280ms/step
- accuracy: 0.9631 - loss: 0.0541
## [1m 50/251[0m [32m[0m[37m[0m [1m56s[0m 280ms/step
- accuracy: 0.9632 - loss: 0.0540
```

```
## [1m 51/251[0m [32m[0m[37m[0m [1m55s[0m 279ms/step
- accuracy: 0.9633 - loss: 0.0539
## [1m 52/251[0m [32m[0m[37m[0m [1m55s[0m 279ms/step
- accuracy: 0.9634 - loss: 0.0538
## [1m 53/251[0m [32m[0m[37m[0m [1m55s[0m 279ms/step
- accuracy: 0.9635 - loss: 0.0537
## [1m 54/251[0m [32m[0m[37m[0m [1m55s[0m 279ms/step
- accuracy: 0.9635 - loss: 0.0537
## [1m 55/251[0m [32m[0m[37m[0m [1m54s[0m 279ms/step
- accuracy: 0.9636 - loss: 0.0536
## [1m 56/251[0m [32m[0m[37m[0m [1m54s[0m 279ms/step
- accuracy: 0.9637 - loss: 0.0535
## [1m 57/251[0m [32m[0m[37m[0m [1m54s[0m 279ms/step
- accuracy: 0.9637 - loss: 0.0534
## [1m 58/251[0m [32m[0m[37m[0m [1m53s[0m 279ms/step
- accuracy: 0.9638 - loss: 0.0533
## [1m 59/251[0m [32m[0m[37m[0m [1m53s[0m 279ms/step
- accuracy: 0.9639 - loss: 0.0532
## [1m 60/251[0m [32m[0m[37m[0m [1m53s[0m 279ms/step
- accuracy: 0.9640 - loss: 0.0531
## [1m 61/251[0m [32m[0m[37m[0m [1m53s[0m 279ms/step
- accuracy: 0.9640 - loss: 0.0531
## [1m 62/251[0m [32m[0m[37m[0m [1m52s[0m 279ms/step
- accuracy: 0.9641 - loss: 0.0530
## [1m 63/251[0m [32m[0m[37m[0m [1m52s[0m 279ms/step
- accuracy: 0.9642 - loss: 0.0529
## [1m 64/251[0m [32m[0m[37m[0m [1m52s[0m 279ms/step
- accuracy: 0.9642 - loss: 0.0528
## [1m 65/251[0m [32m[0m[37m[0m [1m51s[0m 279ms/step
- accuracy: 0.9643 - loss: 0.0527
## [1m 66/251[0m [32m[0m[37m[0m [1m51s[0m 278ms/step
- accuracy: 0.9644 - loss: 0.0527
## [1m 67/251[0m [32m[0m[37m[0m [1m51s[0m 278ms/step
- accuracy: 0.9644 - loss: 0.0526
## [1m 68/251[0m [32m[0m[37m[0m [1m50s[0m 278ms/step
- accuracy: 0.9645 - loss: 0.0525
## [1m 69/251[0m [32m[0m[37m[0m [1m50s[0m 278ms/step
- accuracy: 0.9645 - loss: 0.0525
## [1m 70/251[0m [32m[0m[37m[0m [1m50s[0m 278ms/step
- accuracy: 0.9646 - loss: 0.0524
## [1m 71/251[0m [32m[0m[37m[0m [1m50s[0m 278ms/step
- accuracy: 0.9646 - loss: 0.0524
## [1m 72/251[0m [32m[0m[37m[0m [1m49s[0m 278ms/step
- accuracy: 0.9647 - loss: 0.0523
## [1m 73/251[0m [32m[0m[37m[0m [1m49s[0m 278ms/step
- accuracy: 0.9647 - loss: 0.0523
## [1m 74/251[0m [32m[0m[37m[0m [1m49s[0m 278ms/step
- accuracy: 0.9648 - loss: 0.0522
## [1m 75/251[0m [32m[0m[37m[0m [1m48s[0m 277ms/step
- accuracy: 0.9648 - loss: 0.0522
```

```
## [1m 76/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 277ms/step
- accuracy: 0.9649 - loss: 0.0521
## [1m 77/251[0m [32m-----[0m[37m-----[0m [1m48s[0m 277ms/step
- accuracy: 0.9649 - loss: 0.0521
## [1m 78/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 277ms/step
- accuracy: 0.9650 - loss: 0.0520
## [1m 79/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 276ms/step
- accuracy: 0.9650 - loss: 0.0520
## [1m 80/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 276ms/step
- accuracy: 0.9650 - loss: 0.0520
## [1m 81/251[0m [32m-----[0m[37m-----[0m [1m47s[0m 277ms/step
- accuracy: 0.9650 - loss: 0.0519
## [1m 82/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 277ms/step
- accuracy: 0.9651 - loss: 0.0519
## [1m 83/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 278ms/step
- accuracy: 0.9651 - loss: 0.0518
## [1m 84/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 279ms/step
- accuracy: 0.9651 - loss: 0.0518
## [1m 85/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 279ms/step
- accuracy: 0.9652 - loss: 0.0518
## [1m 86/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 280ms/step
- accuracy: 0.9652 - loss: 0.0517
## [1m 87/251[0m [32m-----[0m[37m-----[0m [1m46s[0m 281ms/step
- accuracy: 0.9653 - loss: 0.0517
## [1m 88/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 282ms/step
- accuracy: 0.9653 - loss: 0.0516
## [1m 89/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 282ms/step
- accuracy: 0.9653 - loss: 0.0516
## [1m 90/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 282ms/step
- accuracy: 0.9653 - loss: 0.0516
## [1m 91/251[0m [32m-----[0m[37m-----[0m [1m45s[0m 282ms/step
- accuracy: 0.9654 - loss: 0.0516
## [1m 92/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 282ms/step
- accuracy: 0.9654 - loss: 0.0516
## [1m 93/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 283ms/step
- accuracy: 0.9654 - loss: 0.0515
## [1m 94/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 283ms/step
- accuracy: 0.9655 - loss: 0.0516
## [1m 95/251[0m [32m-----[0m[37m-----[0m [1m44s[0m 283ms/step
- accuracy: 0.9655 - loss: 0.0516
## [1m 96/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 283ms/step
- accuracy: 0.9655 - loss: 0.0516
## [1m 97/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 283ms/step
- accuracy: 0.9655 - loss: 0.0516
## [1m 98/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 284ms/step
- accuracy: 0.9655 - loss: 0.0517
## [1m 99/251[0m [32m-----[0m[37m-----[0m [1m43s[0m 283ms/step
- accuracy: 0.9656 - loss: 0.0517
## [1m100/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 282ms/step
- accuracy: 0.9656 - loss: 0.0517
```



```
-- [1m101/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 283ms/step  
- accuracy: 0.9656 - loss: 0.0517  
## [1m102/251[0m [32m-----[0m[37m-----[0m [1m42s[0m 283ms/step  
- accuracy: 0.9656 - loss: 0.0517  
## [1m103/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 282ms/step  
- accuracy: 0.9656 - loss: 0.0518  
## [1m104/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 282ms/step  
- accuracy: 0.9657 - loss: 0.0518  
## [1m105/251[0m [32m-----[0m[37m-----[0m [1m41s[0m 282ms/step  
- accuracy: 0.9657 - loss: 0.0518  
## [1m106/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 282ms/step  
- accuracy: 0.9657 - loss: 0.0518  
## [1m107/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 282ms/step  
- accuracy: 0.9657 - loss: 0.0518  
## [1m108/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 282ms/step  
- accuracy: 0.9657 - loss: 0.0519  
## [1m109/251[0m [32m-----[0m[37m-----[0m [1m40s[0m 282ms/step  
- accuracy: 0.9657 - loss: 0.0519  
## [1m110/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 282ms/step  
- accuracy: 0.9658 - loss: 0.0519  
## [1m111/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 282ms/step  
- accuracy: 0.9658 - loss: 0.0519  
## [1m112/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 282ms/step  
- accuracy: 0.9658 - loss: 0.0520  
## [1m113/251[0m [32m-----[0m[37m-----[0m [1m39s[0m 283ms/step  
- accuracy: 0.9658 - loss: 0.0520  
## [1m114/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 283ms/step  
- accuracy: 0.9658 - loss: 0.0520  
## [1m115/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 283ms/step  
- accuracy: 0.9659 - loss: 0.0520  
## [1m116/251[0m [32m-----[0m[37m-----[0m [1m38s[0m 283ms/step  
- accuracy: 0.9659 - loss: 0.0520  
## [1m117/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 283ms/step  
- accuracy: 0.9659 - loss: 0.0520  
## [1m118/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 283ms/step  
- accuracy: 0.9659 - loss: 0.0520  
## [1m119/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 283ms/step  
- accuracy: 0.9660 - loss: 0.0521  
## [1m120/251[0m [32m-----[0m[37m-----[0m [1m37s[0m 283ms/step  
- accuracy: 0.9660 - loss: 0.0521  
## [1m121/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 283ms/step  
- accuracy: 0.9660 - loss: 0.0521  
## [1m122/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 283ms/step  
- accuracy: 0.9660 - loss: 0.0521  
## [1m123/251[0m [32m-----[0m[37m-----[0m [1m36s[0m 283ms/step  
- accuracy: 0.9660 - loss: 0.0521  
## [1m124/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 283ms/step  
- accuracy: 0.9661 - loss: 0.0521  
## [1m125/251[0m [32m-----[0m[37m-----[0m [1m35s[0m 283ms/step  
- accuracy: 0.9661 - loss: 0.0521
```

[illegible]

[illegible]

```
## [1m176/251[0m [32m-----[0m[37m-----[0m [1m21s[0m 286ms/step
- accuracy: 0.9668 - loss: 0.0521
## [1m177/251[0m [32m-----[0m[37m-----[0m [1m21s[0m 286ms/step
- accuracy: 0.9669 - loss: 0.0521
## [1m178/251[0m [32m-----[0m[37m-----[0m [1m20s[0m 286ms/step
- accuracy: 0.9669 - loss: 0.0521
## [1m179/251[0m [32m-----[0m[37m-----[0m [1m20s[0m 286ms/step
- accuracy: 0.9669 - loss: 0.0520
## [1m180/251[0m [32m-----[0m[37m-----[0m [1m20s[0m 287ms/step
- accuracy: 0.9669 - loss: 0.0520
## [1m181/251[0m [32m-----[0m[37m-----[0m [1m20s[0m 287ms/step
- accuracy: 0.9669 - loss: 0.0520
## [1m182/251[0m [32m-----[0m[37m-----[0m [1m19s[0m 288ms/step
- accuracy: 0.9669 - loss: 0.0520
## [1m183/251[0m [32m-----[0m[37m-----[0m [1m19s[0m 288ms/step
- accuracy: 0.9669 - loss: 0.0520
## [1m184/251[0m [32m-----[0m[37m-----[0m [1m19s[0m 288ms/step
- accuracy: 0.9669 - loss: 0.0520
## [1m185/251[0m [32m-----[0m[37m-----[0m [1m19s[0m 288ms/step
- accuracy: 0.9669 - loss: 0.0520
## [1m186/251[0m [32m-----[0m[37m-----[0m [1m18s[0m 288ms/step
- accuracy: 0.9670 - loss: 0.0520
## [1m187/251[0m [32m-----[0m[37m-----[0m [1m18s[0m 288ms/step
- accuracy: 0.9670 - loss: 0.0520
## [1m188/251[0m [32m-----[0m[37m-----[0m [1m18s[0m 288ms/step
- accuracy: 0.9670 - loss: 0.0520
## [1m189/251[0m [32m-----[0m[37m-----[0m [1m17s[0m 288ms/step
- accuracy: 0.9670 - loss: 0.0520
## [1m190/251[0m [32m-----[0m[37m-----[0m [1m17s[0m 288ms/step
- accuracy: 0.9670 - loss: 0.0520
## [1m191/251[0m [32m-----[0m[37m-----[0m [1m17s[0m 288ms/step
- accuracy: 0.9670 - loss: 0.0520
## [1m192/251[0m [32m-----[0m[37m-----[0m [1m17s[0m 289ms/step
- accuracy: 0.9670 - loss: 0.0519
## [1m193/251[0m [32m-----[0m[37m-----[0m [1m16s[0m 289ms/step
- accuracy: 0.9671 - loss: 0.0519
## [1m194/251[0m [32m-----[0m[37m-----[0m [1m16s[0m 289ms/step
- accuracy: 0.9671 - loss: 0.0519
## [1m195/251[0m [32m-----[0m[37m-----[0m [1m16s[0m 289ms/step
- accuracy: 0.9671 - loss: 0.0519
## [1m196/251[0m [32m-----[0m[37m-----[0m [1m15s[0m 289ms/step
- accuracy: 0.9671 - loss: 0.0519
## [1m197/251[0m [32m-----[0m[37m-----[0m [1m15s[0m 289ms/step
- accuracy: 0.9671 - loss: 0.0519
## [1m198/251[0m [32m-----[0m[37m-----[0m [1m15s[0m 290ms/step
- accuracy: 0.9671 - loss: 0.0519
## [1m199/251[0m [32m-----[0m[37m-----[0m [1m15s[0m 290ms/step
- accuracy: 0.9671 - loss: 0.0519
## [1m200/251[0m [32m-----[0m[37m-----[0m [1m14s[0m 290ms/step
- accuracy: 0.9672 - loss: 0.0519
```

[illegible]

[illegible]

```
## [1m251/251[0m [32m-----[0m [1m0s[0m 292ms/step -
accuracy: 0.9676 - loss: 0.0519
## Epoch 9: val_accuracy did not improve from 0.97655
##
## [1m251/251[0m [32m-----[0m [37m[0m [1m80s[0m 317ms/step
- accuracy: 0.9676 - loss: 0.0519 - val_accuracy: 0.9686 - val_loss:
0.0496
```

5. Evaluar el modelo

```
evaluation <- model$evaluate(validation_generator)
```

```
##
## [1m 1/63[0m [37m-----[0m [1m9s[0m 155ms/step -
accuracy: 0.9688 - loss: 0.0759
## [1m 2/63[0m [37m-----[0m [1m7s[0m 118ms/step -
accuracy: 0.9766 - loss: 0.0571
## [1m 3/63[0m [37m-----[0m [1m8s[0m 145ms/step -
accuracy: 0.9809 - loss: 0.0466
## [1m 4/63[0m [32m-[0m [37m-----[0m [1m8s[0m 142ms/step -
accuracy: 0.9837 - loss: 0.0409
## [1m 5/63[0m [32m-[0m [37m-----[0m [1m7s[0m 132ms/step -
accuracy: 0.9857 - loss: 0.0365
## [1m 6/63[0m [32m-[0m [37m-----[0m [1m7s[0m 125ms/step -
accuracy: 0.9872 - loss: 0.0334
## [1m 7/63[0m [32m-[0m [37m-----[0m [1m6s[0m 119ms/step -
accuracy: 0.9884 - loss: 0.0310
## [1m 8/63[0m [32m-[0m [37m-----[0m [1m6s[0m 116ms/step -
accuracy: 0.9889 - loss: 0.0304
## [1m 9/63[0m [32m-[0m [37m-----[0m [1m6s[0m 111ms/step -
accuracy: 0.9886 - loss: 0.0306
## [1m10/63[0m [32m-[0m [37m-----[0m [1m5s[0m 110ms/step -
accuracy: 0.9882 - loss: 0.0309
## [1m11/63[0m [32m-[0m [37m-----[0m [1m5s[0m 110ms/step -
accuracy: 0.9879 - loss: 0.0313
## [1m12/63[0m [32m-[0m [37m-----[0m [1m5s[0m 110ms/step -
accuracy: 0.9872 - loss: 0.0322
## [1m13/63[0m [32m-[0m [37m-----[0m [1m5s[0m 109ms/step -
accuracy: 0.9865 - loss: 0.0329
## [1m14/63[0m [32m-[0m [37m-----[0m [1m5s[0m 108ms/step -
accuracy: 0.9859 - loss: 0.0336
## [1m15/63[0m [32m-[0m [37m-----[0m [1m5s[0m 107ms/step -
accuracy: 0.9853 - loss: 0.0343
## [1m16/63[0m [32m-[0m [37m-----[0m [1m4s[0m 106ms/step -
accuracy: 0.9849 - loss: 0.0348
## [1m17/63[0m [32m-[0m [37m-----[0m [1m4s[0m 105ms/step -
accuracy: 0.9844 - loss: 0.0353
## [1m18/63[0m [32m-[0m [37m-----[0m [1m4s[0m 106ms/step -
accuracy: 0.9839 - loss: 0.0359
## [1m19/63[0m [32m-[0m [37m-----[0m [1m4s[0m 104ms/step -
accuracy: 0.9835 - loss: 0.0363
```

```
## [1m20/63[0m [32m-----[0m[37m-----[0m [1m4s[0m 105ms/step -  
accuracy: 0.9832 - loss: 0.0367  
## [1m21/63[0m [32m-----[0m[37m-----[0m [1m4s[0m 104ms/step -  
accuracy: 0.9827 - loss: 0.0373  
## [1m22/63[0m [32m-----[0m[37m-----[0m [1m4s[0m 103ms/step -  
accuracy: 0.9823 - loss: 0.0378  
## [1m23/63[0m [32m-----[0m[37m-----[0m [1m4s[0m 103ms/step -  
accuracy: 0.9819 - loss: 0.0383  
## [1m24/63[0m [32m-----[0m[37m-----[0m [1m4s[0m 103ms/step -  
accuracy: 0.9816 - loss: 0.0386  
## [1m25/63[0m [32m-----[0m[37m-----[0m [1m3s[0m 103ms/step -  
accuracy: 0.9813 - loss: 0.0390  
## [1m26/63[0m [32m-----[0m[37m-----[0m [1m3s[0m 104ms/step -  
accuracy: 0.9810 - loss: 0.0393  
## [1m27/63[0m [32m-----[0m[37m-----[0m [1m3s[0m 103ms/step -  
accuracy: 0.9808 - loss: 0.0395  
## [1m28/63[0m [32m-----[0m[37m-----[0m [1m3s[0m 104ms/step -  
accuracy: 0.9806 - loss: 0.0397  
## [1m29/63[0m [32m-----[0m[37m-----[0m [1m3s[0m 105ms/step -  
accuracy: 0.9804 - loss: 0.0399  
## [1m30/63[0m [32m-----[0m[37m-----[0m [1m3s[0m 105ms/step -  
accuracy: 0.9803 - loss: 0.0401  
## [1m31/63[0m [32m-----[0m[37m-----[0m [1m3s[0m 105ms/step -  
accuracy: 0.9802 - loss: 0.0402  
## [1m32/63[0m [32m-----[0m[37m-----[0m [1m3s[0m 105ms/step -  
accuracy: 0.9801 - loss: 0.0403  
## [1m33/63[0m [32m-----[0m[37m-----[0m [1m3s[0m 105ms/step -  
accuracy: 0.9800 - loss: 0.0403  
## [1m34/63[0m [32m-----[0m[37m-----[0m [1m3s[0m 105ms/step -  
accuracy: 0.9800 - loss: 0.0403  
## [1m35/63[0m [32m-----[0m[37m-----[0m [1m2s[0m 105ms/step -  
accuracy: 0.9800 - loss: 0.0403  
## [1m36/63[0m [32m-----[0m[37m-----[0m [1m2s[0m 104ms/step -  
accuracy: 0.9799 - loss: 0.0404  
## [1m37/63[0m [32m-----[0m[37m-----[0m [1m2s[0m 104ms/step -  
accuracy: 0.9798 - loss: 0.0405  
## [1m38/63[0m [32m-----[0m[37m-----[0m [1m2s[0m 104ms/step -  
accuracy: 0.9798 - loss: 0.0406  
## [1m39/63[0m [32m-----[0m[37m-----[0m [1m2s[0m 104ms/step -  
accuracy: 0.9797 - loss: 0.0406  
## [1m40/63[0m [32m-----[0m[37m-----[0m [1m2s[0m 104ms/step -  
accuracy: 0.9797 - loss: 0.0407  
## [1m41/63[0m [32m-----[0m[37m-----[0m [1m2s[0m 103ms/step -  
accuracy: 0.9797 - loss: 0.0407  
## [1m42/63[0m [32m-----[0m[37m-----[0m [1m2s[0m 103ms/step -  
accuracy: 0.9796 - loss: 0.0407  
## [1m43/63[0m [32m-----[0m[37m-----[0m [1m2s[0m 103ms/step -  
accuracy: 0.9796 - loss: 0.0407  
## [1m44/63[0m [32m-----[0m[37m-----[0m [1m1s[0m 103ms/step -  
accuracy: 0.9796 - loss: 0.0407
```



```

## [1m45/63[0m [32m-----[0m[37m-----[0m [1m1s[0m 102ms/step -
accuracy: 0.9795 - loss: 0.0407
## [1m46/63[0m [32m-----[0m[37m-----[0m [1m1s[0m 102ms/step -
accuracy: 0.9795 - loss: 0.0408
## [1m47/63[0m [32m-----[0m[37m-----[0m [1m1s[0m 102ms/step -
accuracy: 0.9795 - loss: 0.0408
## [1m48/63[0m [32m-----[0m[37m-----[0m [1m1s[0m 101ms/step -
accuracy: 0.9795 - loss: 0.0408
## [1m49/63[0m [32m-----[0m[37m-----[0m [1m1s[0m 101ms/step -
accuracy: 0.9794 - loss: 0.0409
## [1m50/63[0m [32m-----[0m[37m-----[0m [1m1s[0m 101ms/step -
accuracy: 0.9794 - loss: 0.0410
## [1m51/63[0m [32m-----[0m[37m-----[0m [1m1s[0m 101ms/step -
accuracy: 0.9793 - loss: 0.0410
## [1m52/63[0m [32m-----[0m[37m-----[0m [1m1s[0m 100ms/step -
accuracy: 0.9792 - loss: 0.0411
## [1m53/63[0m [32m-----[0m[37m-----[0m [1m0s[0m 100ms/step -
accuracy: 0.9792 - loss: 0.0412
## [1m54/63[0m [32m-----[0m[37m-----[0m [1m0s[0m 100ms/step -
accuracy: 0.9791 - loss: 0.0412
## [1m55/63[0m [32m-----[0m[37m-----[0m [1m0s[0m 100ms/step -
accuracy: 0.9790 - loss: 0.0413
## [1m56/63[0m [32m-----[0m[37m-----[0m [1m0s[0m 100ms/step -
accuracy: 0.9790 - loss: 0.0414
## [1m57/63[0m [32m-----[0m[37m-----[0m [1m0s[0m 100ms/step -
accuracy: 0.9789 - loss: 0.0415
## [1m58/63[0m [32m-----[0m[37m-----[0m [1m0s[0m 100ms/step -
accuracy: 0.9789 - loss: 0.0416
## [1m59/63[0m [32m-----[0m[37m-----[0m [1m0s[0m 99ms/step -
accuracy: 0.9788 - loss: 0.0416
## [1m60/63[0m [32m-----[0m[37m-----[0m [1m0s[0m 99ms/step -
accuracy: 0.9788 - loss: 0.0417
## [1m61/63[0m [32m-----[0m[37m-----[0m [1m0s[0m 99ms/step -
accuracy: 0.9787 - loss: 0.0417
## [1m62/63[0m [32m-----[0m[37m-----[0m [1m0s[0m 99ms/step -
accuracy: 0.9787 - loss: 0.0417
## [1m63/63[0m [32m-----[0m[37m[0m [1m6s[0m 98ms/step -
accuracy: 0.9786 - loss: 0.0418

```

```

cat(sprintf("Pérdida en validación: %.4f\n", evaluation["loss"]))

```

```

## Pérdida en validación: NA

```

```

cat(sprintf("Precisión en validación: %.4f\n", evaluation["accuracy"]))

```

```

## Precisión en validación: NA

```

```

# 6. Guardar el modelo

```

```

save_model_hdf5(model, "modelo_cajas.h5")

```

```

# 7. Visualizaciones

```

```

# Función para visualizar La arquitectura del modelo
plot_model_architecture <- function() {
  # Crear un gráfico simple que represente La arquitectura
  layers <- c(
    "Input (128x128x1)",
    "Conv2D (32 filtros)\n+ ReLU",
    "MaxPooling2D",
    "Conv2D (64 filtros)\n+ ReLU",
    "MaxPooling2D",
    "Conv2D (128 filtros)\n+ ReLU",
    "MaxPooling2D",
    "Flatten",
    "Dense (128)\n+ ReLU",
    "Dense (1)\n+ Sigmoid"
  )

  df <- data.frame(
    x = rep(1, length(layers)),
    y = seq(length(layers), 1),
    label = layers
  )

  ggplot(df, aes(x, y, label = label)) +
    geom_point(size = 10, color = "steelblue") +
    geom_text(color = "white", size = 3) +
    geom_segment(aes(x = x, xend = x, y = y-0.3, yend = y+0.3),
      arrow = arrow(length = unit(0.2, "cm")),
      color = "gray") +
    theme_void() +
    labs(title = "Arquitectura de la Red Neuronal") +
    theme(plot.title = element_text(hjust = 0.5))
}

# Gráficos de entrenamiento
plot_training_history <- function(history) {
  # Extraer métricas de La nueva estructura
  metrics <- history$history

  # Crear dataframe para ggplot
  df <- data.frame(
    epoch = rep(1:length(metrics$loss), 2),
    value = c(metrics$loss, metrics$val_loss,
      metrics$accuracy, metrics$val_accuracy),
    metric = rep(c("Loss", "Accuracy"), each = length(metrics$loss)),
    dataset = rep(rep(c("Train", "Validation"), each =
length(metrics$loss)), 2)
  )
}

```

```

# Gráfico de pérdida
p_loss <- ggplot(subset(df, metric == "Loss"), aes(epoch, value, color
= dataset)) +
  geom_line() + geom_point() +
  labs(title = "Training and Validation Loss", y = "Loss") +
  theme_minimal()

# Gráfico de precisión
p_acc <- ggplot(subset(df, metric == "Accuracy"), aes(epoch, value,
color = dataset)) +
  geom_line() + geom_point() +
  labs(title = "Training and Validation Accuracy", y = "Accuracy") +
  theme_minimal()

grid.arrange(p_loss, p_acc, ncol = 1)
}

# Visualizar algunas imágenes de ejemplo
plot_sample_images <- function(generator, n = 6) {
  batch <- generator_next(generator)
  images <- batch[[1]]
  labels <- batch[[2]]

  # Configurar área de gráfico
  par(mfrow = c(2, 3), mar = c(1, 1, 2, 1))

  for (i in 1:n) {
    # Extraer y normalizar la imagen (convertir a rango [0,1])
    img <- images[i,,,1]
    img_norm <- (img - min(img)) / (max(img) - min(img))

    # Crear plot con funciones base de R
    plot(NA, xlim = c(0, 1), ylim = c(0, 1), xaxt = 'n', yaxt = 'n', ann
= FALSE)
    rasterImage(img_norm, 0, 0, 1, 1)
    title(main = ifelse(labels[i] == 1, classes[2], classes[1]), line =
0.5)
    box()
  }
}

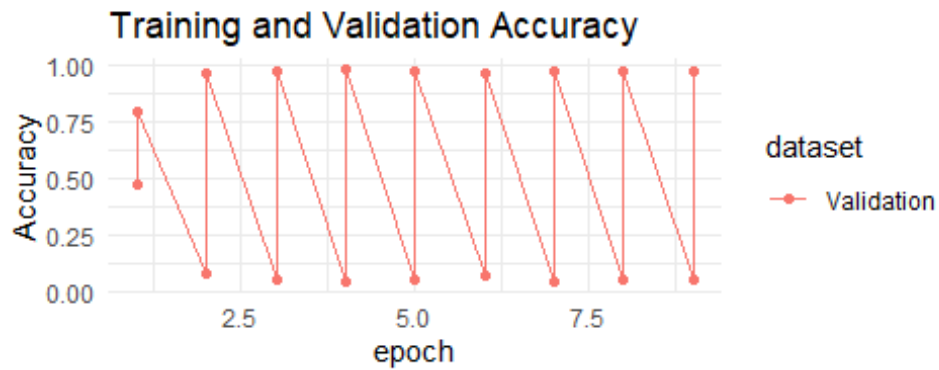
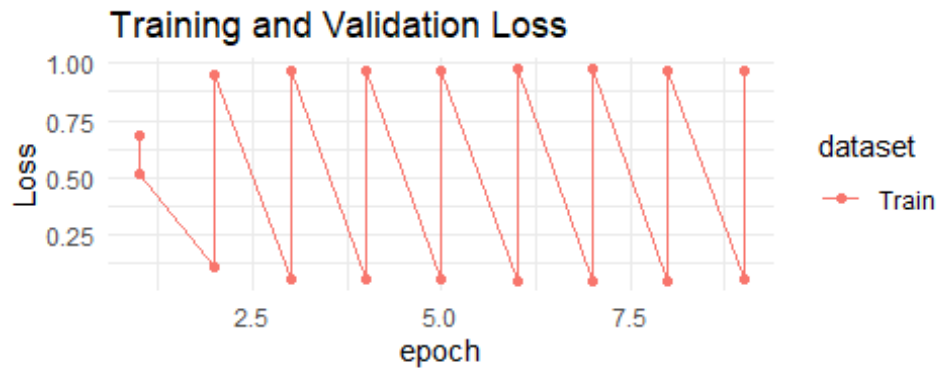
# Mostrar todas las visualizaciones
plot_model_architecture()

```

Arquitectura de la Red Neuronal



```
plot_training_history(history)
```



```
plot_sample_images(train_generator)
```



red_neuronal_convolutacional_cajas en Python

Importación de librerías necesarias

import tensorflow as tf # Framework principal para deep learning

from tensorflow.keras.preprocessing.image import ImageDataGenerator #
Generador de datos aumentados

from tensorflow.keras import layers # Capas para construir modelos

import matplotlib.pyplot as plt # Visualización de gráficos

import os # Operaciones del sistema de archivos

import pandas as pd # Manipulación de datos tabulares

import numpy as np # Cálculos numéricos

import seaborn as sns # Visualización de datos

from sklearn.model_selection import train_test_split # División de datos

```
from sklearn.metrics import confusion_matrix # Matriz de confusión

from tensorflow import keras # API de alto nivel para TensorFlow

from tensorflow.keras.layers import Dense, Conv2D, MaxPooling2D, Flatten, Dropout
# Tipos de capas

from tensorflow.keras.models import Sequential # Modelo secuencial

from tensorflow.keras.optimizers import Adamax # Optimizador

from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint #
Callbacks para entrenamiento

from tensorflow.keras.layers import BatchNormalization # Normalización por lotes

from tqdm import tqdm # Barra de progreso

from PIL import Image # Procesamiento de imágenes

import warnings

warnings.filterwarnings('ignore') # Ignorar advertencias


# Configuración de directorio para guardar modelos
saved_models = "/content/saved_models"


# Parámetros del dataset y modelo
dataset_path = path + "/dataset" # Ruta del conjunto de datos
target_size = (128, 128) # Dimensiones de las imágenes (alto, ancho)
batch_size = 32 # Número de muestras por lote
epochs = 100 # Número de épocas de entrenamiento
classes = ['con_marca', 'sin_marca'] # Clases de clasificación


# Configuración del generador de datos con aumento de datos
datagen = ImageDataGenerator(
    rescale=1.0/255.0, # Normalización de valores de píxeles [0-1]
    rotation_range=20, # Rango de rotación aleatoria (grados)
```

```
width_shift_range=0.2, # Rango de desplazamiento horizontal aleatorio
height_shift_range=0.2, # Rango de desplazamiento vertical aleatorio
horizontal_flip=True, # Volteo horizontal aleatorio
validation_split=0.2 # Porcentaje para validación
)
```

```
# Generador de datos de entrenamiento
```

```
train_generator = datagen.flow_from_directory(
    dataset_path,
    target_size=target_size, # Redimensionamiento de imágenes
    color_mode='grayscale', # Conversión a escala de grises
    batch_size=batch_size, # Tamaño del lote
    class_mode='binary', # Modo de clasificación binaria
    classes=classes, # Clases a considerar
    subset='training' # Subconjunto de entrenamiento
)
```

```
# Generador de datos de validación
```

```
validation_generator = datagen.flow_from_directory(
    dataset_path,
    target_size=target_size,
    color_mode='grayscale',
    batch_size=batch_size,
    class_mode='binary',
    classes=classes,
    subset='validation' # Subconjunto de validación
)
```

```

# Construcción del modelo CNN (Red Neuronal Convolucional)

model = tf.keras.Sequential([

    # Capa convolucional con 32 filtros y activación ReLU

    layers.Conv2D(32, (3, 3), activation='relu', input_shape=(target_size[0],
target_size[1], 1)),

    # Capa de max pooling para reducción dimensional

    layers.MaxPooling2D((2, 2)),

    # Segunda capa convolucional con 64 filtros

    layers.Conv2D(64, (3, 3), activation='relu'),

    layers.MaxPooling2D((2, 2)),

    # Tercera capa convolucional con 128 filtros

    layers.Conv2D(128, (3, 3), activation='relu'),

    layers.MaxPooling2D((2, 2)),

    # Aplanamiento para transición a capas densas

    layers.Flatten(),

    # Capa densa completamente conectada

    layers.Dense(128, activation='relu'),

    # Capa de salida con activación sigmoide (clasificación binaria)

    layers.Dense(1, activation='sigmoid')

])

# Compilación del modelo

model.compile(optimizer='adam', # Optimizador Adam

              loss='binary_crossentropy', # Función de pérdida para clasificación binaria

              metrics=['accuracy']) # Métrica a monitorear

# Resumen de la arquitectura del modelo

```



```
model.summary()
```

```
# Configuración del directorio para guardar modelos
```

```
if not os.path.exists(saved_models):
```

```
    os.makedirs(saved_models)
```

```
# Configuración del callback para guardar los mejores pesos
```

```
filepath = "/content/saved_models/weights-improvement-{epoch:02d}-  
{val_accuracy:.2f}.keras"
```

```
checkpoint = ModelCheckpoint(filepath, monitor='val_accuracy', verbose=1,  
save_best_only=True, mode='max')
```

```
# Configuración de early stopping para evitar sobreajuste
```

```
early_stop = EarlyStopping(  
    monitor='val_loss', # Métrica a monitorear
```

```
    patience=25, # Número de épocas sin mejora antes de detener
```

```
    restore_best_weights=True # Restaurar los mejores pesos encontrados
```

```
)
```

```
# Lista de callbacks a usar durante el entrenamiento
```

```
callbacks_list = [checkpoint, early_stop]
```

```
# Entrenamiento del modelo
```

```
history = model.fit(  
    train_generator, # Datos de entrenamiento
```

```
    epochs=epochs, # Número de épocas
```

```
    validation_data=validation_generator, # Datos de validación
```

```
    callbacks=callbacks_list, # Callbacks configurados
```

```
verbose=1 # Mostrar progreso
)
```

```
# Evaluación del modelo con datos de validación
```

```
loss, accuracy = model.evaluate(validation_generator)
```

```
print(f"Pérdida en validación: {loss}")
```

```
print(f"Precisión en validación: {accuracy}")
```

```
# Guardado del modelo entrenado
```

```
model.save("modelo_cajas.keras")
```

Found 8016 images belonging to 2 classes.

Found 2004 images belonging to 2 classes.

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 126, 126, 32)	320
max_pooling2d (MaxPooling2D)	(None, 63, 63, 32)	0
conv2d_1 (Conv2D)	(None, 61, 61, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 30, 30, 64)	0

conv2d_2 (Conv2D)	(None, 28, 28, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 14, 14, 128)	0
flatten (Flatten)	(None, 25088)	0
dense (Dense)	(None, 128)	3,211,392
dense_1 (Dense)	(None, 1)	129

Total params: 3,304,193 (12.60 MB)

Trainable params: 3,304,193 (12.60 MB)

Non-trainable params: 0 (0.00 B)

Epoch 1/100

251/251 ————— **0s** 103ms/step - accuracy: 0.5370 - loss: 0.6534

Epoch 1: val_accuracy improved from -inf to 0.93663, saving model to /content/saved_models/weights-improvement-01-0.94.keras

251/251 ————— **38s** 126ms/step - accuracy: 0.5376 - loss: 0.6528 - val_accuracy: 0.9366 - val_loss: 0.1330

Epoch 2/100

251/251 ————— **0s** 68ms/step - accuracy: 0.9567 - loss: 0.0898

Epoch 2: val_accuracy improved from 0.93663 to 0.96856, saving model to
/content/saved_models/weights-improvement-02-0.97.keras

251/251 ————— **22s** 89ms/step - accuracy:
0.9568 - loss: 0.0897 - val_accuracy: 0.9686 - val_loss: 0.0633

Epoch 3/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9660 - loss: 0.0609

Epoch 3: val_accuracy improved from 0.96856 to 0.97255, saving model to
/content/saved_models/weights-improvement-03-0.97.keras

251/251 ————— **20s** 79ms/step - accuracy:
0.9659 - loss: 0.0609 - val_accuracy: 0.9726 - val_loss: 0.0553

Epoch 4/100

251/251 ————— **0s** 68ms/step - accuracy:
0.9668 - loss: 0.0548

Epoch 4: val_accuracy improved from 0.97255 to 0.97305, saving model to
/content/saved_models/weights-improvement-04-0.97.keras

251/251 ————— **20s** 81ms/step - accuracy:
0.9668 - loss: 0.0548 - val_accuracy: 0.9731 - val_loss: 0.0447

Epoch 5/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9690 - loss: 0.0504

Epoch 5: val_accuracy improved from 0.97305 to 0.97705, saving model to
/content/saved_models/weights-improvement-05-0.98.keras

251/251 ————— **21s** 82ms/step - accuracy:
0.9690 - loss: 0.0504 - val_accuracy: 0.9770 - val_loss: 0.0447

Epoch 6/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9611 - loss: 0.0668

Epoch 6: val_accuracy did not improve from 0.97705

251/251 ————— **20s** 79ms/step - accuracy:
0.9611 - loss: 0.0668 - val_accuracy: 0.9561 - val_loss: 0.0727

Epoch 7/100

251/251 ————— **0s** 68ms/step - accuracy:
0.9696 - loss: 0.0515

Epoch 7: val_accuracy did not improve from 0.97705

251/251 ————— **20s** 81ms/step - accuracy:
0.9695 - loss: 0.0515 - val_accuracy: 0.9755 - val_loss: 0.0430

Epoch 8/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9707 - loss: 0.0487

Epoch 8: val_accuracy did not improve from 0.97705

251/251 ————— **21s** 82ms/step - accuracy:
0.9707 - loss: 0.0487 - val_accuracy: 0.9636 - val_loss: 0.0552

Epoch 9/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9687 - loss: 0.0459

Epoch 9: val_accuracy did not improve from 0.97705

251/251 ————— **20s** 79ms/step - accuracy:
0.9687 - loss: 0.0459 - val_accuracy: 0.9681 - val_loss: 0.0579

Epoch 10/100

251/251 ————— **0s** 69ms/step - accuracy:
0.9742 - loss: 0.0496

Epoch 10: val_accuracy did not improve from 0.97705

251/251 ————— **21s** 82ms/step - accuracy:
0.9742 - loss: 0.0496 - val_accuracy: 0.9706 - val_loss: 0.0484

Epoch 11/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9713 - loss: 0.0468

Epoch 11: val_accuracy did not improve from 0.97705

251/251 ————— **21s** 82ms/step - accuracy:
0.9713 - loss: 0.0468 - val_accuracy: 0.9731 - val_loss: 0.0435

Epoch 12/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9735 - loss: 0.0443

Epoch 12: val_accuracy did not improve from 0.97705

251/251 ————— **20s** 79ms/step - accuracy:
0.9735 - loss: 0.0443 - val_accuracy: 0.9716 - val_loss: 0.0450

Epoch 13/100

251/251 ————— **0s** 68ms/step - accuracy:
0.9769 - loss: 0.0401

Epoch 13: val_accuracy did not improve from 0.97705

251/251 ————— **20s** 81ms/step - accuracy:
0.9769 - loss: 0.0402 - val_accuracy: 0.9726 - val_loss: 0.0531

Epoch 14/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9713 - loss: 0.0505

Epoch 14: val_accuracy did not improve from 0.97705

251/251 ————— **22s** 86ms/step - accuracy:
0.9714 - loss: 0.0505 - val_accuracy: 0.9686 - val_loss: 0.0628

Epoch 15/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9734 - loss: 0.0481

Epoch 15: val_accuracy did not improve from 0.97705

251/251 ————— **20s** 80ms/step - accuracy:
0.9734 - loss: 0.0481 - val_accuracy: 0.9736 - val_loss: 0.0436

Epoch 16/100

251/251 ————— **0s** 69ms/step - accuracy:
0.9689 - loss: 0.0485

Epoch 16: val_accuracy did not improve from 0.97705

251/251 ————— **21s** 82ms/step - accuracy:
0.9689 - loss: 0.0485 - val_accuracy: 0.9731 - val_loss: 0.0411

Epoch 17/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9756 - loss: 0.0403

Epoch 17: val_accuracy did not improve from 0.97705

251/251 ————— **21s** 83ms/step - accuracy:
0.9756 - loss: 0.0403 - val_accuracy: 0.9701 - val_loss: 0.0463

Epoch 18/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9769 - loss: 0.0374

Epoch 18: val_accuracy did not improve from 0.97705

251/251 ————— **20s** 79ms/step - accuracy:
0.9769 - loss: 0.0374 - val_accuracy: 0.9746 - val_loss: 0.0448

Epoch 19/100

251/251 ————— **0s** 68ms/step - accuracy:
0.9758 - loss: 0.0401

Epoch 19: val_accuracy did not improve from 0.97705

251/251 ————— **21s** 82ms/step - accuracy:
0.9758 - loss: 0.0401 - val_accuracy: 0.9726 - val_loss: 0.0392

Epoch 20/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9716 - loss: 0.0421

Epoch 20: val_accuracy did not improve from 0.97705

251/251 ————— **21s** 82ms/step - accuracy:
0.9716 - loss: 0.0421 - val_accuracy: 0.9746 - val_loss: 0.0379

Epoch 21/100

251/251 ————— **0s** 69ms/step - accuracy:
0.9781 - loss: 0.0346

Epoch 21: val_accuracy improved from 0.97705 to 0.97754, saving model to
/content/saved_models/weights-improvement-21-0.98.keras

251/251 ————— **41s** 83ms/step - accuracy:
0.9781 - loss: 0.0346 - val_accuracy: 0.9775 - val_loss: 0.0339

Epoch 22/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9793 - loss: 0.0401

Epoch 22: val_accuracy did not improve from 0.97754

251/251 ————— **20s** 81ms/step - accuracy:
0.9793 - loss: 0.0401 - val_accuracy: 0.9736 - val_loss: 0.0386

Epoch 23/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9749 - loss: 0.0456

Epoch 23: val_accuracy improved from 0.97754 to 0.98154, saving model to
/content/saved_models/weights-improvement-23-0.98.keras

251/251 ————— **20s** 80ms/step - accuracy:
0.9749 - loss: 0.0456 - val_accuracy: 0.9815 - val_loss: 0.0332

Epoch 24/100

251/251 ————— **0s** 68ms/step - accuracy:
0.9816 - loss: 0.0385

Epoch 24: val_accuracy did not improve from 0.98154

251/251 ————— **20s** 81ms/step - accuracy:
0.9816 - loss: 0.0385 - val_accuracy: 0.9746 - val_loss: 0.0417

Epoch 25/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9785 - loss: 0.0366

Epoch 25: val_accuracy did not improve from 0.98154

251/251 ————— **20s** 81ms/step - accuracy:
0.9785 - loss: 0.0366 - val_accuracy: 0.9726 - val_loss: 0.0379

Epoch 26/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9745 - loss: 0.0413

Epoch 26: val_accuracy did not improve from 0.98154

251/251 ————— **20s** 78ms/step - accuracy:
0.9745 - loss: 0.0413 - val_accuracy: 0.9760 - val_loss: 0.0416

Epoch 27/100

251/251 ————— **0s** 68ms/step - accuracy:
0.9716 - loss: 0.0422

Epoch 27: val_accuracy did not improve from 0.98154

251/251 ————— **20s** 81ms/step - accuracy:
0.9716 - loss: 0.0422 - val_accuracy: 0.9810 - val_loss: 0.0325

Epoch 28/100

251/251 ————— **0s** 64ms/step - accuracy:
0.9689 - loss: 0.0435

Epoch 28: val_accuracy did not improve from 0.98154

251/251 ————— **21s** 85ms/step - accuracy:
0.9689 - loss: 0.0435 - val_accuracy: 0.9711 - val_loss: 0.0396

Epoch 29/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9767 - loss: 0.0397

Epoch 29: val_accuracy did not improve from 0.98154

251/251 ————— **20s** 78ms/step - accuracy:
0.9767 - loss: 0.0397 - val_accuracy: 0.9765 - val_loss: 0.0361

Epoch 30/100

251/251 ————— **0s** 68ms/step - accuracy:
0.9722 - loss: 0.0644

Epoch 30: val_accuracy did not improve from 0.98154

251/251 ————— **20s** 81ms/step - accuracy:
0.9722 - loss: 0.0645 - val_accuracy: 0.9721 - val_loss: 0.0503

Epoch 31/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9741 - loss: 0.0448

Epoch 31: val_accuracy did not improve from 0.98154

251/251 ————— **21s** 85ms/step - accuracy:
0.9741 - loss: 0.0448 - val_accuracy: 0.9726 - val_loss: 0.0491

Epoch 32/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9757 - loss: 0.0478

Epoch 32: val_accuracy did not improve from 0.98154

251/251 ————— **21s** 86ms/step - accuracy:
0.9757 - loss: 0.0478 - val_accuracy: 0.9716 - val_loss: 0.0419

Epoch 33/100

251/251 ————— **0s** 67ms/step - accuracy:
0.9732 - loss: 0.0453

Epoch 33: val_accuracy did not improve from 0.98154

251/251 ————— **40s** 81ms/step - accuracy:
0.9732 - loss: 0.0453 - val_accuracy: 0.9750 - val_loss: 0.0379

Epoch 34/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9741 - loss: 0.0430

Epoch 34: val_accuracy did not improve from 0.98154

251/251 ————— **20s** 79ms/step - accuracy:
0.9741 - loss: 0.0430 - val_accuracy: 0.9780 - val_loss: 0.0351

Epoch 35/100

251/251 ————— **0s** 67ms/step - accuracy:
0.9776 - loss: 0.0377

Epoch 35: val_accuracy did not improve from 0.98154

251/251 ————— **21s** 80ms/step - accuracy:
0.9776 - loss: 0.0377 - val_accuracy: 0.9785 - val_loss: 0.0395

Epoch 36/100

251/251 ————— **0s** 68ms/step - accuracy:
0.9743 - loss: 0.0384

Epoch 36: val_accuracy did not improve from 0.98154

251/251 ————— **22s** 88ms/step - accuracy:
0.9743 - loss: 0.0384 - val_accuracy: 0.9815 - val_loss: 0.0335

Epoch 37/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9746 - loss: 0.0379

Epoch 37: val_accuracy did not improve from 0.98154

251/251 ————— **21s** 82ms/step - accuracy:
0.9746 - loss: 0.0379 - val_accuracy: 0.9800 - val_loss: 0.0337

Epoch 38/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9815 - loss: 0.0370

Epoch 38: val_accuracy did not improve from 0.98154

251/251 ————— **20s** 79ms/step - accuracy:
0.9815 - loss: 0.0370 - val_accuracy: 0.9785 - val_loss: 0.0321

Epoch 39/100

251/251 ————— **0s** 69ms/step - accuracy:
0.9784 - loss: 0.0362

Epoch 39: val_accuracy did not improve from 0.98154

251/251 ————— **21s** 83ms/step - accuracy:
0.9784 - loss: 0.0362 - val_accuracy: 0.9795 - val_loss: 0.0387

Epoch 40/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9759 - loss: 0.0389

Epoch 40: val_accuracy improved from 0.98154 to 0.98353, saving model to
/content/saved_models/weights-improvement-40-0.98.keras

251/251 ————— **22s** 87ms/step - accuracy:
0.9759 - loss: 0.0389 - val_accuracy: 0.9835 - val_loss: 0.0344

Epoch 41/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9803 - loss: 0.0344

Epoch 41: val_accuracy did not improve from 0.98353

251/251 ————— **20s** 79ms/step - accuracy:
0.9803 - loss: 0.0344 - val_accuracy: 0.9741 - val_loss: 0.0385

Epoch 42/100

251/251 ————— **0s** 69ms/step - accuracy:
0.9785 - loss: 0.0368

Epoch 42: val_accuracy did not improve from 0.98353

251/251 ————— **21s** 83ms/step - accuracy:
0.9785 - loss: 0.0368 - val_accuracy: 0.9830 - val_loss: 0.0351

Epoch 43/100

251/251 ————— **0s** 67ms/step - accuracy:
0.9833 - loss: 0.0317

Epoch 43: val_accuracy improved from 0.98353 to 0.98603, saving model to
/content/saved_models/weights-improvement-43-0.99.keras

251/251 ————— **21s** 84ms/step - accuracy:
0.9833 - loss: 0.0318 - val_accuracy: 0.9860 - val_loss: 0.0275

Epoch 44/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9804 - loss: 0.0368

Epoch 44: val_accuracy improved from 0.98603 to 0.99102, saving model to
/content/saved_models/weights-improvement-44-0.99.keras

251/251 ————— **20s** 80ms/step - accuracy:
0.9804 - loss: 0.0368 - val_accuracy: 0.9910 - val_loss: 0.0255

Epoch 45/100

251/251 ————— **0s** 69ms/step - accuracy:
0.9834 - loss: 0.0317

Epoch 45: val_accuracy did not improve from 0.99102

251/251 ————— **21s** 82ms/step - accuracy:
0.9834 - loss: 0.0317 - val_accuracy: 0.9810 - val_loss: 0.0303

Epoch 46/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9833 - loss: 0.0312

Epoch 46: val_accuracy did not improve from 0.99102

251/251 ————— **21s** 82ms/step - accuracy:
0.9833 - loss: 0.0312 - val_accuracy: 0.9731 - val_loss: 0.0430

Epoch 47/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9773 - loss: 0.0497

Epoch 47: val_accuracy did not improve from 0.99102

251/251 ————— **20s** 79ms/step - accuracy:
0.9773 - loss: 0.0498 - val_accuracy: 0.9795 - val_loss: 0.0358

Epoch 48/100

251/251 ————— **0s** 69ms/step - accuracy:
0.9802 - loss: 0.0370

Epoch 48: val_accuracy did not improve from 0.99102

251/251 ————— **21s** 83ms/step - accuracy:
0.9802 - loss: 0.0370 - val_accuracy: 0.9790 - val_loss: 0.0373

Epoch 49/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9808 - loss: 0.0327

Epoch 49: val_accuracy did not improve from 0.99102

251/251 ————— **21s** 82ms/step - accuracy:
0.9808 - loss: 0.0327 - val_accuracy: 0.9785 - val_loss: 0.0412

Epoch 50/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9801 - loss: 0.0355

Epoch 50: val_accuracy did not improve from 0.99102

251/251 ————— **20s** 79ms/step - accuracy:
0.9801 - loss: 0.0355 - val_accuracy: 0.9810 - val_loss: 0.0347

Epoch 51/100

251/251 ————— **0s** 70ms/step - accuracy:
0.9801 - loss: 0.0338

Epoch 51: val_accuracy did not improve from 0.99102

251/251 ————— **21s** 83ms/step - accuracy:
0.9801 - loss: 0.0339 - val_accuracy: 0.9805 - val_loss: 0.0332

Epoch 52/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9842 - loss: 0.0301

Epoch 52: val_accuracy did not improve from 0.99102

251/251 ————— **21s** 82ms/step - accuracy:
0.9842 - loss: 0.0301 - val_accuracy: 0.9880 - val_loss: 0.0260

Epoch 53/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9859 - loss: 0.0325

Epoch 53: val_accuracy did not improve from 0.99102

251/251 ————— **20s** 78ms/step - accuracy:
0.9858 - loss: 0.0325 - val_accuracy: 0.9716 - val_loss: 0.0491

Epoch 54/100

251/251 ————— **0s** 69ms/step - accuracy:
0.9836 - loss: 0.0296

Epoch 54: val_accuracy did not improve from 0.99102

251/251 ————— **21s** 82ms/step - accuracy:
0.9836 - loss: 0.0296 - val_accuracy: 0.9890 - val_loss: 0.0245

Epoch 55/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9850 - loss: 0.0301

Epoch 55: val_accuracy did not improve from 0.99102

251/251 ————— **22s** 86ms/step - accuracy:
0.9850 - loss: 0.0301 - val_accuracy: 0.9840 - val_loss: 0.0286

Epoch 56/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9855 - loss: 0.0260

Epoch 56: val_accuracy did not improve from 0.99102

251/251 ————— **22s** 86ms/step - accuracy:
0.9855 - loss: 0.0260 - val_accuracy: 0.9860 - val_loss: 0.0235

Epoch 57/100

251/251 ————— **0s** 68ms/step - accuracy:
0.9845 - loss: 0.0327

Epoch 57: val_accuracy did not improve from 0.99102

251/251 ————— **20s** 81ms/step - accuracy:
0.9845 - loss: 0.0326 - val_accuracy: 0.9875 - val_loss: 0.0186

Epoch 58/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9913 - loss: 0.0205

Epoch 58: val_accuracy did not improve from 0.99102

251/251 ————— **21s** 82ms/step - accuracy:
0.9913 - loss: 0.0205 - val_accuracy: 0.9795 - val_loss: 0.0361

Epoch 59/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9835 - loss: 0.0266

Epoch 59: val_accuracy did not improve from 0.99102

251/251 ————— **40s** 79ms/step - accuracy:
0.9835 - loss: 0.0266 - val_accuracy: 0.9840 - val_loss: 0.0274

Epoch 60/100

251/251 ————— **0s** 67ms/step - accuracy:
0.9854 - loss: 0.0380

Epoch 60: val_accuracy did not improve from 0.99102

251/251 ————— **20s** 81ms/step - accuracy:
0.9854 - loss: 0.0380 - val_accuracy: 0.9885 - val_loss: 0.0257

Epoch 61/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9904 - loss: 0.0225

Epoch 61: val_accuracy did not improve from 0.99102

251/251 ————— **22s** 86ms/step - accuracy:
0.9904 - loss: 0.0225 - val_accuracy: 0.9845 - val_loss: 0.0213

Epoch 62/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9848 - loss: 0.0308

Epoch 62: val_accuracy improved from 0.99102 to 0.99301, saving model to
/content/saved_models/weights-improvement-62-0.99.keras

251/251 ————— **22s** 87ms/step - accuracy:
0.9847 - loss: 0.0308 - val_accuracy: 0.9930 - val_loss: 0.0201

Epoch 63/100

251/251 ————— **0s** 67ms/step - accuracy:
0.9899 - loss: 0.0214

Epoch 63: val_accuracy did not improve from 0.99301

251/251 ————— **40s** 82ms/step - accuracy:
0.9899 - loss: 0.0214 - val_accuracy: 0.9885 - val_loss: 0.0211

Epoch 64/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9890 - loss: 0.0207

Epoch 64: val_accuracy did not improve from 0.99301

251/251 ————— **20s** 79ms/step - accuracy:
0.9890 - loss: 0.0208 - val_accuracy: 0.9890 - val_loss: 0.0186

Epoch 65/100

251/251 ————— **0s** 68ms/step - accuracy:
0.9840 - loss: 0.0325

Epoch 65: val_accuracy did not improve from 0.99301

251/251 ————— **21s** 81ms/step - accuracy:
0.9840 - loss: 0.0325 - val_accuracy: 0.9895 - val_loss: 0.0221

Epoch 66/100

251/251 ————— **0s** 69ms/step - accuracy:
0.9895 - loss: 0.0200

Epoch 66: val_accuracy did not improve from 0.99301

251/251 ————— **21s** 83ms/step - accuracy:
0.9895 - loss: 0.0200 - val_accuracy: 0.9885 - val_loss: 0.0183

Epoch 67/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9934 - loss: 0.0173

Epoch 67: val_accuracy did not improve from 0.99301

251/251 ————— **22s** 86ms/step - accuracy:
0.9934 - loss: 0.0173 - val_accuracy: 0.9855 - val_loss: 0.0248

Epoch 68/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9874 - loss: 0.0208

Epoch 68: val_accuracy did not improve from 0.99301

251/251 ————— **20s** 79ms/step - accuracy:
0.9874 - loss: 0.0208 - val_accuracy: 0.9895 - val_loss: 0.0168

Epoch 69/100

251/251 ————— **0s** 69ms/step - accuracy:
0.9871 - loss: 0.0229

Epoch 69: val_accuracy did not improve from 0.99301

251/251 ————— **21s** 82ms/step - accuracy:
0.9871 - loss: 0.0229 - val_accuracy: 0.9701 - val_loss: 0.0744

Epoch 70/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9821 - loss: 0.0420

Epoch 70: val_accuracy did not improve from 0.99301

251/251 ————— **22s** 86ms/step - accuracy:
0.9821 - loss: 0.0419 - val_accuracy: 0.9850 - val_loss: 0.0260

Epoch 71/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9881 - loss: 0.0209

Epoch 71: val_accuracy did not improve from 0.99301

251/251 ————— **20s** 79ms/step - accuracy:
0.9881 - loss: 0.0209 - val_accuracy: 0.9890 - val_loss: 0.0204

Epoch 72/100

251/251 ————— **0s** 70ms/step - accuracy:
0.9894 - loss: 0.0225

Epoch 72: val_accuracy did not improve from 0.99301

251/251 ————— **21s** 84ms/step - accuracy:
0.9895 - loss: 0.0224 - val_accuracy: 0.9905 - val_loss: 0.0195

Epoch 73/100

251/251 ————— **0s** 71ms/step - accuracy:
0.9843 - loss: 0.0313

Epoch 73: val_accuracy did not improve from 0.99301

251/251 ————— **22s** 88ms/step - accuracy:
0.9843 - loss: 0.0313 - val_accuracy: 0.9870 - val_loss: 0.0215

Epoch 74/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9911 - loss: 0.0197

Epoch 74: val_accuracy did not improve from 0.99301

251/251 ————— **20s** 79ms/step - accuracy:
0.9911 - loss: 0.0197 - val_accuracy: 0.9920 - val_loss: 0.0167

Epoch 75/100

251/251 ————— **0s** 69ms/step - accuracy:
0.9873 - loss: 0.0331

Epoch 75: val_accuracy improved from 0.99301 to 0.99551, saving model to
/content/saved_models/weights-improvement-75-1.00.keras

251/251 ————— **21s** 83ms/step - accuracy:
0.9873 - loss: 0.0331 - val_accuracy: 0.9955 - val_loss: 0.0174

Epoch 76/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9878 - loss: 0.0229

Epoch 76: val_accuracy did not improve from 0.99551

251/251 ————— **21s** 83ms/step - accuracy:
0.9878 - loss: 0.0229 - val_accuracy: 0.9930 - val_loss: 0.0144

Epoch 77/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9901 - loss: 0.0203

Epoch 77: val_accuracy did not improve from 0.99551

251/251 ————— **20s** 78ms/step - accuracy:
0.9901 - loss: 0.0203 - val_accuracy: 0.9820 - val_loss: 0.0303

Epoch 78/100

251/251 ————— **0s** 70ms/step - accuracy:
0.9824 - loss: 0.0342

Epoch 78: val_accuracy did not improve from 0.99551

251/251 ————— **21s** 83ms/step - accuracy:
0.9824 - loss: 0.0342 - val_accuracy: 0.9910 - val_loss: 0.0173

Epoch 79/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9880 - loss: 0.0192

Epoch 79: val_accuracy did not improve from 0.99551

251/251 ————— **21s** 83ms/step - accuracy:
0.9880 - loss: 0.0192 - val_accuracy: 0.9880 - val_loss: 0.0153

Epoch 80/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9871 - loss: 0.0227

Epoch 80: val_accuracy did not improve from 0.99551

251/251 ————— **20s** 79ms/step - accuracy:
0.9871 - loss: 0.0227 - val_accuracy: 0.9875 - val_loss: 0.0222

Epoch 81/100

251/251 ————— **0s** 69ms/step - accuracy:
0.9886 - loss: 0.0191

Epoch 81: val_accuracy did not improve from 0.99551

251/251 ————— **21s** 81ms/step - accuracy:
0.9886 - loss: 0.0191 - val_accuracy: 0.9895 - val_loss: 0.0199

Epoch 82/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9921 - loss: 0.0166

Epoch 82: val_accuracy did not improve from 0.99551

251/251 ————— **21s** 82ms/step - accuracy:
0.9921 - loss: 0.0166 - val_accuracy: 0.9900 - val_loss: 0.0243

Epoch 83/100

251/251 ————— **0s** 65ms/step - accuracy:
0.9896 - loss: 0.0168

Epoch 83: val_accuracy did not improve from 0.99551

251/251 ————— **22s** 86ms/step - accuracy:
0.9895 - loss: 0.0169 - val_accuracy: 0.9736 - val_loss: 0.0439

Epoch 84/100

251/251 ————— **0s** 69ms/step - accuracy:
0.9803 - loss: 0.0378

Epoch 84: val_accuracy did not improve from 0.99551

251/251 ————— **21s** 82ms/step - accuracy:
0.9803 - loss: 0.0378 - val_accuracy: 0.9810 - val_loss: 0.0332

Epoch 85/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9862 - loss: 0.0222

Epoch 85: val_accuracy did not improve from 0.99551

251/251 ————— **42s** 86ms/step - accuracy:
0.9862 - loss: 0.0222 - val_accuracy: 0.9865 - val_loss: 0.0201

Epoch 86/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9920 - loss: 0.0154

Epoch 86: val_accuracy did not improve from 0.99551

251/251 ————— **20s** 79ms/step - accuracy:
0.9920 - loss: 0.0155 - val_accuracy: 0.9895 - val_loss: 0.0145

Epoch 87/100

251/251 ————— **0s** 68ms/step - accuracy:
0.9915 - loss: 0.0142

Epoch 87: val_accuracy did not improve from 0.99551

251/251 ————— **20s** 81ms/step - accuracy:
0.9915 - loss: 0.0142 - val_accuracy: 0.9900 - val_loss: 0.0157

Epoch 88/100

251/251 ————— **0s** 66ms/step - accuracy:
0.9907 - loss: 0.0167

Epoch 88: val_accuracy did not improve from 0.99551

251/251 ————— **21s** 82ms/step - accuracy:
0.9907 - loss: 0.0168 - val_accuracy: 0.9905 - val_loss: 0.0196

Epoch 89/100

251/251 ————— **0s** 67ms/step - accuracy:
0.9823 - loss: 0.0660

Epoch 89: val_accuracy did not improve from 0.99551

251/251 ————— **20s** 80ms/step - accuracy:
0.9823 - loss: 0.0659 - val_accuracy: 0.9865 - val_loss: 0.0254

Epoch 90/100

251/251 ————— **0s** 69ms/step - accuracy:
0.9842 - loss: 0.0311

Epoch 90: val_accuracy did not improve from 0.99551

251/251 ————— **21s** 83ms/step - accuracy:
0.9842 - loss: 0.0311 - val_accuracy: 0.9900 - val_loss: 0.0177

Epoch 91/100

251/251 ————— **0s** 67ms/step - accuracy:
0.9897 - loss: 0.0217

Epoch 91: val_accuracy did not improve from 0.99551

251/251 ————— **21s** 84ms/step - accuracy:
0.9897 - loss: 0.0217 - val_accuracy: 0.9895 - val_loss: 0.0197

Epoch 92/100

251/251 ————— **0s** 67ms/step - accuracy:
0.9905 - loss: 0.0173

Epoch 92: val_accuracy did not improve from 0.99551

251/251 ————— **20s** 81ms/step - accuracy:
0.9905 - loss: 0.0173 - val_accuracy: 0.9950 - val_loss: 0.0145

Epoch 93/100

251/251 ————— **0s** 69ms/step - accuracy:
0.9798 - loss: 0.0421

Epoch 93: val_accuracy did not improve from 0.99551

251/251 ————— **23s** 89ms/step - accuracy:
0.9798 - loss: 0.0421 - val_accuracy: 0.9890 - val_loss: 0.0209

Epoch 94/100

251/251 ————— **0s** 70ms/step - accuracy:
0.9905 - loss: 0.0215

Epoch 94: val_accuracy did not improve from 0.99551

251/251 ————— **21s** 83ms/step - accuracy:
0.9905 - loss: 0.0215 - val_accuracy: 0.9895 - val_loss: 0.0191

Epoch 95/100

251/251 ————— **0s** 67ms/step - accuracy:
0.9896 - loss: 0.0203

Epoch 95: val_accuracy did not improve from 0.99551

251/251 ————— **21s** 83ms/step - accuracy:
0.9896 - loss: 0.0203 - val_accuracy: 0.9895 - val_loss: 0.0202

Epoch 96/100

251/251 ————— **0s** 67ms/step - accuracy:
0.9927 - loss: 0.0168

Epoch 96: val_accuracy did not improve from 0.99551

251/251 ————— **20s** 80ms/step - accuracy:
0.9927 - loss: 0.0168 - val_accuracy: 0.9905 - val_loss: 0.0183

Epoch 97/100

251/251 ————— **0s** 70ms/step - accuracy:
0.9914 - loss: 0.0175

Epoch 97: val_accuracy did not improve from 0.99551

251/251 ————— **21s** 83ms/step - accuracy:
0.9914 - loss: 0.0175 - val_accuracy: 0.9910 - val_loss: 0.0170

Epoch 98/100

251/251 ————— **0s** 70ms/step - accuracy:
0.9893 - loss: 0.0199

Epoch 98: val_accuracy did not improve from 0.99551

251/251 ————— **23s** 91ms/step - accuracy:
0.9892 - loss: 0.0200 - val_accuracy: 0.9721 - val_loss: 0.0501

Epoch 99/100

251/251 ————— **0s** 67ms/step - accuracy:
0.9866 - loss: 0.0280

Epoch 99: val_accuracy did not improve from 0.99551

251/251 ————— **21s** 83ms/step - accuracy:
0.9866 - loss: 0.0280 - val_accuracy: 0.9770 - val_loss: 0.0427

Epoch 100/100

251/251 ————— **0s** 69ms/step - accuracy:
0.9866 - loss: 0.0255

Epoch 100: val_accuracy did not improve from 0.99551

251/251 ————— **22s** 89ms/step - accuracy:
0.9866 - loss: 0.0254 - val_accuracy: 0.9910 - val_loss: 0.0147

63/63 ————— **4s** 59ms/step - accuracy:
0.9870 - loss: 0.0217

Pérdida en validación: 0.01993519254028797

Precisión en validación: 0.9890219569206238

validation_generator.reset()

try:

 x_batch, y_batch = next(validation_generator)

```

pred_batch = model.predict(x_batch, verbose=0)

# Conversión robusta a etiquetas de clase
pred_labels = np.squeeze((pred_batch > 0.5).astype(int)) # Quitar dimensiones
extra
true_labels = np.squeeze(y_batch.astype(int)) # Asegurar enteros

# Visualización
plt.figure(figsize=(15, 7))
for i in range(min(5, len(x_batch))):
    plt.subplot(1, 5, i+1)
    plt.imshow(x_batch[i].squeeze(), cmap='gray')

# Manejo seguro de índices
true_idx = min(max(int(true_labels[i]), 0), len(classes)-1)
pred_idx = min(max(int(pred_labels[i]), 0), len(classes)-1)

color = 'green' if true_idx == pred_idx else 'red'
plt.title(f'Real: {classes[true_idx]}\nPred: {classes[pred_idx]}, color=color)
plt.axis('off')

plt.tight_layout()
plt.savefig('predictions_examples.png')
plt.show()

except Exception as e:
    print(f"Error al generar predicciones: {e}")
    print(f"Forma de y_batch: {y_batch.shape if 'y_batch' in locals() else 'N/A'}")

```



```
print(f"Forma de pred_batch: {pred_batch.shape if 'pred_batch' in locals() else 'N/A'}")
```



```
from sklearn.metrics import confusion_matrix, classification_report
```

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
from tensorflow.keras.utils import plot_model
```

```
# Configurar generador para clasificación binaria (sin one-hot encoding)
```

```
valid_gen = datagen.flow_from_directory(
```

```
    dataset_path,
```

```
    target_size=target_size,
```

```
    batch_size=batch_size,
```

```
    color_mode='grayscale',
```

```
    class_mode='binary', # Cambiado a 'binary' en lugar de 'categorical'
```

```
    classes=classes,
```

```
    subset='validation',
```

```
    shuffle=False # Importante para la matriz de confusión
```

```
)
```

```
# 1. Curvas de Entrenamiento (Accuracy/Loss)
```

```
plt.figure(figsize=(12, 5))
```

```
plt.subplot(1, 2, 1)
plt.plot(history.history['accuracy'], label='Entrenamiento')
plt.plot(history.history['val_accuracy'], label='Validación')
plt.title('Precisión por Época')
plt.ylabel('Precisión')
plt.xlabel('Época')
plt.legend()
```

```
plt.subplot(1, 2, 2)
plt.plot(history.history['loss'], label='Entrenamiento')
plt.plot(history.history['val_loss'], label='Validación')
plt.title('Pérdida por Época')
plt.ylabel('Pérdida')
plt.xlabel('Época')
plt.legend()
```

```
plt.tight_layout()
plt.savefig('training_curves.png')
plt.show()
```

2. Matriz de Confusión y Reporte de Clasificación

```
valid_gen.reset()
y_pred = model.predict(valid_gen, verbose=1)
y_pred_classes = (y_pred > 0.5).astype(int) # Umbral de decisión 0.5 para binario

# Etiquetas reales (ya vienen como 0/1 en modo binary)
y_true_classes = valid_gen.labels
```

```
# Matriz de confusión
```

```
cm = confusion_matrix(y_true_classes, y_pred_classes)
```

```
plt.figure(figsize=(8, 6))
```

```
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',  
             xticklabels=classes, yticklabels=classes)
```

```
plt.title('Matriz de Confusión')
```

```
plt.xlabel('Predicciones')
```

```
plt.ylabel('Reales')
```

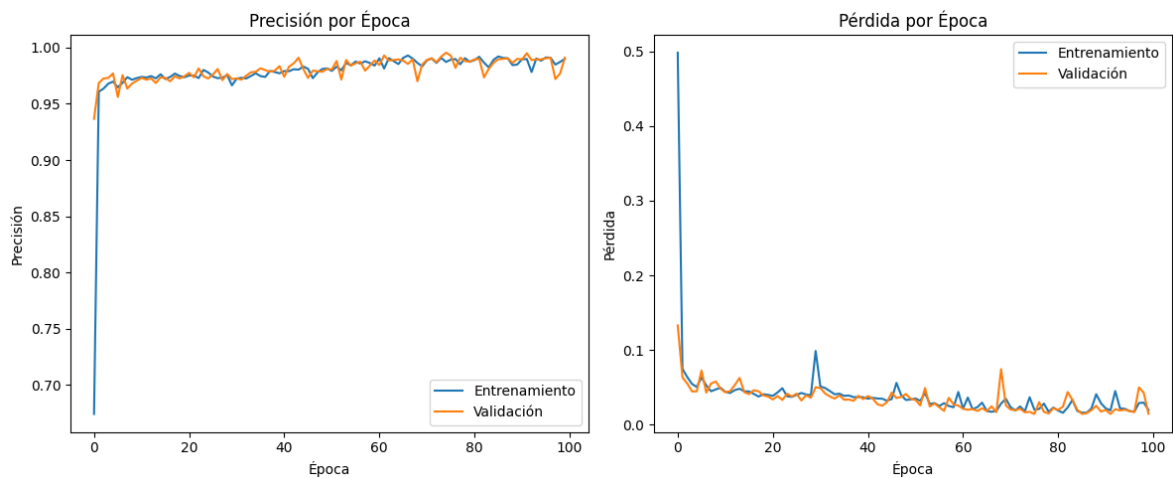
```
plt.savefig('confusion_matrix.png')
```

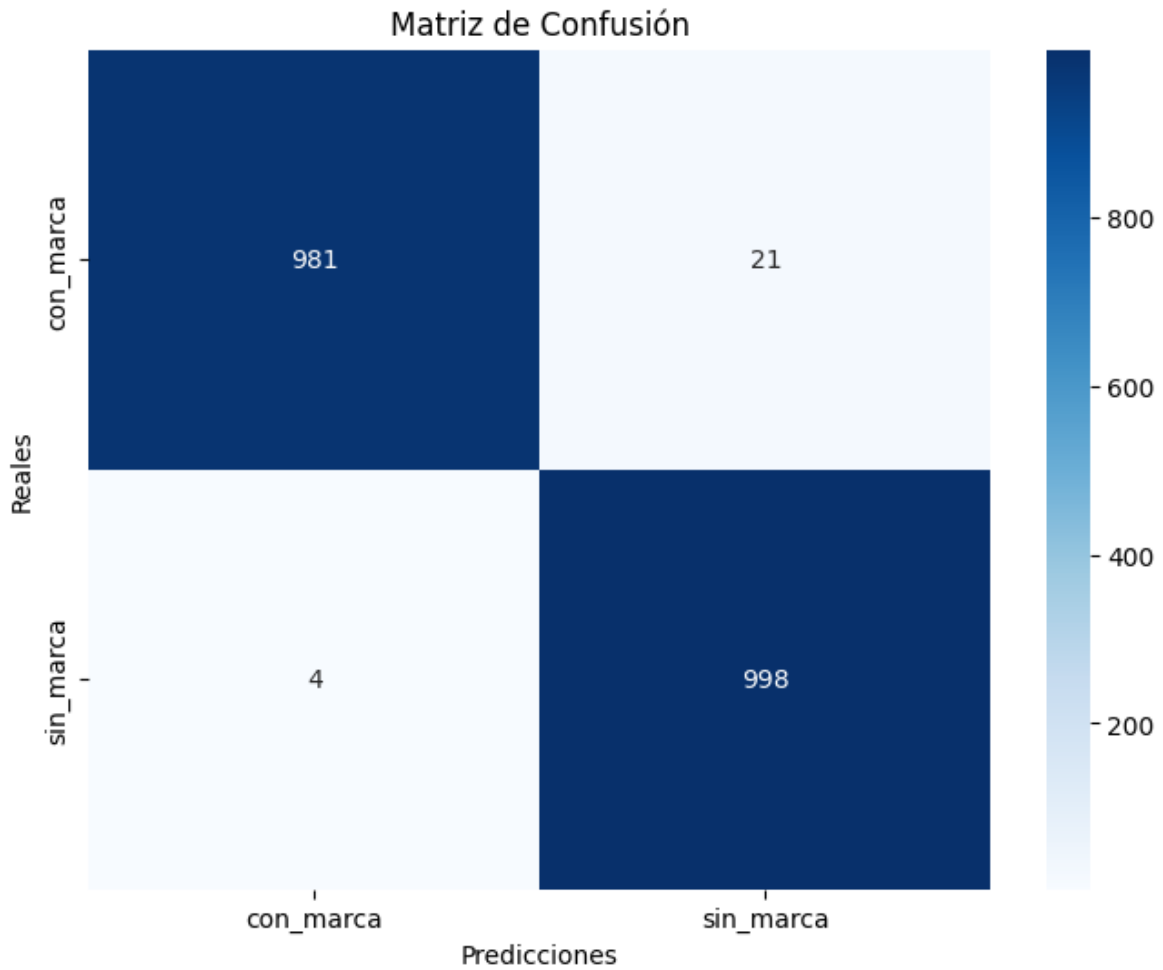
```
plt.show()
```

```
# Reporte de clasificación
```

```
print("\nReporte de Clasificación:")
```

```
print(classification_report(y_true_classes, y_pred_classes, target_names=classes))
```





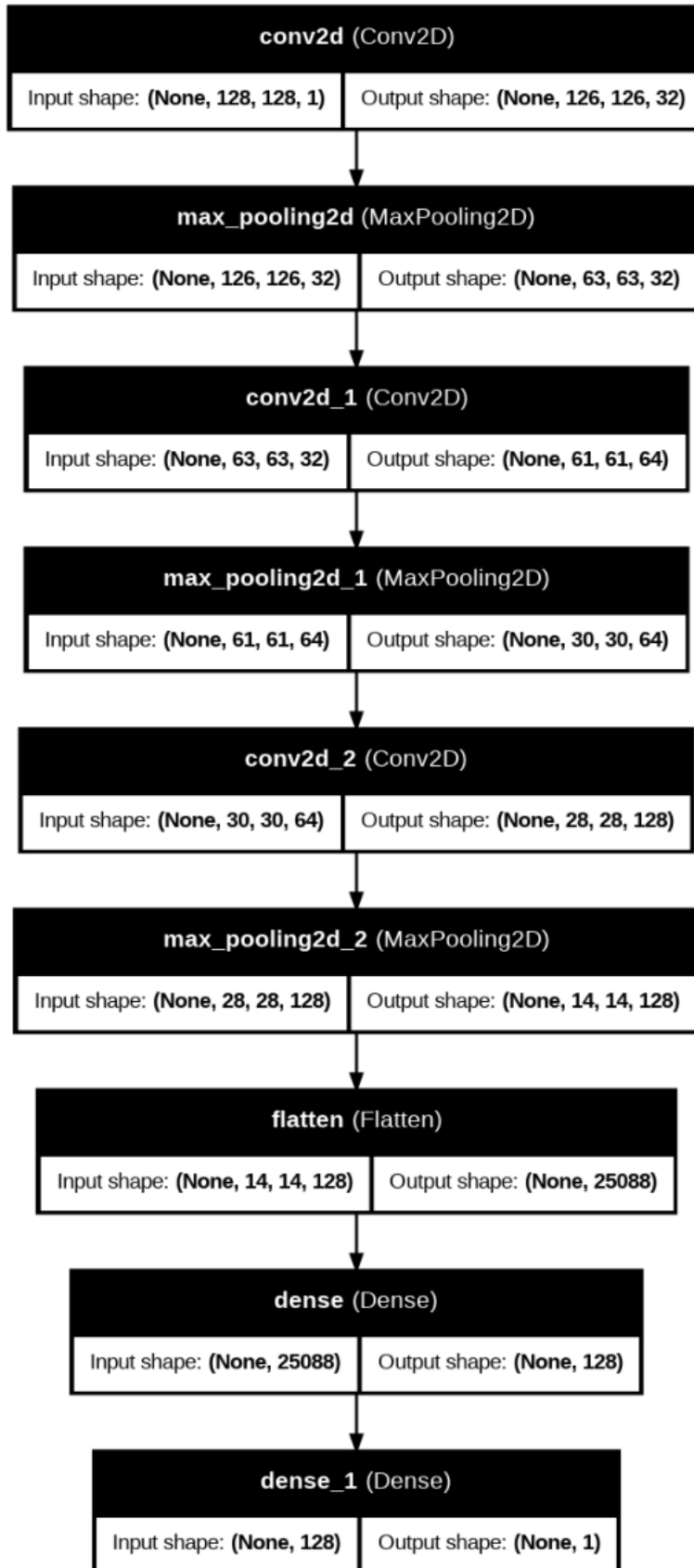
try:

```
# Generar y guardar el diagrama
plot_model(
    model, # Tu modelo (¡con mayúscula!)
    to_file='arquitectura_red_neuronal.png',
    show_shapes=True, # Muestra las formas de entrada/salida
    show_layer_names=True, # Muestra nombres de capas
    dpi=96, # Resolución de la imagen
    expand_nested=True # Detalla capas dentro de bloques (ej. Secuenciales)
)
```

```
# Mostrar la imagen en el notebook o entorno local  
img = plt.imread('arquitectura_red_neuronal.png')  
plt.figure(figsize=(20, 15))  
plt.imshow(img)  
plt.axis('off')  
plt.show()
```

except ImportError:

```
print("Instala las dependencias: pip install pydot graphviz")
```



```
import visualkeras
```

```
# Para modelos secuenciales
```

```
visualkeras.layered_view(  
    model,  
    legend=True, # mostrar leyenda  
    to_file='visualkeras.png',  
    spacing=30 # espacio entre capas  
)
```

